Foreman notebook, §§15–22

The centre ray read off the list; the five-mark lists along their hinge trees; the masters of $\binom{[5]}{2}$ and two marks doubled, closed; the reading of a ray through the origin; the three-mark centre, set as an examination; the law of the stride; the gauge end; the torus. With a ledger of what is sorted, open and corrected

22–24 October 2026

This file brings together the entries of 22–24 October 2026 and replaces them: §§15–19 with the questions of §20 (22–23 October), the answers to those questions (23 October), §21 "Which negatives are absolute" and §22 "The negatives hashed out" (24 October). §15 follows §14.17 of the running notebook. Where two entries proved or computed the same thing, one statement stands: §20 prints each question at the head of its answer, with the later readings of the five-mark frame, the transform, the six-mark list and the merit placed where they answer it; §21 is the gauge end, its first form, its form for every number of marks and the readings of both entries merged; §22 is the torus. Corrections made by later entries are applied in place. The ongoing things, the sortings of negatives and hypotheses, the open lists, the corrections and the receipts, are merged in the Ledger at the end, which also gives the old numbers against the new; numbers 15.x–19.x *are unchanged. Handbook citations are by printed number and page of the handbook PDF of 20 October 2026; "the note" is the tree-coaction note of 22 October 2026. Every object is a member of the serif family or a construction of the handbook, written with its slots; classical names appear only as aliases, in the last section.*

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15 The centre ray read off the list

The neighbours of the five-mark centre are the values of the centre ray of the list (Def. 35.9, p. 309), the period $P_{\mathcal{B}}(\sigma\mathbf{1};\sigma,\sigma)$. The interest of the list is one Gramma (Thm. 8.3, p. 54), the period is its residue on the balance wall, and the residue is the Gramma of the letter columns (Thm. 2.9, p. 15). Its share side (Prop. 4.2, p. 25) is the working form of this entry: a fibre integral of Gammas, one per basis for each letter, pinned by equal incidence. The pin is what the list contributes. Closing the fibre at the negative strengths (Thm. 4.5, p. 26) lands on the pairs of k-tuples of bases with equal incidence; the residue at zero shares is the corner of Thm. 38.5(i); the native shares (Thm. 4.3, p. 25) are uniform, the share side of the table's gauge; and the banana is the list whose pin identifies the two letters mark by mark, which turns its share side into the Mellin read of a power of one Gramma.

15.1 The centre period on its share side

Proposition 15.1 (The share side of the centre)**.** [derived]*; receipts* [computed] *Let* $(\mathcal{E},\mathcal{B})$ *be a list,* $n=|\mathcal{E}|\geq 2$*,* $e_0\in\mathcal{E}$*, and* $\operatorname{Re}\sigma>0$*. Then*

$$\Gamma(\sigma)^2\,P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma)=\int_{\substack{|\mathbf{a}|=\sigma,\ |\mathbf{b}|=\sigma\\ \sum_T a_T\mathbf{1}_T=\sum_T b_T\mathbf{1}_T}}\ \prod_{T\in\mathcal{B}}\Gamma(a_T)\,\Gamma(b_T),$$

the fibre integral of Def. 4.1, p. 25, with one share a_T *for the letter* K *and one share* b_T *for the letter* Ψ *at every basis, pinned by the two letter totals and by the incidence at the marks* $e\neq e_0$*; it has dimension* $2|\mathcal{B}|-n-1$*, and the uniform shares* $a_T=b_T=\sigma/|\mathcal{B}|$ *are an admissible real point.*

Proof. By Thm. 8.3, p. 54, $\Gamma(s)\Gamma(t)\,J_{\mathcal{B}}(\mathbf{m};s,t)=\mathsf{G}_{(\boldsymbol{\mu};s;t);t_{\mathcal{B}}}(\mathbf{1})$ with the unit columns $(\mathbf{e}_e;0;0)$ and the letter columns $(\chi_T/2;1;0)$, $(-\chi_T/2;0;1)$. The linear form $\ell(x;v;w)=\sum_e x_e-\frac{2r-n}{2}(v-w)$ is 1 on the unit columns and 0 on the letter columns, and at the index it is $\kappa_{\mathcal{E}}$ (Thm. 8.2(iii), p. 54): the balance wall is the first wall of this Gramma along ℓ, with the letter columns as facet terms. By Thm. 8.10, p. 56, the period is the residue of $J_{\mathcal{B}}$ there, and by Thm. 2.9, p. 15, at $p=0$ that residue is the Gramma of the facet terms. Its share side (Prop. 4.2) has one share per letter column; the pins of the two letter scales are $|\mathbf{a}|=s$, $|\mathbf{b}|=t$, and those of the marks at $\boldsymbol{\mu}=0$ are $\sum_T(a_T-b_T)\chi_T/2\in\mathbb{C}\mathbf{1}$, which with $|\mathbf{a}|=|\mathbf{b}|$ is $\sum_T(a_T-b_T)\mathbf{1}_T=0$. At the centre $s=t=\sigma$. The uniform shares satisfy the pins and have positive real parts. □

Receipts. For $\binom{[2]}{1}$ the fibre has dimension one and the integral is four Gammas on a line (Cor. 4.7, p. 26),

$$\frac{1}{2\pi i}\int\Gamma(a)^2\,\Gamma(\sigma-a)^2\,da=\Gamma^{\,4}_{\,1}\!\begin{bmatrix}\sigma,\ \sigma,\ \sigma,\ \sigma\\ 2\sigma\end{bmatrix},\qquad\text{so}\qquad P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma)=\Gamma^{\,2}_{\,1}\!\begin{bmatrix}\sigma,\ \sigma\\ 2\sigma\end{bmatrix}\quad\text{[derived]}.$$

For $\binom{[3]}{1}$ at $\sigma=1$ the two-dimensional fibre integral is $2.343907238689463\ldots$, the value of the receipts table below, to fourteen digits [computed].

Proposition 15.2 (The banana's pin)**.** [derived]*; receipt* [computed] *For* $\binom{\mathcal{E}}{1}$ *the pin is* $a_e=b_e$ *at every mark, so each mark carries* $\Gamma(a_e)^2$*, which is the share kernel of the Gramma* $\mathsf{G}_{0;(1,-1)}(1,y)=\int_0^\infty x^{-1}e^{-x-y/x}dx$ *(Prop. 4.2), and*

$$\Gamma(\sigma)^2\,P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma)=\int_{|\mathbf{a}|=\sigma}\prod_{e\in\mathcal{E}}\Gamma(a_e)^2=\int_0^\infty y^{\sigma-1}\,\mathsf{G}_{0;(1,-1)}(1,y)^n\,dy.$$

At $n=5$*,* $\sigma=1$ *the right side is* $39.945588011996245699\ldots$ [computed].

Proof. With $T=\{e\}$ the pins are $a_e-b_e=0$ for $e\neq e_0$ and the two totals, so $a=b$ and the fibre is the share integral $|\mathbf{a}|=\sigma$ (Def. 4.1). The share side of $\mathsf{G}_{0;(1,-1)}(1,y)$ is $\frac{1}{2\pi i}\int\Gamma(c)^2y^{-c}dc$, and the read $\int_0^\infty y^{\sigma-1}(\cdot)\,dy$ of a product of n such fibre integrals pins the sum of their shares at σ. □

So the "Bessel moments" are the banana's share side: the pin of a single mark turns the two letters into one Gramma per mark (alias $\mathsf{G}_{0;(1,-1)}(1,y)=2K_0(2\sqrt{y})$). A general list keeps a share per basis and per letter and couples them only through the incidence; that coupling is the broader arena.

Proposition 15.3 (The corner is the residue at zero shares)**.** [derived] *With $U_k=\{\pi(k),\dots,\pi(n)\}$ the tails of an order π and c the width (Def. 7.10, p. 49),*

$$\lim_{\sigma\downarrow 0}\sigma^{n-1}P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma)=\sum_{\pi}\prod_{k=2}^{n}c(U_k)^{-1},$$

and $\sigma^{n-1}P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma)$ has no term of order σ. This is Thm. 38.5(i), p. 330, at the centre question, whose walls are $\sigma c(S)$ (Prop. 8.21(ii), p. 59); the term of order σ there is a sum of $\log\theta$ at greedy bases, zero for a list. On the share side, $\sigma^{n+1}\Gamma(\sigma)^2P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma)$ tends to the same number: it is the residue of the fibre integral of Proposition 15.1 at the origin, where all $2|\mathcal{B}|$ Gammas have their first pole. For $\binom{\mathcal{E}}{1}$ the corner is $n!$, the $5!$ of §14.17; for $\binom{[5]}{2}$ the tails of every order have widths $1,2,2,1$ and it is 30.

Proposition 15.4 (Closing the centre at the negative strengths)**.** [derived] *For $k\in\mathbb{N}$, at $\sigma=-k$ the share side closes (Thm. 4.5, p. 26) onto the lattice of pairs $(\boldsymbol{\alpha},\boldsymbol{\beta})\in\mathbb{N}^{\mathcal{B}}\times\mathbb{N}^{\mathcal{B}}$ with $|\boldsymbol{\alpha}|=|\boldsymbol{\beta}|=k$ and $\sum_T\alpha_T\mathbf{1}_T=\sum_T\beta_T\mathbf{1}_T$, where every Gamma is at a pole, $a_T=-\alpha_T$, $b_T=-\beta_T$, with the residues $\prod_T(-1)^{\alpha_T+\beta_T}/(\alpha_T!\,\beta_T!)$ (Prop. 1.3(iv), p. 11). The continued period has a pole of order at most $n-1$ at $\sigma=-k$, and*

$$\begin{aligned}\lim_{\sigma\to-k}(\sigma+k)^{n-1}\,P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma)&=\sum_{\pi}\prod_{k=2}^{n}c(U_k)^{-1}\cdot\sum_{(\boldsymbol{\alpha},\boldsymbol{\beta})}\frac{(k!)^2}{\prod_T\alpha_T!\,\beta_T!}\\&=\sum_{\pi}\prod_{k=2}^{n}c(U_k)^{-1}\cdot[\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k,\end{aligned}$$

the sum over that lattice: it counts the pairs of k-tuples of bases with equal incidence.

Proof. On the balance wall the period is $\sum_\pi\int_{[0,1]^{n-1}}\prod_{k'=2}^{n}q_{k'}^{\kappa_{U_{k'}}-1}h_\pi(q)\,dq$ in the sector coordinates $\alpha_{\pi(k')}=\rho\,q_2\cdots q_{k'}$ (Thm. 8.7(ii), p. 55, and Thm. 8.10). At the centre $\kappa_{U_{k'}}=\sigma c(U_{k'})$ and $h_\pi=(k_\pi\psi_\pi)^{-\sigma}$, and $k_\pi\psi_\pi=\prod_{k'}q_{k'}^{c(U_{k'})}K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}$ is a polynomial in q with nonnegative coefficients and constant term one (Lemma 8.5, p. 55, weights one). By Lemma 8.6, p. 55, each cube integral continues with at most a simple pole in each $\kappa_{U_{k'}}$ at $-\mathbb{N}$, and at $\sigma=-k$ the $n-1$ poles meet only through the term q^{kc}, $c=(c(U_{k'}))_{k'}$, of $(k_\pi\psi_\pi)^k(k_\pi\psi_\pi)^{-(\sigma+k)}$, with coefficient $\prod_{k'}c(U_{k'})^{-1}[q^{kc}](k_\pi\psi_\pi)^k$. The map $\alpha^v\mapsto q^{(v(U_{k'}))_{k'}}$ is injective on exponents of degree zero, so $[q^{kc}](k_\pi\psi_\pi)^k=[\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k$. Expanding each letter's k-th power (the share kernel of Cor. 4.4, p. 26, closed at the exponent $-k$) gives the lattice sum. On the share side the factor $(k!)^{-2}$ is the residue of $\Gamma(\sigma)^2$ at $-k$, and the common factor of the lattice points is the residue at zero shares, the corner (Proposition 15.3). □

Corollary 15.5 (One law for the period and its closing)**.** [derived] *If polynomials $c_0,\dots,c_d$ satisfy*

$$\sum_{i=0}^{d}c_i(\sigma)\,P_{\mathcal{B}}((\sigma+i)\,\mathbf{1};(\sigma+i),(\sigma+i))=0$$

identically, then for every $k \in \mathbb{N}$

$$\sum_{i=0}^{\min(d,k)} c_i(-k)\,[\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^{k-i} = 0.$$

So the least law of the period in its strength is at least as long as the least recurrence of its closings, and the closings obey the reflected law from $k = 0$*, with nothing below.*

Proof. Multiply by $(\sigma+k)^{n-1}$ and let $\sigma \to -k$: the terms with $i \le k$ tend to $c_i(-k)$ times the corner times the closing at $k-i$ by Proposition 15.4, and the others are regular there. □

The laws come from the family's own relations. By the letter presentation (Prop. 3.6, p. 20) the centre period is a Breta with one bunch for the shape and one singleton bunch per letter; its letter splits and scale exchanges (Thm. 3.5, p. 19) are the letter and mark relations of the interest (Thm. 8.3(i)–(iii)), and along the centre ray they are the relations of Thm. 35.3, p. 308, whose eliminations are the laws of Cor. 35.6. The converse of Corollary 15.5, that the least recurrence of the closings, read at $k = -\sigma$, is a law of the period, is not derived here; for the banana through seven marks it is not needed, the laws being derived by adding marks (Theorem 15.25). For every list below the minimal solution it gives agrees with an independent quadrature to fourteen digits, and the closings obey it from $k = 0$ with nothing below [computed].

Procedure 15.6 (The centre ray from the list)**.** (i) Close the share side at $\sigma = -k$: $x_k = [\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k$, $k \le k_{\max}$, the letter x_k abbreviating them in this procedure. (ii) Find their least recurrence $\sum_{j=0}^{d} p_j(k)x_{k+j} = 0$ by exact linear algebra and check it on the surplus equations. (iii) Read it at $k = -\sigma$: $c_i(\sigma) = p_{d-i}(-\sigma-d)$, $\sum_i c_i(\sigma)P_{\mathcal{B}}((\sigma+i)\,\mathbf{1};(\sigma+i),(\sigma+i)) = 0$. (iv) The period is its minimal solution: $P_{\mathcal{B}}((\sigma+1)\,\mathbf{1};(\sigma+1),(\sigma+1))/P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma) \to K(\mathbf{1})^{-2}$ (Prop. 35.7, p. 309), the native value of least modulus. (v) Scale it by the residue at zero shares: c_0 has a zero of order $n-1$ at 0, and $[\sigma^{n-1}]c_0 \cdot \sum_\pi \prod_{k=2}^{n} c(U_k)^{-1} + \sum_{i\ge1} c_i(0)P_{\mathcal{B}}(i\,\mathbf{1};i,i) = 0$.

Receipts [computed], no quadrature in the procedure:

list	$P_{\mathcal{B}}(\mathbf{1};1,1)$	reading
$\binom{[3]}{1}$	2.34390723868945889060156 2 …	$3L(\chi_{-3},2)$ (alias)
$\binom{[4]}{1}$	8.41439832211715999779816 7 …	$(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty)) = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$
$\binom{[4]}{2}$	2.34390723868945889060156 2 …	as $\binom{[3]}{1}$: Cor. 19.17 (p. 143) and duality
$\binom{[5]}{1}$	39.94558801199624569894402 …	$M_5(1)$ of §14.17

The four-mark values are those of the note (Ex. 6.3 and 6.2, p. 39). There they are closings of the share side at positive strength: the Gammas of a basis meet in double poles, whose Laurent data are Trailmoments of the counting input and of lattice inputs (Prop. 5.14, p. 33; Lemma 35.20, p. 314).

15.2 The masters are the signs

A native gauge of the centre ray is a point where its direction $(\mathbf{1};1,1)$ is native (Def. 35.4, p. 308): the defect $1 - R_e(\alpha) - R_e^*(\alpha)$ vanishes at every mark, and since $R_e^*(\alpha) = 1 - R_e(\alpha^{-1})$ (Thm. 7.14, p. 50) this reads

$$R_e(\alpha) = R_e(\alpha^{-1}) \qquad (e \in \mathcal{E}):$$

inverting the gauge leaves the occupation unchanged. Its native value is $\big(K(\alpha)K(\alpha^{-1})\big)^{-1}$.

Proposition 15.7 (The banana has only signs)**.** [derived] *The native gauges of the centre ray of* $\binom{\mathcal{E}}{1}$ *are exactly the signs* $\eta \in \{\pm 1\}^{\mathcal{E}}$*,* $\eta_{e_0} = 1$*, with* $K(\eta) = \sum_e \eta_e \neq 0$*; their native values are* $K(\eta)^{-2}$ *(Thm. 35.12, p. 310).*

Proof. For $K = \sum_e \alpha_e$ the occupation is $R_e(\alpha) = \alpha_e / K(\alpha)$, so $R_e(\alpha) = R_e(\alpha^{-1})$ for every e says $\alpha_e^2 = K(\alpha)/K(\alpha^{-1})$ for every e. □

So along the banana sequence the native values are $(n-2j)^{-2}$, j the number of minus signs, $\lceil n/2 \rceil$ of them, all simple. At $n = 5$ they are $\frac{1}{25}, \frac{1}{9}, 1$, the symbol of the ray law vanishes there (Cor. 35.6, p. 308), the order is three, and the corner leaves two masters. Every list has the signs with $K(\eta) \neq 0$ among the native gauges of its centre ray; what is special about $\binom{\mathcal{E}}{1}$ is that it has nothing else.

Remark 15.8 (The lowered settings of §14.17). With $g = \mathsf{G}_{0;(1,-1)}(1, \cdot)$ and $h(y) = \sum_m y^m / \Gamma(1+m)^2$, the settings $x^{(j)}(s) = \int_0^\infty y^{s-1} g^{5-j} h^j \, dy$, $j = 0, 1, 2$, satisfy the ray law in the normalization $M(s) = \Gamma(s)^2 P_{\mathcal{B}}(s\,\mathbf{1}; s, s)$, and $x^{(0)} = M$ is Proposition 15.2. Their corners are $120, 0, 0$ [computed], and $x^{(j)}(s+1)/\big(s^2 x^{(j)}(s)\big) \to (5-2j)^{-2}$: they are the sign classes with j minus signs. "The g's outnumber the h's" is $K(\eta) \neq 0$; the tie at four marks is $K(\eta) = 0$, which is not a native gauge.

Proposition 15.9 (The lowered-setting block is the frame)**.** [derived]*; receipt* [computed] *Let* $\Phi(s) = \det\big[x^{(i)}(s+l)\big]_{i,l=0}^{2}$*. By Prop. 35.8, p. 309, applied to the ray law of* M *(end coefficients* s^6 *and* -225*),* $\Phi(s+1) = \frac{s^6}{225}\Phi(s)$ *and* $\Phi(s) = \Phi(1)\,\Gamma(s)^6\, 225^{1-s}$*. The amplitudes at the three sign classes,* $x^{(j)}(s) \sim 2\pi^{(5-2j)/2} 2^{-j} \rho_j^{5/2}\, \Gamma(2s - \frac{5}{2})\, \rho_j^{-2s}$ *with* $\rho_j = 2(5-2j)$*, give*

$$\Phi(1) = \frac{128\sqrt{15}}{16875}\,\pi^3 = 0.91088056479820303289039718797 5\ldots,$$

which the settings computed directly (quadrature, as a check) reproduce to thirty digits. The corners reduce the third column, so

$$\Phi(1) = \frac{120}{225}\Big(x^{(1)}(1)\,x^{(2)}(2) - x^{(1)}(2)\,x^{(2)}(1)\Big), \qquad x^{(1)}(1)x^{(2)}(2) - x^{(1)}(2)x^{(2)}(1) = \frac{16\sqrt{15}}{1125}\,\Gamma(\tfrac{1}{2})^6 :$$

the block of §14.17 is the frame of the ray over its corner. The masters do not enter Φ*; their coefficients are minors that the zero corners of the lowered settings kill. So the frame does not shorten the last minor of §14.17.*

15.3 A parallel class closes through the banana

Proposition 15.10. [derived] *Let* $\mathcal{B}$ *be obtained from a list* $\mathcal{H}$ *on the marks* $1, \ldots, m$ *by replacing mark* i *with a parallel class* P_i *of* s_i *marks, so that* $K_{\mathcal{B}}(\alpha) = K_{\mathcal{H}}\big((\alpha(P_i))_i\big)$ *with* $\alpha(P_i) = \sum_{e \in P_i} \alpha_e$*. Then* $[\alpha^v] K_{\mathcal{B}}^k = \big([A^w] K_{\mathcal{H}}^k\big) \prod_i \big(w_i! / \prod_{e \in P_i} v_e!\big)$ *with* $w_i = v(P_i)$*, and the closings of* $\mathcal{B}$ *are*

$$\sum_w \big([A^w] K_{\mathcal{H}}(A)^k\big)^2 \prod_i [\alpha^0] \Big(K_{P_i}(\alpha)\Psi_{P_i}(\alpha)/\alpha^{P_i}\Big)^{w_i},$$

with $K_{P_i} = \sum_{e \in P_i} \alpha_e$ *and* Ψ_{P_i} *the letters of the banana* $\binom{P_i}{1}$*: its closings at* $-w_i$*.*

The banana enters the closing only through its own closings, as the parallel node enters the network through its full leans (Cor. 25.9, p. 209).

15.4 The five-mark centres

All connected lists on five marks, up to duality, which leaves the centre unchanged (Prop. 8.14, p. 57); "doubled" and "tripled" are parallel classes. [computed]

list	$\lvert\mathcal{B}\rvert$	$K(\eta)^2$ at the signs (mult.)	roots of the symbol	order	corner	masters
$\binom{[5]}{1}$	5	25 (1), 9 (5), 1 (10)	$25, 9, 1$	3	120	2
$\binom{[4]}{2}$, a mark doubled	9	81 (1), 9 (6), 1 (9)	$81, 9, 1$	3	36	2
$\binom{[5]}{2}$	10	100 (1), 4 (15)	$100, 4, 1, -8$	4	30	3
$\binom{[3]}{2}$, two marks doubled	8	64 (1), 16 (2), 4 (8), 0 (5)	$64, 16, 4, 4, 1$	5	44	$3^\dagger$
$\binom{[3]}{2}$, a mark tripled	7	49 (1), 25 (1), 9 (3), 1 (11)	$49, 25, 9, 1, 1, 1$	6	60	$4^\dagger$

list	$P_{\mathcal{B}}(\mathbf{1}; 1, 1)$
$\binom{[5]}{1}$	39.945588011996245698944028498423995891849782924 77 . . .
$\binom{[4]}{2}$, a mark doubled	5.493901143580844012444257607820691195853638640680 1 . . .
$\binom{[5]}{2}$	4.029714170711049607320695734428406933139867563639 6 . . .
$\binom{[3]}{2}$, two marks doubled	7.865837368364021592421974697446168210982382809458 1 . . .
$\binom{[3]}{2}$, a mark tripled	12.848722141915555957312339931253534636619123934709 . . .

The roots of the symbol are those of the recurrence of the closings, the values $K(\alpha)K(\alpha^{-1})$ at the native gauges; the ray law of the period has their reciprocals. "Masters" is $m-1$ for the least m with a rational relation among $1, P_{\mathcal{B}}(\mathbf{1}; 1, 1), \ldots, P_{\mathcal{B}}(m\,\mathbf{1}; m, m)$ (PSLQ at 120 digits, coefficients below 10^{15}, values good to 150 digits): over $\mathbb{Q}$, for the symmetric insertion, on the centre ray. In the first three rows the one relation is the corner. Over $\mathbb{Q} + \mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ the list $\binom{[5]}{2}$ has two masters, and over $\mathbb{Q} + \mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) + \mathbb{Q}\,\Gamma(\frac{1}{2})^4$ the triangle with two marks doubled has one (Theorems 17.8 and 17.12).

† In the last two rows c_0 and c_1 both carry $(\sigma+1)^2$, so the ray law at $\sigma = -1$ reaches the period at neither -1 nor 0, and it is a second relation, a *second corner*: the boundary term, the constant term at $\sigma = -1$ (Thm. 1.6, p. 12),

$$\operatorname*{CT}_{\sigma=-1}\Big(c_0(\sigma)P_{\mathcal{B}}(\sigma\,\mathbf{1};\sigma,\sigma)+c_1(\sigma)P_{\mathcal{B}}((\sigma{+}1)\,\mathbf{1};(\sigma{+}1),(\sigma{+}1))\Big) = -\sum_{i\geq 2} c_i(-1)\,P_{\mathcal{B}}((i{-}1)\,\mathbf{1};(i{-}1),(i{-}1)),$$

is rational [computed]. In lowest terms the two relations read

$$\begin{aligned} &2503 - 566\,P_{\mathcal{B}}(\mathbf{1};1,1) + 88248\,P_{\mathcal{B}}(2\,\mathbf{1};2,2) - 908872\,P_{\mathcal{B}}(3\,\mathbf{1};3,3) + 1425600\,P_{\mathcal{B}}(4\,\mathbf{1};4,4) = 0,\\ &13800002 - 1848045\,P_{\mathcal{B}}(\mathbf{1};1,1) + 227950815\,P_{\mathcal{B}}(2\,\mathbf{1};2,2) - 3601110627\,P_{\mathcal{B}}(3\,\mathbf{1};3,3)\\ &\qquad + 11177654265\,P_{\mathcal{B}}(4\,\mathbf{1};4,4) - 8565102000\,P_{\mathcal{B}}(5\,\mathbf{1};5,5) = 0, \end{aligned}$$

so the masters are $5-2=3$ and $6-2=4$. Write $c_0 = (\sigma+1)^2 A$ and $c_1 = (\sigma+1)^2 B$. Proposition 15.4 at $\sigma = -1$ requires $A(-1)\,\lvert\mathcal{B}\rvert + B(-1) = 0$, the closing at 1 being $\lvert\mathcal{B}\rvert$; it holds, $B(-1)/A(-1) = -8$ and -7 [computed]. There the recursion of Corollary 15.5 leaves the corner at -1 free, so this is a check of Proposition 15.4 that the recursion does not supply.

$\binom{[5]}{2}$ has native gauges off the signs: on the pattern $\alpha \propto (a, a, b, b, 1)$ the native equations have solutions with $K(\alpha)K(\alpha^{-1}) = 1$ and -8. There is no rational relation among 1, the two masters of $\binom{[5]}{1}$ and the three of $\binom{[5]}{2}$ (coefficients below 10^8, 150 digits), nor with π^2 adjoined.

15.5 The native shares, the roots, and an exact pairing

Every root of the symbol of the closings' recurrence is the reciprocal native value $K(\alpha_c)K(\alpha_c^{-1})$ of a native gauge α_c of the centre ray (Cor. 35.6, p. 308).

Proposition 15.11 (Both reads sit at the native shares)**.** [derived]*; receipts* [computed] *Let the list be connected, and let* $H = 2\big(\kappa_{\mathbf{1}}(X_e, X_{e'})\big)_{e,e'\neq e_0}$*, twice the covariances of the indicators under the weight law (Def. 9.2, p. 61), the Hessian of Thm. 35.11, p. 310.*

(i) On the real fibre of Proposition 15.1 the native shares of Thm. 4.3(ii), p. 25, are the uniform shares $a_T = b_T = \sigma/|\mathcal{B}|$*: every basis takes the same share from each letter. This is the table's gauge* $\mathbf{1}$ *read on the share side.*

(ii) The period reads the fibre along its imaginary directions. By Thm. 4.3(iii), which at the period is Thm. 38.5(ii), p. 330,

$$P_{\mathcal{B}}(k\,\mathbf{1}; k, k) \sim K(\mathbf{1})^{-2k}\Big(\frac{2\pi}{k}\Big)^{\frac{n-1}{2}}(\det H)^{-\frac{1}{2}}.$$

(iii) The closing reads the same fibre on its integer points. Dropping the pin, the multinomial theorem gives $\sum_{|\boldsymbol{\alpha}|=|\boldsymbol{\beta}|=k}(k!)^2/\prod_T \alpha_T!\beta_T! = K(\mathbf{1})^k\Psi(\mathbf{1})^k = |\mathcal{B}|^{2k}$*; the pin keeps the Gaussian of the fibre at the native shares scaled by* k*, and*

$$[\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k \sim K(\mathbf{1})^{2k}(2\pi k)^{-\frac{n-1}{2}}(\det H)^{-\frac{1}{2}}.$$

(iv) So $k^{n-1}\,P_{\mathcal{B}}(k\,\mathbf{1}; k, k)\,[\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k \to (\det H)^{-1}$*: one Gaussian, read once along a contour and once on a lattice, so that the powers of* 2π *cancel and the determinant appears twice.*

Proof. (i) The uniform point satisfies the pins, and $\nabla S = -\log \mathbf{w}$ is constant there, a combination of the two rows of the letter totals; S is strictly concave on the real fibre (Thm. 4.3(ii)). (ii) Thm. 38.5(ii) at the centre: $X(0) = -2\log K(\mathbf{1})$ and $-\nabla_z^2 X(0) = 8\kappa_{\mathbf{1}} = 4H$ on the chart, $y = 2z$; Thm. 35.11 is the same edge read on the generating function of the ray. (iii) The terms $(k!)^2/\prod \alpha_T!\beta_T!$ on the pinned lattice peak at the uniform shares scaled by k; their Gaussian there is the quadratic form of $\sum \log\Gamma$ at the native shares restricted to the fibre, the same form as in (ii), now summed over the integer points of the fibre; $\nabla_z^2 E(0) = 8\kappa_{\mathbf{1}}$ (Thm. 7.14, p. 50) identifies it with H. □

Receipts [computed]: $(\det H)^{-1} = \frac{3125}{16}, \frac{729}{16}, \frac{3125}{81}, \frac{512}{9}, \frac{16807}{192}$ in the order of the table of §15.4. At $k = 70$ the product agrees with these to about 10^{-4}, the error falling like k^{-2}. For $\binom{[2]}{1}$ the product is exact at every k: $k\,\Gamma_{\binom{1}{2}}\begin{bmatrix} k,\ k \\ 2k \end{bmatrix}\binom{2k}{k} = 2 = (\det H)^{-1}$ [derived].

Remark 15.12 (The signs are the pin read modulo two)*.* Averaging over the signs detects the incidence modulo two: for every k,

$$2^{-n}\sum_{\eta\in\{\pm1\}^{\mathcal{E}}} K(\eta)^{2k} = \sum_{\substack{|\boldsymbol{\alpha}|=|\boldsymbol{\beta}|=k \\ \sum_T \alpha_T\mathbf{1}_T \equiv \sum_T \beta_T\mathbf{1}_T \ (2)}} \frac{(k!)^2}{\prod_T \alpha_T!\,\beta_T!} \ \geq\ [\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k \quad \text{[derived]},$$

since $K(\eta)\Psi(\eta)/\eta^{\mathcal{E}} = K(\eta)^2$. So the sign values $K(\eta)^2$ are the growth rates of the pin's reduction modulo two, with $K(\mathbf{1})^2$ the largest. The full pin keeps them as roots (Thm. 35.12) and adds the native gauges off the signs, below.

Proposition 15.13 (An exact pairing of the period with its closing for the five-mark banana)**.** [derived] *from the ray law; receipt* [computed]*. For* $\binom{[5]}{1}$ *the ray law* $\sum_{i=0}^{3} c_i(\sigma) P_{\mathcal{B}}((\sigma+i)\,\mathbf{1}; (\sigma+i), (\sigma+i)) = 0$ *has*

$$c_0 = -\sigma^4, \qquad c_1 = 35\sigma^4 + 70\sigma^3 + 63\sigma^2 + 28\sigma + 5,$$
$$c_2 = -(\sigma+1)^2(259\sigma^2 + 518\sigma + 285), \qquad c_3 = 225(\sigma+1)^2(\sigma+2)^2,$$

and it is self-adjoint under the reflection of the strength: $c_i(-\sigma - i) = c_i(\sigma)$ *for every* i*. So the closings, read as a function of* σ*, solve the adjoint law, and for every* $\sigma \in \{1, 2, 3, \dots\}$

$$\sum_{i=1}^{3} \sum_{j=1}^{i} c_i(\sigma - j)\ [\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^{\sigma-j}\ P_{\mathcal{B}}((\sigma - j + i)\,\mathbf{1}; (\sigma - j + i), (\sigma - j + i)) = 120,$$

the residue at zero shares, with the closings at negative indices read as 0*.*

Proof. The left side, $B(\sigma)$ in this proof, is the bilinear concomitant of the law and its adjoint: $B(\sigma+1) - B(\sigma)$ is the law applied to the period times the closing at σ, less the period at σ times the adjoint law applied to the closings. The first vanishes by the law. The adjoint law at σ is $\sum_i c_i(\sigma - i) x_{\sigma-i}$, which by $c_i(\sigma - i) = c_i(-\sigma)$ is the recurrence of Corollary 15.5 at $k = \sigma$, zero from $k = 0$ on. At $\sigma = 1$ only $j = 1$ survives, and $B(1) = \sum_i c_i(0) P_{\mathcal{B}}(i\,\mathbf{1}; i, i) = 120$ is the corner relation of Procedure 15.6(v). The self-adjointness is an identity of the four polynomials; $B(\sigma) = 120$ holds to seventy digits for $\sigma \le 55$ [computed]. □

At large σ the closings and the period move by the factors 25 and $\frac{1}{25}$ per step, so $B(\sigma) \to \lambda\chi'(\lambda) \cdot \lim_k k^4\, P_{\mathcal{B}}(k\,\mathbf{1}; k, k)\,[\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^k$, with $\chi(\lambda) = \sum_i [\sigma^4] c_i\, \lambda^i = -1 + 35\lambda - 259\lambda^2 + 225\lambda^3$ the symbol and $\lambda = \frac{1}{25}$. With Proposition 15.11:

$$120 \cdot \det H = \lambda\chi'(\lambda)\big|_{\lambda = 1/25} = \frac{384}{625}, \qquad \det H = \frac{16}{3125} \quad \text{[derived]}.$$

The residue at zero shares (the low end of strength, Thm. 38.5(i)) and the Gaussian at the native shares (the high end, Thm. 38.5(ii)) are the two readings of one exact identity between the fibre and its lattice. For the other four five-mark lists the closings solve no adjoint of the ray law in the forms tried: no combination $\sum_{l<d} r_l(\sigma) [\alpha^0]\big(K(\alpha)\Psi(\alpha)/\alpha^{\mathcal{E}}\big)^{\sigma+l}$ with polynomial r_l of degree at most 20, nor with the apparent factors of c_0 as denominators up to degree 14, solves it [computed]. In that framework the pairing of the period with its closing is not a single product there; it must carry other solution classes.

Proposition 15.14 (Signs through parallel classes)**.** [derived] *In the setting of Proposition 15.10,* $K_{\mathcal{B}}(\eta) = K_{\mathcal{H}}\big(s_1(\eta), \dots, s_m(\eta)\big)$ *with* $s_i(\eta) = \sum_{e \in P_i} \eta_e \in \{s_i, s_i - 2, \dots, -s_i\}$*: the sign values of the bananas* $\binom{P_i}{1}$ *feed the letter of* $\mathcal{H}$*.*

For the triangle $\mathcal{H} = \binom{[3]}{2}$, $K_{\mathcal{H}} = e_2$, and with one mark tripled, $s = s_1(\eta) \in \{3, 1, -1, -3\}$:

$$e_2(s, 1, 1) = 2s + 1 \in \{7, 3, -1, -5\}, \qquad e_2(s, 1, -1) = -1 \quad \text{for every } s,$$

which is the census 49 (1), 25 (1), 9 (3), 1 (11) of §15.4. The second form does not see s.

Proposition 15.15 (Letters that forget a class)**.** [derived] *Write* $K_{\mathcal{B}} = A_i\, K_{\mathcal{H}/i}(A) + K_{\mathcal{H}\setminus i}(A)$ *with* $A_i = \alpha(P_i)$ *(Def. 7.3, p. 48), and* $A_i' = \sum_{e \in P_i} \alpha_e^{-1}$*.*

(i) Tripled triangle. On $\alpha_3 = -\alpha_2$ the contraction letter $K_{\mathcal{H}/1} = A_2 + A_3$ vanishes, so every copy has occupation zero at α and at α^{-1}, and $K(\alpha) = -\alpha_2^2$, $K(\alpha^{-1}) = -\alpha_2^{-2}$ whatever the copies are. The native gauges there are

$$\{\alpha_3 = -\alpha_2,\ A_1 = \alpha_2^2 A_1'\},$$

a family of dimension two modulo scale, all of native value 1, containing the eight sign classes with $\eta_3 = -\eta_2$.

(ii) Two marks doubled. On $P_2 = (t, -t)$ the class letter vanishes, $A_2 = A_2' = 0$, and $K = A_1\alpha_3$ is the letter of $\mathcal{H} \setminus 2$, in which both remaining classes are coloops. The native gauges there are

$$\Big\{P_1 = (c, c),\ P_2 = (t, -t),\ t^2 = \frac{c\,\alpha_3\,(2\alpha_3 + c)}{2c + \alpha_3}\Big\},$$

a curve of native value $\frac{1}{4}$ which on the torus $|\alpha_e| = 1$ is a circle through the signs with $(s_1, s_2) = (\pm 2, 0)$.

Proof. Nativity is $R_e(\alpha) = R_e(\alpha^{-1})$ at every mark. (i) At a copy $x \in P_1$, $R_x(\alpha) = \alpha_x(A_2 + A_3)/K(\alpha)$ and $R_x(\alpha^{-1}) = \alpha_x^{-1}(A_2' + A_3')/K(\alpha^{-1})$, both zero on $\alpha_3 = -\alpha_2$; at α_2 and α_3 the two sides differ by $\mp(A_1/\alpha_2 - \alpha_2 A_1')$. (ii) At α_3 both occupations are 1 on $A_2 = A_2' = 0$; at P_1 the equation holds when $\alpha_{1a}^2 = \alpha_{1b}^2 = A_1/A_1'$, and $\alpha_{1b} = -\alpha_{1a}$ gives $K = 0$; at P_2 it holds exactly at the stated t^2. When $|c| = |\alpha_3| = 1$, $|2\alpha_3 + c| = |2c + \alpha_3|$, so $|t| = 1$. □

Remark 15.16 (Local exponents). Put $x_k \approx \lambda^k k^\beta$ into the recurrence of the closings [computed]. At every simple root of every row, $\beta = -2 = -\frac{n-1}{2}$, the exponent of a nondegenerate native gauge (Thm. 35.5(i), p. 308). At the repeated roots the indicial polynomials are

$$(\beta+1)(\beta+2)(\beta+3) \quad \text{(tripled triangle, } \lambda = 1), \qquad (2\beta+3)(2\beta+5) \quad \text{(two marks doubled, } \lambda = 4).$$

A family of real dimension d on the torus $|\alpha_e| = 1$, transversally nondegenerate, gives $\beta = -\frac{n-1-d}{2}$. The Hessian of $\log\big(K(\alpha)K(\alpha^{-1})\big)$ there has rank two along the family of (i) and rank three along the circle of (ii) [computed], so these are -1 and $-\frac{3}{2}$. The isolated sign orbit at $\lambda = 1$ (one copy plus, two minus, $\eta_2 = \eta_3$) has full rank and gives -2. The exponents -3 and $-\frac{5}{2}$: [not yet computed]. These families are integral directions in the exceptional set of Thm. 35.19(i), p. 313: the order of the ray law counts native values with these multiplicities, and the families raise the order, and with it the masters, past the number of native values.

Remark 15.17 (Native gauges at complex points off the signs [computed]). $\binom{[5]}{2}$: $K(\alpha)K(\alpha^{-1}) = 1$ at $(\omega^2, \omega^2, \omega, \omega, 1)$, $\omega = e^{2\pi \mathrm{i}/3}$, where $K = K(\alpha^{-1}) = 1$; and -8 at $(1, 1, b, b, 1)$, $b = -(2 + \sqrt{3})$, and at its inverse, where $K = -2 - 2\sqrt{3}$ and $K(\alpha^{-1}) = -2 + 2\sqrt{3}$. Two marks doubled: 1 at $(\omega, \omega^2, \omega, \omega^2, 1)$. The signs are the points of order two of the torus; points of order three are the next layer that occurs here.

15.6 Closing the banana at strength one

Proposition 15.18 (Four marks close on the half-lattice)**.** [derived] *For $\binom{[4]}{1}$, merging the marks in pairs by four Gammas on a line (Cor. 4.7, p. 26), $\int_{a_1+a_2=c} \Gamma(a_1)^2\Gamma(a_2)^2 = \Gamma_{\,1}^{\,4}\begin{bmatrix} c,\ c,\ c,\ c \\ 2c \end{bmatrix}$, leaves a fibre of dimension one, and*

$$\begin{aligned} P_{\mathcal{B}}(\mathbf{1}; 1, 1) &= \frac{1}{2\pi i}\int_{(1/2)} \Gamma_{\,1}^{\,4}\begin{bmatrix} c,\ c,\ c,\ c \\ 2c \end{bmatrix} \Gamma_{\,1}^{\,4}\begin{bmatrix} 1-c,\ 1-c,\ 1-c,\ 1-c \\ 2-2c \end{bmatrix} dc \\ &= (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)). \end{aligned}$$

Proof. By Lemma 5.16, p. 34, and the shift and reflection of the stacked Gamma (Prop. 1.3(iii),(v), p. 11) the integrand is

$$\Gamma_{\substack{4\\0}}\big[c,\ c,\ 1-c,\ 1-c\big]^2\ \frac{\sin 2\pi c}{\pi(1-2c)} = \frac{2\pi^3\cos\pi c}{(1-2c)\sin^3\pi c}.$$

On the lines $\operatorname{Re} c = N+\frac12$ it is $O\big(N^{-1}e^{-2\pi|\operatorname{Im} c|}\big)$, so it closes to the right (Thm. 4.5, p. 26). At $c=m\ge 1$, $2\pi^3\cos\pi c/\sin^3\pi c = 2(c-m)^{-3}+O(c-m)$, with no parts of orders one and two, as in Lemma 5.16; the half-integers are zeros. So the residue is $2\,[\varepsilon^2](1-2m-2\varepsilon)^{-1} = -8/(2m-1)^3$, and the closing is $\sum_{m\ge1} 8(2m-1)^{-3} = \sum_{x\in\frac12+\mathbb{Z},\,x>0} x^{-3}$. The last equality is the coset relation at the step 2 (Thm. 5.12(i), p. 32). The half-lattice is the doubled share $2c$ of a merged pair, the lower entry of its stacked Gamma. □

This is the note's value (Ex. 6.2, p. 39), now read as a closing of the share side.

Proposition 15.19 (A mark added is a critical-line kernel)**.** [derived]*; receipts* [computed] *For every* $n\ge1$*,*

$$P_{\binom{[n+1]}{1}}(\mathbf{1};1,1) = -\sum_{m\ge1}\partial_\sigma P_{\binom{[n]}{1}}(\sigma\mathbf{1};\sigma,\sigma)\big|_{\sigma=m}.$$

Proof. In the share side of $\binom{[n+1]}{1}$ (Proposition 15.2) merge the first n marks: their fibre at the total d is $\Gamma(d)^2 P_{\binom{[n]}{1}}(d\mathbf{1};d,d)$, by the same proposition at n marks. The last mark carries $\Gamma(1-d)^2$, and $\Gamma(d)^2\Gamma(1-d)^2 = \Gamma_{\substack{4\\0}}\big[d,\ d,\ 1-d,\ 1-d\big]$, whose right closing reads the derivative of the rest at the positive integers (Lemma 5.16(i)). The rest is holomorphic on $\operatorname{Re} d>0$, bounded on each vertical line by its value at the real part, and decays as $\operatorname{Re} d\to\infty$, so the closing applies (Thm. 4.5). □

Receipts [computed]. At $n=2$, with $\partial_\sigma\Gamma_{\substack{2\\1}}\left[\begin{smallmatrix}\sigma,\ \sigma\\ 2\sigma\end{smallmatrix}\right] = 2\Gamma_{\substack{2\\1}}\left[\begin{smallmatrix}\sigma,\ \sigma\\ 2\sigma\end{smallmatrix}\right]\big((\log\Gamma)'(\sigma)-(\log\Gamma)'(2\sigma)\big)$ and $(\log\Gamma)'(j+1) = -(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))$ (Prop. 5.14, p. 33),

$$P_{\binom{[3]}{1}}(\mathbf{1};1,1) = 2\sum_{m\ge1}\Gamma_{\substack{2\\1}}\left[\begin{smallmatrix}m,\ m\\ 2m\end{smallmatrix}\right]\ (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((m-1,2m-1]) = 2.343907238689458890601562\ldots,$$

the period of two marks times a head of the counting input, to 25 digits. At $n=3$ the sum is $7(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and at $n=4$ it is $39.945588011996245699\ldots$, both to 20 digits.

Remark 15.20 (The five-mark centre is a Trailmoment of a live input). By Proposition 15.19,

$$P_{\binom{[5]}{1}}(\mathbf{1};1,1) = \big(\mathsf{T}_{0;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}}\big)((0,\infty)),\qquad f(m) = -\partial_\sigma P_{\binom{[4]}{1}}(\sigma\mathbf{1};\sigma,\sigma)\big|_{\sigma=m},$$

a live input (Chapter 6): its density is read from the four-mark centre period, and the four-mark ray law differentiated in σ gives it a recurrence of order two whose inhomogeneity is the values. At $\sigma=0$ the differentiated law is a relation among the first two slopes, the corner and the values, the term of order σ of Proposition 15.3 being zero. So every $f(m)$ lies in $\mathbb{Q}+\mathbb{Q}(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))+\mathbb{Q}f(1)$ [derived]. For example $P_{\binom{[4]}{1}}(2\,\mathbf{1};2,2) = (7\zeta(3)-6)/16$ and $f(2) = -\big(36+21\zeta(3)-2f(1)\big)/32$ (alias $\zeta(3) = (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$) [computed]. The first slope is

$$-f(1) = 48\operatorname{Li}_4(\tfrac12)+2\log^4 2-2\pi^2\log^2 2-\tfrac{17}{30}\pi^4 = -39.381582123122165624939809345 08\ldots$$

by PSLQ at 60 digits [computed]. In the family these aliases are Trailmoments of index at most four of the counting input at the frequencies 0 and $\frac12$ and of the lattice input of density 2^{-x} (Def. 5.2, p. 28). So the four-mark slope sits on the half-lattice of Proposition 15.18, at index four. The masters of the five-mark centre are values of this live Trailmoment, and this is where the closing stops landing on plain lattice inputs.

15.7 Closing the live input of five marks

Throughout this subsection $p(\sigma) = P_{\binom{[4]}{1}}(\sigma\mathbf{1}; \sigma, \sigma)$ for real $\sigma > 0$ and $f = -p'$; $p\,\mathbb{I}_{\mathbb{Z}}$ and $f\,\mathbb{I}_{\mathbb{Z}}$ are the *values input* and the *slope input* (Def. 5.4, p. 29). The four-mark ray law is

$$\sum_{i=0}^{2} c_i(\sigma)\, p(\sigma + i) = 0, \qquad c_0 = \sigma^3, \quad c_1 = -2(2\sigma+1)(5\sigma^2+5\sigma+2), \quad c_2 = 64(\sigma+1)^3,$$

read off the closings by Procedure 15.6 and derived in Theorem 15.25 below. For a function R on a coset of $\mathbb{Z}$ put $(L^*R)(x) = \sum_i c_i(x-i)\,R(x-i)$, and let L'^* be the same with the derivatives c_i' in place of the c_i.

Proposition 15.21 (The live recurrence)**.** [derived] *The slope input has a recurrence of order two whose inhomogeneity is the values input:*

$$\sum_{i=0}^{2} c_i(\sigma)\, f(\sigma + i) = \sum_{i=0}^{2} c_i'(\sigma)\, p(\sigma + i), \qquad c_0' = 3\sigma^2, \quad c_1' = -6(10\sigma^2+10\sigma+3), \quad c_2' = 192(\sigma+1)^2.$$

Proof. The law holds identically in σ (Theorem 15.25); differentiate it. □

Proposition 15.22 (A mark added, at every strength)**.** [derived]*; receipts* [computed] *For real $s > 0$ let $W_s(d) = \Gamma_2^2\begin{bmatrix} d,\ d \\ 1+d-s,\ 1+d-s \end{bmatrix}$. Then*

$$\Gamma(s)^2\, P_{\binom{[5]}{1}}(s\mathbf{1}; s, s) = -\big(\mathsf{T}_{0;(1)}[x]\,(W_s p)'\,\mathbb{I}_{s+\mathbb{Z}}\big)((s-1,\infty)),$$
$$x^{(1)}(s) = \quad \big(\mathsf{T}_{0;(1)}[x]\,W_s p\,\mathbb{I}_{s+\mathbb{Z}}\big)((s-1,\infty)),$$

with $x^{(1)}$ the lowered setting with one h of Remark 15.8. The period reads the slopes of the word $W_s p$ on the coset of its strength, and the lowered setting reads its values. At $s \in \{1, 2, \dots\}$ the word $W_s(x) = \prod_{j=1}^{s-1}(x-j)^2$ is a polynomial with double zeros at $1, \dots, s-1$, the cut moves to 0, and

$$\Gamma(s)^2 P_{\binom{[5]}{1}}(s\mathbf{1}; s, s) = (\mathsf{T}_{0;(1)}[x]\,(W_s f - W_s' p)\,\mathbb{I}_{\mathbb{Z}})((0,\infty)), \qquad x^{(1)}(s) = (\mathsf{T}_{0;(1)}[x]\,W_s p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

Proof. Merge the first four marks in the share side of five (Proposition 15.2): $\Gamma(s)^2 P_{\binom{[5]}{1}}(s\mathbf{1}; s, s) = \frac{1}{2\pi i}\int_{(c)} \Gamma(d)^2 p(d)\,\Gamma(s-d)^2\,dd$ with $0 < c < s$, as in the proof of Proposition 15.19. By the reflection (Prop. 1.3(v), p. 11), $\Gamma(s-d)\Gamma(1-s+d) = \pi/\sin\pi(s-d)$, so

$$\Gamma(d)^2\,\Gamma(s-d)^2 = W_s(d)\,\Gamma_0^4\big[s-d,\ s-d,\ 1-s+d,\ 1-s+d\big],$$

the stacked Gamma of Lemma 5.16, p. 34, at $a = s-d$: double poles at $d \in s+\mathbb{Z}$, with no simple parts. The double zeros of W_s at $d = s-1, s-2, \dots$ remove the poles between the contour and s. The rest is as in Proposition 15.19: $W_s p$ is holomorphic on $\operatorname{Re} d > 0$, of polynomial growth along vertical lines, where the kernel decays like $e^{-2\pi|\operatorname{Im} d|}$, and it decays like $16^{-\operatorname{Re} d}$ to the right; closing to the right (Thm. 4.5, p. 26) reads $-(W_s p)'$ at $d = s, s+1, \dots$. For $x^{(1)}$, expand $h = \sum_k y^k/\Gamma(1+k)^2$ against g^4 (Proposition 15.2 at four marks): $x^{(1)}(s) = \sum_{k\ge 0}\Gamma(s+k)^2 p(s+k)/\Gamma(1+k)^2$, and $\Gamma(s+k)^2/\Gamma(1+k)^2 = W_s(s+k)$. □

Receipts [computed]: at $s = \frac{3}{2}$ both sides of the first identity are 2.27912905076377, and at $s = \frac{9}{4}$ both are 0.129204079141924 (the left side by one-dimensional quadrature of the share side; fifteen digits).

Proposition 15.23 (Summation by parts at the cut: what lands on lattice inputs)**.** [derived] *For every polynomial R,*

$$(\mathsf{T}_{0;(1)}[x]\,(L^*R)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -24\,R(0),$$
$$(\mathsf{T}_{0;(1)}[x]\,(L^*R)\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (\mathsf{T}_{0;(1)}[x]\,(L'^*R)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

Proof. By the shifted inputs (Thm. 6.2, p. 42) a shift of the input is a shift of the word, and the cut relation (Prop. 5.3, p. 29) returns the points that cross the cut: for a density φ,

$$\sum_{x\geq 1} R(x)\sum_i c_i(x)\,\varphi(x+i) = (\mathsf{T}_{0;(1)}[x]\,(L^*R)\,\varphi\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - R(0)\sum_{i\geq 1} c_i(0)\,\varphi(i) - R(-1)\,c_2(-1)\,\varphi(1),$$

and $c_2(-1) = c_2'(-1) = 0$. For $\varphi = p$ the left side vanishes by the law. For $\varphi = f$ it is $\sum_x R(x)\sum_i c_i'(x)\,p(x+i)$ by Proposition 15.21, to which the same move applies. So the boundaries are

$$\sum_{i\geq 1} c_i(0)\,p(i) = -4p(1)+64p(2), \qquad \sum_{i\geq 1}\big(c_i(0)f(i) - c_i'(0)p(i)\big) = -4f(1)+64f(2)+18p(1)-192p(2).$$

The first is $-[\sigma^3]c_0$ times the corner 24 (Procedure 15.6(v)); the second is the term of order σ of $\sigma^3 p(\sigma)$, which vanishes (Proposition 15.3). With the values of Remark 15.20 they are -24 and 0: the lattice Trailmoments in them, $p(1) = 7(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $f(1)$ and $f(2)$, cancel identically. □

So what the closing of the live Trailmoment puts on lattice inputs is the residue at zero shares, and nothing else: neither $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ nor the four-mark slope $f(1)$ survives.

Proposition 15.24 (The fixed block)**.** [derived]*; receipts* [computed] *L^* raises the degree of a polynomial by three, with leading factor $1-20+64 = 45$, the symbol of the law at $\lambda = 1$, which is not a native value of four marks; so the polynomials are $L^*(\mathbb{Q}[x]) \oplus \operatorname{span}\{1,\, x-1,\, (x-1)^2\}$. By Proposition 15.23, every Trailmoment at index zero of a polynomial word against the values input or the slope input, cut at 0, is a rational number plus a rational combination of the six*

$$(\mathsf{T}_{0;(1)}[x]\,(x-1)^j\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)), \qquad (\mathsf{T}_{0;(1)}[x]\,(x-1)^j\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) \qquad (j=0,1,2).$$

The share side adds one relation:

$$5\,(\mathsf{T}_{0;(1)}[x]\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - 5\,(\mathsf{T}_{0;(1)}[x]\,(x-1)\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (\mathsf{T}_{0;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

Proof. The splitting: $\deg L^*R = \deg R + 3$, so L^* is injective on polynomials and its image misses exactly three dimensions in each degree. The relation says that each mark takes on average a fifth of the strength. On the fibre of Proposition 15.2 the shares sum to σ, and the integrand and the pins are symmetric in the five marks; so the fibre integral with one share a_e inserted is the same for every e, and the five sum to $\sigma\,\Gamma(\sigma)^2 P_{\binom{[5]}{1}}(\sigma\mathbf{1};\sigma,\sigma)$. Merging the first four marks, the fifth share is $\sigma - d$, and at $\sigma = 1$ the closing of Proposition 15.22 reads its insertion as $-\sum_{x\geq 1}\partial_d\big((1-d)p(d)\big)\big|_{d=x} = (\mathsf{T}_{0;(1)}[x]\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{0;(1)}[x]\,(x-1)f\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$; the right side is $\frac{1}{5}P_{\binom{[5]}{1}}(\mathbf{1};1,1) = \frac{1}{5}(\mathsf{T}_{0;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. □

Receipts [computed]. At the strengths $2,\dots,5$ the same insertion reduces to rational multiples of this relation. There is no rational relation among 1, $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $f(1)$ and the five Trailmoments other than $(\mathsf{T}_{0;(1)}[x]\,(x-1)f\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ with coefficients below 10^{30} (PSLQ, 360 digits): in that framework, over $\mathbb{Q}$, the block is five-dimensional. Its six values:

j	$(\mathsf{T}_{0;(1)}[x]\,(x-1)^j\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$	$(\mathsf{T}_{0;(1)}[x]\,(x-1)^j\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$
0	8.57028044337446126814969582424653728 6…	39.945588011996245698944028498423995 89…
1	0.16106970227667485885131066624851419 8…	0.58116284097521212836089012456173810 7…
2	0.17187458138353690484768349157529391 6…	0.61683551197162484909292454115302812 6…

Theorem 15.25 (A mark added carries the ray law)**.** [derived]*; receipts* [computed] *Normalize the ray law of* $\binom{[n]}{1}$*,* $\sum_{i=0}^{d_n} c_i^{(n)}(\sigma)\,P_{\binom{[n]}{1}}((\sigma+i)\mathbf{1};\sigma+i,\sigma+i)=0$*, so that* $c_0^{(n)}=(-1)^n\sigma^{n-1}$*, as for four marks above and for five marks in Proposition 15.13, and let* L_n^* *be its adjoint. For* $n=1,\dots,6$*,*

$$\sum_{i=0}^{d_{n+1}} c_i^{(n+1)}(s)\binom{x-1}{s-1+i}^2=\big(L_n^*A_s\big)(x),\qquad A_s(x)=\big((n+1)x-(n+2)s\big)\binom{x}{s}^2,$$

an identity of rational functions of x *and* s *once both sides are divided by* $\binom{x}{s}^2$*. Read one way it produces the law: the words* $\binom{x-1}{s-1+i}^2$ *have distinct degrees, so the coefficients of* $L_n^*A_s$ *in them are the* $c_i^{(n+1)}(s)$*. Read the other way it proves it: for* $2\le n\le 6$*, if the ray law of* $\binom{[n]}{1}$ *holds identically in* σ*, so does that of* $\binom{[n+1]}{1}$*, and the lowered setting* $x^{(1)}$ *of* $n+1$ *marks, divided by* $\Gamma(s)^2$*, obeys it too. The two-mark period* $P_{\binom{[2]}{1}}(\sigma\mathbf{1};\sigma,\sigma)=\Gamma\begin{bmatrix}\sigma,\ \sigma\\ 2\sigma\end{bmatrix}_2$ *obeys* $\sigma P(\sigma)-2(2\sigma+1)P(\sigma+1)=0$ *by the shift of the stacked Gamma (Prop. 1.3(iii), p. 11). So the ray laws of* $\binom{[n]}{1}$*,* $n\le 7$*, are derived from the stacked Gamma, one mark at a time.*

Proof. Write p_n for the centre period of $\binom{[n]}{1}$, and let $s>0$ be non-integral. Proposition 15.22, with n marks merged in place of four, gives $P_{\binom{[n+1]}{1}}((s+i)\mathbf{1};s+i,s+i)=-\sum_{d\in s+\mathbb{N}}(w_{s,i}\,p_n)'(d)$ with $w_{s,i}=W_{s+i}/\Gamma(s+i)^2$, the terms $d=s,\dots,s+i-1$ being zero by the double zeros of W_{s+i}. On the coset $w_{s,i}(d)$ is $\binom{d-1}{s-1+i}^2$ written with Gammas, so the displayed identity holds there. On $s+\mathbb{N}$ the move of Proposition 15.23 has no boundary for the words A_s and A_s', which vanish at $d=s-1,s-2,\dots$: $\sum_d(L_n^*A)\,\varphi=\sum_d A\,(L_n\varphi)$, and likewise with L_n'. With $(L_n^*A)'=L_n'^*A+L_n^*(A')$ and $L_np_n'=-L_n'p_n$ (the law differentiated),

$$\sum_i c_i^{(n+1)}(s)\,P_{\binom{[n+1]}{1}}((s+i)\mathbf{1};s+i,s+i)=-\sum_{d\in s+\mathbb{N}}\Big((L_n'^*A_s)\,p_n+A_s'\,L_np_n+A_s\,L_np_n'\Big)(d)$$
$$=-\sum_d(L_n'^*A_s)\,p_n+\sum_d A_s\,L_n'p_n=0,$$

and $\sum_i c_i^{(n+1)}(s)\,\Gamma(s+i)^{-2}x^{(1)}(s+i)=\sum_d(L_n^*A_s)\,p_n=\sum_d A_s\,L_np_n=0$. Both sides are holomorphic in s, so the laws hold identically. The displayed identity is a finite computation with rational functions of (x,s). □

Receipts [computed]. The certificate A_s was found by exact linear algebra at each $n\le 6$ and $s\le 5$, the same at every n up to the factor $(n+1)x-(n+2)s$, and the identity was then checked with s indeterminate. At $n=1$ ($p_1=1$, $c^{(1)}=(-1,1)$) it holds as an identity, but the sums of the proof diverge. The five-mark law at $s=\frac12$ and $s=\frac{3}{10}$, from the share side by quadrature: residuals $4.6\cdot10^{-14}$ and $2.3\cdot10^{-14}$ against terms of size 10^2. At $s=0$, with the cut at 0, the same reduction returns the corner relation of $n+1$ marks as $A_0'(0)=n+1$ times the boundary of Proposition 15.23: the corners $(n+1)!=(n+1)\cdot n!$ of Proposition 15.3, here $120=5\cdot24$.

Corollary 15.26 (Two masters over the corner, two lowered numbers)**.** [derived] *For every* $s \in \{1,2,\dots\}$ *there are rationals* a_s, b_s, q_s *with*

$$\Gamma(s)^2 P_{\binom{[5]}{1}}(s\mathbf{1}; s, s) = q_s + a_s\, P_{\binom{[5]}{1}}(\mathbf{1}; 1, 1) + b_s\, P_{\binom{[5]}{1}}(2\mathbf{1}; 2, 2),$$
$$x^{(1)}(s) = a_s\, x^{(1)}(1) + b_s\, x^{(1)}(2),$$

and in the block

$$P_{\binom{[5]}{1}}(\mathbf{1}; 1, 1) = (\mathsf{T}_{0;(1)}[x]\, f\, \mathbb{I}_{\mathbb{Z}})((0,\infty)),$$
$$P_{\binom{[5]}{1}}(2\mathbf{1}; 2, 2) = (\mathsf{T}_{0;(1)}[x]\, (x-1)^2 f\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) - 2\,(\mathsf{T}_{0;(1)}[x]\, (x-1)\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)),$$
$$x^{(1)}(1) = (\mathsf{T}_{0;(1)}[x]\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)), \qquad x^{(1)}(2) = (\mathsf{T}_{0;(1)}[x]\, (x-1)^2 p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

Modulo $L^*(x\,\mathbb{Q}[x])$ *every word* W_s *lies in* $\operatorname{span}\{1, (x-1)^2\}$*: the ray reaches the direction* $x-1$ *of the block only through the value term* $-2(\mathsf{T}_{0;(1)}[x]\,(x-1)p\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ *of its second master. For example* $(a_3, b_3, q_3) = \big(-\frac{1}{45}, \frac{19}{15}, \frac{8}{15}\big)$ *and* $(a_4, b_4, q_4) = \big(-\frac{113}{1125}, \frac{1907}{375}, \frac{944}{375}\big)$*.*

Proof. The four block identities are Proposition 15.22 at $s = 1, 2$, with $W_1 = 1$, $W_2 = (x-1)^2$, $W_2' = 2(x-1)$. Write $W_s/\Gamma(s)^2 = L^*B_s + r_s$ with $\deg r_s \le 2$. Since the display of Theorem 15.25 is an identity of polynomials and the splitting of Proposition 15.24 is direct, the r_s obey the five-mark recurrence for $s \ge 1$, and at $s = 0$, where $A_0 = 5x$, they obey $\sum_{i\ge1} c_i^{(5)}(0)\, r_i = 0$. As $r_1 = 1$ and $r_2 = (x-1)^2$, every r_s lies in their span, and the same recursion, from $B_1 = B_2 = 0$, gives $B_s(0) = 0$. Then $x^{(1)}(s) = \Gamma(s)^2(\mathsf{T}_{0;(1)}[x]\, r_s\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$, and by Proposition 15.23 $P_{\binom{[5]}{1}}(s\mathbf{1}; s, s) = (\mathsf{T}_{0;(1)}[x]\,(r_s f - r_s' p)\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + 24B_s'(0)$, where $\sum_{i\ge1} c_i^{(5)}(0)B_i'(0) = A_0'(0) = 5$ carries the corner. □

Proposition 15.27 (The lowered half is a stacked Gamma)**.** [computed]

$$x^{(1)}(1) = (\mathsf{T}_{0;(1)}[x]\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) = \frac{1}{120}\Gamma_4^8\left[\begin{matrix}\frac{1}{15}, & \frac{2}{15}, & \frac{4}{15}, & \frac{8}{15}, & \frac{1}{5}, & \frac{2}{5}, & \frac{3}{5}, & \frac{4}{5} \\ & & \frac{1}{2}, & \frac{1}{2}, & \frac{1}{2}, & \frac{1}{2} & & \end{matrix}\right]$$

to 364 *digits, and*

$$x^{(1)}(2) = \frac{65\, x^{(1)}(1) - 32\,\Gamma(\frac12)^8/x^{(1)}(1)}{1125} \quad (360\ \textit{digits}),$$
$$x^{(2)}(1) = \frac{\sqrt{15}}{4\,\Gamma(\frac12)^2}\, x^{(1)}(1), \qquad x^{(2)}(2) = \frac{\sqrt{15}}{4\,\Gamma(\frac12)^2} \cdot \frac{65\, x^{(1)}(1) + 32\,\Gamma(\frac12)^8/x^{(1)}(1)}{1125} \quad (100\ \textit{digits}).$$

So the block of Proposition 15.9 is $x^{(1)}(1)x^{(2)}(2) - x^{(1)}(2)x^{(2)}(1) = \frac{\sqrt{15}}{4\Gamma(1/2)^2} \cdot \frac{64\,\Gamma(1/2)^8}{1125} = \frac{16\sqrt{15}}{1125}\Gamma(\frac12)^6$*: the stacked Gamma cancels from the frame, which is why the frame saw only* $\Gamma(\frac12)$*.*

The values half that the ray reaches is two stacked-Gamma numbers, $x^{(1)}(1)$ and $\Gamma(\frac12)^8/x^{(1)}(1)$. The direction $x-1$ is not among them: $(\mathsf{T}_{0;(1)}[x]\,(x-1)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ has no rational relation with 1, $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\Gamma(\frac12)^4$, $x^{(1)}(1)$ and $\Gamma(\frac12)^8/x^{(1)}(1)$, nor with ten products of $x^{(1)}(1)^{\pm1}$, powers of $\Gamma(\frac12)^2$ and $\sqrt{15}$, with coefficients below 10^{12} [computed].

Proposition 15.28 (The block on the four-mark chart and its pencil)**.** [derived]*; receipts* [computed] *On the chart* $y_{e_0} = 0$ *of four marks let* $X = K(\alpha)K(\alpha^{-1})$*, the reciprocal of the stride of the centre ray (Def. 35.9, p. 309), so that* $p(\sigma) = \int X^{-\sigma}dy$*, the value along the ray from the real origin* $(\sigma\mathbf{1}; \sigma, \sigma)$ *(Def. 35.1, p. 307), with* $X \ge 16$ *on the chart (Thm. 35.10(i), p. 309). For a polynomial* R*,*

$$(\mathsf{T}_{0;(1)}[x]\, R\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) = \int \rho_R(X)\, dy, \qquad (\mathsf{T}_{0;(1)}[x]\, R\, f\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) = \int \log X\, \rho_R(X)\, dy,$$

$$\rho_R(X) = \sum_{m\geq 1} R(m) X^{-m},$$

a rational function of X whose only pole is $X = 1$. In particular

$$P_{\binom{[5]}{1}}(\mathbf{1}; 1, 1) = \int \frac{\log X}{X-1}\, dy, \qquad P_{\binom{[5]}{1}}(2\mathbf{1}; 2, 2) = \int \frac{(X+1)\log X - 2(X-1)}{(X-1)^3}\, dy,$$
$$x^{(1)}(1) = \int \frac{dy}{X-1}, \qquad x^{(1)}(2) = \int \frac{X+1}{(X-1)^3}\, dy, \qquad (\mathsf{T}_{0;(1)}[x]\,(x-1)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \int \frac{dy}{(X-1)^2}.$$

The first is the left closing of Lemma 5.16(ii), p. 34, read on the chart. With the generating function of the four-mark ray from the origin $(\varepsilon\mathbf{1}; \varepsilon, \varepsilon)$, $G_\varepsilon(q) = \sum_{k\geq 1} p(k+\varepsilon)\, q^{k-1} = \int X^{-\varepsilon}\, dy/(X-q)$ for $|q| < 16$, carried by the pencil $K\Psi - q\alpha^{\mathbf{1}}$ (Thm. 35.10(ii)),

$$(\mathsf{T}_{0;(1)}[x]\,(x-1)^j\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (q\partial_q)^j G_0(q)\big|_{q=1},$$
$$(\mathsf{T}_{0;(1)}[x]\,(x-1)^j\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\partial_\varepsilon (q\partial_q)^j G_\varepsilon(q)\big|_{q=1,\ \varepsilon=0}:$$

the block is the jet at $q = 1$ of the four-mark generating function and of its slope in the origin.

Proof. Def. 35.1 along the ray from $(\sigma\mathbf{1}; \sigma, \sigma)$, the positivity of every term, and $\sum_{m\geq 1} R(m)X^{-m}$ summed for $X > 1$: $\rho_1 = 1/(X-1)$, $\rho_{x-1} = 1/(X-1)^2$, $\rho_{(x-1)^2} = (X+1)/(X-1)^3$. For the masters use Corollary 15.26; for the pencil, Thm. 35.10(ii) with $k_0 = 1$ and the origin form $X^{-\varepsilon}dy$. □

Receipts [computed], by three-dimensional quadrature on the chart, to twelve digits:

$$\int \frac{dy}{X} = 8.414398322115, \qquad \int \frac{\log X}{X-1}\, dy = 39.945588011991, \qquad \int \frac{dy}{X-1} = 8.570280443372,$$

against $7(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, the five-mark centre at strength one, and $x^{(1)}(1)$.

Remark 15.29 (What the closing found). (i) The live Trailmoment lands on lattice inputs only through the residue at zero shares (Proposition 15.23).

(ii) The words of the five-mark centre ray reach a plane of the five-dimensional block on each side: on the values side the lowered setting, two stacked-Gamma numbers (Proposition 15.27); on the slopes side the period, whose two masters are

$$P_{\binom{[5]}{1}}(\mathbf{1}; 1, 1) = (\mathsf{T}_{0;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),$$
$$P_{\binom{[5]}{1}}(2\mathbf{1}; 2, 2) = (\mathsf{T}_{0;(1)}[x]\,(x-1)^2 f\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - 2\,(\mathsf{T}_{0;(1)}[x]\,(x-1)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)):$$

the slope in the origin of the four-mark generating function at $q = 1$, and its second jet less twice the first jet of the generating function itself. They have no rational relation with 1, $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $f(1)$, $\Gamma(\frac{1}{2})^4$, $\Gamma(\frac{1}{2})^8$, $(\mathsf{T}_{5;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\Gamma(\frac{1}{2})^4(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, the two stacked-Gamma numbers and the direction $x-1$ of the block, with coefficients below 10^{20} (PSLQ, 360 digits) [computed]: in that framework, over $\mathbb{Q}$, they are new numbers.

(iii) The last minor of §14.17 is now written in the masters over the stacked Gamma (Proposition 15.27):

$$P_{\binom{[5]}{1}}(\mathbf{1}; 1, 1)\, x^{(1)}(2) - P_{\binom{[5]}{1}}(2\mathbf{1}; 2, 2)\, x^{(1)}(1)$$
$$= \frac{65\, x^{(1)}(1) - 32\,\Gamma(\frac{1}{2})^8/x^{(1)}(1)}{1125}\, P_{\binom{[5]}{1}}(\mathbf{1}; 1, 1) - x^{(1)}(1)\, P_{\binom{[5]}{1}}(2\mathbf{1}; 2, 2)$$

$= 4.340002931535662034 07\ldots,$

and it is not a rational combination of 1, $\Gamma(\frac{1}{2})^4$, $\Gamma(\frac{1}{2})^8$, $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and seven products of $x^{(1)}(1)^{\pm 1}$ or $x^{(1)}(1)^{\pm 2}$ with powers of $\Gamma(\frac{1}{2})^4$, with coefficients below 10^{15} [computed]. Its scale is the masters' scale; nothing smaller is missing.

(iv) The rational words and the pole columns of Chapter 6 did not enter: every word of the centre ray is a polynomial, read at index zero. What closed the live Trailmoment is the shifted inputs (Thm. 6.2), the cut relation (Prop. 5.3), summation by parts in its adjoint form (Thm. 6.5, p. 43), and a fixed block at nonpositive index, the counterpart of Thm. 6.9, p. 45.

16 The five-mark lists along their hinge trees

A connected list is glued from prime tables and series and parallel nodes at its bunches of width one (Thm. 20.9, p. 166), and its canonical hinge tree (Def. 22.14, p. 189) is read off the Skeleton: the width $c(S) = r(S) + r(\mathcal{E} \setminus S) - r(\mathcal{E})$ of Def. 7.10, p. 49, is a count of the circuit and cocircuit layers, and a bunch of width one is where the list composes itself out of two pieces. The period is read from the leaves up (Thm. 22.15, p. 190). A cell enters only through its hinge wall and its law (Thm. 22.7, p. 185), a series node multiplies the transforms of its cells and a parallel node their dual transforms (Thm. 22.13, p. 188), and a mark left in place is a one-mark cell whose two transforms at the centre are the Gramma $\mathsf{G}_{0;(1,-1)}(1,y)$ (Def. 22.17(1), p. 191). This entry reads the five lists of §15.4 that way. Three are series-parallel, one carries a prime node that acts as a mark, and one is prime.

list	canonical hinge tree [derived]
$\binom{[5]}{1}$	one parallel node
$\binom{[4]}{2}$, a mark doubled	the prime node $\binom{[4]}{2}$, a parallel node of three marks at one marker
$\binom{[5]}{2}$	one prime node
$\binom{[3]}{2}$, two marks doubled	a series node of three marks, parallel nodes of three marks at two markers
$\binom{[3]}{2}$, a mark tripled	a series node of three marks, a parallel node of four marks at one marker

The bunches of width one with two marks on each side are the parallel classes, of width $1 + r(\mathcal{E} \setminus P) - r(\mathcal{E}) = 1$; in $\binom{[4]}{2}$ and $\binom{[5]}{2}$ every set of two marks has width 2, so both are prime.

16.1 A prime node that is a mark

Proposition 16.1 (The doubled mark of $\binom{[4]}{2}$)**.** [derived]*; receipts* [computed] *Replace a mark i of $\binom{[4]}{2}$ by a parallel class of $k \geq 1$ marks. Then*

$$P_{\binom{[4]}{2}[i\to k]}(\sigma\mathbf{1};\sigma,\sigma) = P_{\binom{[3]}{1}}(\sigma\mathbf{1};\sigma,\sigma)\; P_{\binom{[k+1]}{1}}(\sigma\mathbf{1};\sigma,\sigma).$$

At $k = 1$ this is $P_{\binom{[4]}{2}} = P_{\binom{[3]}{1}}\,\Gamma_{1}^{2}\begin{bmatrix}\sigma, & \sigma\\ & 2\sigma\end{bmatrix}$, the agreement of the table of §15.1 at every strength; at $k = 2$ it is the second row of §15.4,

$$P_{\binom{[4]}{2},\ \text{a mark doubled}}(\sigma\mathbf{1};\sigma,\sigma) = P_{\binom{[3]}{1}}(\sigma\mathbf{1};\sigma,\sigma)^2.$$

Proof. $\binom{[4]}{2}$ is identically self-dual. The cell of $\binom{[4]}{2}$ hinged at i has the hinge minors $\binom{[4]}{2} \setminus i = \binom{[3]}{2}$ and $\binom{[4]}{2}/i = \binom{[3]}{1}$, dual with the identity and the same weights, so its ratio is identically one, and in every gluing it acts as $\phi(\Delta)$ times one mark, with $\phi(\Delta) = P_{\binom{[3]}{2}}(\sigma\mathbf{1};\kappa,2\sigma-\kappa)$ (Cor. 22.8(i),(ii),

p. 186). Its hinge wall is $\kappa = \sigma$, balanced against the parallel cell of the class, whose hinge wall is $k\sigma - (k-1)\sigma = \sigma$ (Thm. 22.13(ii)). By Prop. 8.14, p. 57, $P_{\binom{[3]}{2}}(\sigma\mathbf{1};\sigma,\sigma) = P_{\binom{[3]}{1}}(\sigma\mathbf{1};\sigma,\sigma)$. What is left is the class glued to one mark, the banana $\binom{[k+1]}{1}$. □

Receipts [computed]: Procedure 15.6 for $\binom{[4]}{2}$ with a mark doubled agrees with $P_{\binom{[3]}{1}}(\sigma\mathbf{1};\sigma,\sigma)^2$ to 60 digits at $\sigma = 1,\dots,4$, and for $\binom{[4]}{2}$ with $P_{\binom{[3]}{1}}\Gamma_{1}^{2}\begin{bmatrix}\sigma,\ \sigma\\ 2\sigma\end{bmatrix}$ at $\sigma = 1,2,3$.

Corollary 16.2. [derived] *The ray law of $\binom{[4]}{2}$ with a mark doubled is the symmetric square of the three-mark law: its native values are the products of $\frac{1}{9}$ and 1 in pairs, the roots $81, 9, 1$ of §15.4, and its corner is $6^2 = 36$. Every value lies in $\mathbb{Q} + \mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1) + \mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)^2$: its two masters are the three-mark centre and its square.*

The same argument at the centre of any identically self-dual list splits off a mark (Cor. 22.8(iii) and Prop. 8.14) [derived]:

$$P_{\binom{[2r]}{r}}(\sigma\mathbf{1};\sigma,\sigma) = \Gamma_{1}^{2}\begin{bmatrix}\sigma,\ \sigma\\ 2\sigma\end{bmatrix}\ P_{\binom{[2r-1]}{r-1}}(\sigma\mathbf{1};\sigma,\sigma).$$

So $\binom{[6]}{3}$ carries exactly the masters of $\binom{[5]}{2}$.

16.2 A series root over banana leaves

Proposition 16.3 (The triangle rows)**.** [derived]*; receipts* [computed] *Let f_{P_k} be the transform at the centre of the parallel cell of a class of k marks; $\int_0^\infty y^{w-1} f_{P_k}(y)\,dy = \Gamma(w)^2 P_{\binom{[k]}{1}}((\sigma-w)\mathbf{1};\sigma-w,\sigma-w)$ for $0 < \operatorname{Re} w < \sigma$. Then*

$$\begin{aligned}\Gamma(\sigma)^2 P_{\binom{[3]}{2},\ \text{a mark tripled}}(\sigma\mathbf{1};\sigma,\sigma) &= \int_0^\infty y^{\sigma-1}\,\mathsf{G}_{0;(1,-1)}(1,y)^2\,f_{P_3}(y)\,dy\\ &= \frac{1}{2\pi i}\int_{(c)} \Gamma(d)^2\Gamma(\sigma-d)^2\,P_{\binom{[4]}{2}}(d\mathbf{1};d,d)\,dd,\\ \Gamma(\sigma)^2 P_{\binom{[3]}{2},\ \text{two marks doubled}}(\sigma\mathbf{1};\sigma,\sigma) &= \int_0^\infty y^{\sigma-1}\,\mathsf{G}_{0;(1,-1)}(1,y)\,f_{P_2}(y)^2\,dy.\end{aligned}$$

So the tripled triangle is the closing of Proposition 15.22 with the ray of the four-mark banana replaced by the ray of $\binom{[4]}{2}$.

Proof. The triangle is a series root (Def. 22.17(5), p. 191): its cells are the parallel cells of the classes and the marks left in place, the latter with the transform $\mathsf{G}_{0;(1,-1)}(1,y)$. The dual transform of a parallel cell is the product of the k Grammas of its marks (Thm. 22.13(ii)), whose Mellin read is $\Gamma(z)^2 P_{\binom{[k]}{1}}(z\mathbf{1};z,z)$ (Proposition 15.2); its hinge wall is σ, and at $s = t = \kappa = \sigma$ the change of transform reflects $z \mapsto \sigma - z$ (Prop. 22.5(iii),(v), p. 184). The Mellin read of a product pins the sum of the shares, as in the proof of Proposition 15.2, and $P_{\binom{[2]}{1}} P_{\binom{[3]}{1}} = P_{\binom{[4]}{2}}$ is Proposition 16.1 at $k = 1$. □

Receipts [computed]: the closing $-\sum_{d\ge1}(W_s\,P_{\binom{[4]}{2}})'(d)/\Gamma(s)^2$ of Proposition 15.22 agrees with Procedure 15.6 for the tripled triangle to 60 digits at $s = 1,2,3$; for two marks doubled, opening the list at its unmarked mark (Prop. 19.19, read as in Proposition 16.5 below: the contraction is the four-mark banana, the deletion two parallel pairs) gives 7.8658373683630 by three-dimensional quadrature, against 7.86583736836402159 ….

Proposition 16.4 (Closing the tripled triangle over its input)**.** [derived]*; receipts* [computed] *Let* $q(\sigma) = P_{\binom{[4]}{2}}(\sigma\mathbf{1};\sigma,\sigma)$ *and* $f_q = -q'$.

(i) q *obeys the three-mark law times the shift of the stacked Gamma,*

$$\sigma^3 q(\sigma) - 2(2\sigma+1)(10\sigma^2+10\sigma+3)\,q(\sigma+1) + 36(\sigma+1)(2\sigma+1)(2\sigma+3)\,q(\sigma+2) = 0,$$

with corner $12 = 2\cdot 6$ *and no term of order* σ*, and its symbol at* $\lambda = 1$ *is* $105 \neq 0$.

(ii) Summation by parts at the cut 0*, as in Proposition 15.23: for every polynomial* R*,*

$$(\mathsf{T}_{0;(1)}[x]\,(L_q^* R)\,q\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -12\,R(0),$$
$$(\mathsf{T}_{0;(1)}[x]\,(L_q^* R)\,f_q\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (\mathsf{T}_{0;(1)}[x]\,(L_q'^* R)\,q\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + 36\,R(-1)\,P_{\binom{[3]}{1}}(\mathbf{1};1,1).$$

The last coefficient of the law has only a simple zero at $\sigma = -1$*: the shift of the stacked Gamma,* $P_{\binom{[2]}{1}}(\sigma+2)/P_{\binom{[2]}{1}}(\sigma+1) = (\sigma+1)/(2(2\sigma+3))$*, cancels one of the two factors* $\sigma+1$ *of the three-mark coefficient* $9(\sigma+1)^2$*. So the slope boundary there reads* $q(1) = P_{\binom{[3]}{1}}(\mathbf{1};1,1)$.

(iii) Every value of the tripled triangle is a rational number, plus a rational multiple of $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$*, plus a rational combination of the three slope Trailmoments* $-(\mathsf{T}_{0;(1)}[x]\,(Rq)'\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$*,* $R \in \{1,\, x,\, x^2/2\}$*. The words* W_s*,* $s \le 8$*, reach all three; the five-mark banana reached two (Corollary 15.26).*

(iv) In these coordinates the rational relation of §15.4 for this row, the second corner, is the relation among $1, P(1), \dots, P(5)$*, coordinate by coordinate.*

Proof. (i) $\sigma\,P_{\binom{[2]}{1}}(\sigma) = 2(2\sigma+1)\,P_{\binom{[2]}{1}}(\sigma+1)$ (Prop. 1.3(iii)) multiplies the three-mark law coefficient by coefficient into σ times the display. The corner is the product of the corners $\lim \sigma P_{\binom{[2]}{1}} = 2$ and 6; $\sigma\Gamma_{\substack{2\\1}}\left[\begin{smallmatrix}\sigma,\ \sigma\\ 2\sigma\end{smallmatrix}\right] = 2\Gamma(1+\sigma)^2/\Gamma(1+2\sigma)$ and $\sigma^2 P_{\binom{[3]}{1}}$ have no terms of order σ (Proposition 15.3). (ii) The move of Proposition 15.23: the value boundary is $-6q(1)+108q(2) = -12$; for the slopes the boundary at the cut 0 is the term of order σ, zero, and the one at -1 is $-R(-1)\,c_2'(-1)\,q(1)$ with $c_2'(-1) = -36$. (iii) As in Proposition 15.24, with $\operatorname{span}\{1,x,x^2\}$ as the complement of $L_q^*(\mathbb{Q}[x])$, and Proposition 15.22 for the values. (iv) Exact arithmetic on the eight coordinates. □

Receipts [computed]. The coordinates reproduce Procedure 15.6 to 60 digits at $s \le 5$. There is no rational relation among 1, the three-mark centre $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, its slope $\partial_\sigma P_{\binom{[3]}{1}}(\sigma\mathbf{1};\sigma,\sigma)|_{\sigma=1}$ and the six Trailmoments

$$(\mathsf{T}_{0;(1)}[x]\,x^j q\,\mathbb{I}_{\mathbb{Z}})((0,\infty)), \qquad (\mathsf{T}_{0;(1)}[x]\,x^j f_q\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) \qquad (j \le 2)$$

with coefficients below 10^{25}, and none among 1, the three-mark centre and the three slope Trailmoments of (iii) below 10^{40} (PSLQ, 300 digits). So the four masters of the tripled triangle are the three-mark centre, entering at the boundary $\sigma = -1$, and the three slope Trailmoments.

16.3 The prime node at a mark

Proposition 16.5 ($\binom{[5]}{2}$ opened at a mark)**.** [derived]*; receipt* [computed] *For a mark* e *of* $\binom{[5]}{2}$ *the deletion is* $\binom{[4]}{2}$ *and the contraction* $\binom{[4]}{1}$*. Write* $X_\backslash = X_{\binom{[4]}{2}}$ *and* $X_/ = X_{\binom{[4]}{1}}$ *for their strides on the chart of the other four marks. Then*

$$P_{\binom{[5]}{2}}(\sigma\mathbf{1};\sigma,\sigma) = \int dy \int_0^1 w^{\sigma-1}(1-w)^{\sigma-1}\big(w\,X_\backslash + (1-w)\,X_/\big)^{-\sigma} dw,$$

and on the share side of the shape,

$$\Gamma(\sigma)^2 P_{\binom{[5]}{2}}(\sigma\mathbf{1};\sigma,\sigma) = \frac{1}{2\pi i}\int_{(c)} \Gamma(d)^2\Gamma(\sigma-d)^2\,\Phi_\sigma(d)\,dd, \qquad \Phi_\sigma(d) = \int X_/^{-d} X_\backslash^{-(\sigma-d)}\,dy,$$

$0 < c < \sigma$, *with* $\Phi_\sigma(d) = P_F\big(\sigma\mathbf{1};(d,\sigma-d),(d,\sigma-d)\big)$ *the centre of the family* F *of the two children (Def. 8.1, p. 53). At* $\sigma = 1$,

$$P_{\binom{[5]}{2}}(\mathbf{1};1,1) = \int \frac{\log X_\backslash - \log X_/}{X_\backslash - X_/}\,dy.$$

Proof. The mark is the Breta of its children's even letters (Prop. 19.19, p. 144): at the centre the prefactor and the stacked Gamma are one, and $E_{\backslash e} = \alpha^{\mathcal{E}\backslash e} X_\backslash$, $E_{/e} = \alpha^{\mathcal{E}\backslash e} X_/$. Splitting the coefficient $wX_\backslash + (1-w)X_/$ (Cor. 4.4, p. 26) and integrating the shape gives the Gammas, one pair for each child. □

Receipt [computed]: the last integral is 4.0297141707108 by three-dimensional quadrature, against 4.02971417071104960732… of §15.4.

Proposition 16.6 (The contraction under the deletion)**.** [derived] *On the chart* $X_\backslash - X_/ = (e_2^2 - e_1e_3)/e_4$ *with*

$$e_2^2 - e_1e_3 = \sum_{i<j}\alpha_i^2\alpha_j^2 + \sum_i \sum_{\{j,k\}\not\ni i} \alpha_i^2\alpha_j\alpha_k + 2\,\alpha_1\alpha_2\alpha_3\alpha_4,$$

a polynomial with positive coefficients, so $X_/ < X_\backslash$ *everywhere. The walls of* F *at the strengths* $(d, 1-d)$ *are* 1 *at* $|S|$ *odd and* $2-d$ *at* $|S| = 2$, *so* $\Phi = \Phi_1$ *is holomorphic on* $\operatorname{Re} d < 2$, *and closing to the left (Thm. 4.5, Lemma 5.16),*

$$P_{\binom{[5]}{2}}(\mathbf{1};1,1) = \sum_{m\ge 0}\Phi'(-m), \qquad \Phi(-m) = \int X_/^{\,m}\, X_\backslash^{-1-m}\,dy :$$

the slopes of the deletion's centre ray with the contraction's even letter inserted m *times.*

Proof. $X_/ = e_1e_3/e_4$, $X_\backslash = e_2^2/e_4$, and the polynomial is read off the monomials of e_2^2 and e_1e_3. The widths are 1 for every S in $\binom{[4]}{1}$ and $1,2,1$ at $|S| = 1,2,3$ in $\binom{[4]}{2}$. On the line $\operatorname{Re} d = -N - \frac12$,

$$|\Phi(d)| \le \int \big(X_/ / X_\backslash\big)^{N+1/2} X_\backslash^{-1}\,dy \to 0,$$

so the left closing applies; at $d = -m$ the kernel $\Gamma(d)^2\Gamma(1-d)^2$ of Lemma 5.16 has the principal part $(d+m)^{-2}$ and no simple part. Equivalently, expand $1/(X_\backslash - X_/) = \sum_m X_/^{\,m} X_\backslash^{-m-1}$ in the last display of Proposition 16.5. □

The ratio $X_/ / X_\backslash$ is largest at the table's gauge, where it is $\frac{16}{36} = \frac49$, so the closing converges like $(\frac49)^m$ (alias: Newton's inequality $e_1e_3 \le \frac49 e_2^2$) [recalled].

Proposition 16.7 (The values at the lattice points are leans of the three-mark banana)**.** [derived]*; receipts* [computed] *For* $m \in \mathbb{N}$, *with* $(e_1e_3)^m = \sum_v c_v\alpha^v$,

$$\Phi(-m) = \sum_v c_v\, P_{\binom{[4]}{2}}\big(\mathbf{1}+v;1+m,1+m\big),$$

and for masses $\boldsymbol{\mu}$ on the balance wall and every mark g,

$$P_{\binom{[4]}{2}}(\boldsymbol{\mu}; s, s) = \frac{1}{\Gamma(2s)} \int_0^\infty y^{\mu_g - 1} \prod_{e \neq g} \mathsf{G}_{2s-\mu_g-\mu_e;(1,-1)}(1, y)\, dy :$$

$\Phi(-m)$ is a finite sum of leans of the three-mark banana, Grammas of integer index read three at a time. The first values are

$$\Phi(0) = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1), \qquad \Phi(-1) = \frac{2 + 2P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)}{9},$$
$$\Phi(-2) = \frac{71 + 56P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)}{810} \quad \text{[computed]}.$$

Proof. On the chart $X_{/}^m X_{\backslash}^{-1-m} = \alpha^{\mathbf{1}} (e_1 e_3)^m e_2^{-2(1+m)}$, and e_2 is both letters of $\binom{[4]}{2}$; each monomial α^v is an insertion moving the masses (Lemma 35.2(iii), p. 308). Then Cor. 22.8(iii) at g, $P_{\binom{[4]}{2}}(\boldsymbol{\mu}; s, s) = \Gamma_{\binom{2}{1}}\begin{bmatrix} \mu_g, & 2s - \mu_g \\ & 2s \end{bmatrix} P_{\binom{[3]}{2}}(\boldsymbol{\mu}|; 2s - \mu_g, \mu_g)$; Prop. 8.14 turns $\binom{[3]}{2}$ into $\binom{[3]}{1}$ at the exponents $(\mu_g, 2s - \mu_g)$; and the share side of the three-mark banana at a lean, $\Gamma(a)\Gamma(b) P_{\binom{[3]}{1}}(\boldsymbol{\mu}'; a, b) = \int_0^\infty y^{a-1} \prod_e \mathsf{G}_{b-\mu'_e;(1,-1)}(1, y)\, dy$ (Prop. 4.2 with the pins $a_e - b_e = \mu'_e - b$, as in Proposition 15.2), whose Gammas cancel the Beta corner. □

Receipts [computed]: $\Phi(-1)$ by three-dimensional quadrature is 0.7430904974869, against $\frac{2}{9}(1 + P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)) = 0.74309049748654642013\ldots$; the leans themselves by one-dimensional quadrature to 40 digits, with PSLQ for the rational coefficients; on the way, $\int_0^\infty \mathsf{G}_{0;(1,-1)}(1, y)\, \mathsf{G}_{1;(1,-1)}(1, y)^2\, dy = \frac{1}{3}$.

Remark 16.8 (What the prime node reads). The prime node reads the three-mark banana at leans. Its values at the lattice points land on $\mathbb{Q} + \mathbb{Q}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$, and its period at strength one is the sum of the slopes there: the division of Propositions 15.23 and 15.24, values on lattice inputs and slopes carrying the masters, now along the deletion's ray with the contraction's even letter as its stride, of native value $\frac{4}{9}$ at the table's gauge. Two of the three masters of $\binom{[5]}{2}$ are slope Trailmoments of that ray, as the two of $\binom{[5]}{1}$ were of the four-mark ray, and the third is $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ (Theorem 17.8); the law in m and its closing are Propositions 17.2 and 17.1.

Remark 16.9 (What the tree decided). A parallel node multiplies Grammas: the banana. An identically self-dual prime node is a mark of ratio one, and the period factors: the doubled mark of $\binom{[4]}{2}$ is the square of the three-mark ray, with no new number. A series root over parallel cells multiplies transforms; its closing has the banana's shape over a product input, and the stacked Gamma in that input's law opens the boundary at $\sigma = -1$, which is where the second corner of the tripled triangle comes from (for two marks doubled it is the boundary at zero shares, Remark 17.13(ii)). A prime node that is not self-dual is opened at a mark into its children, weighed by the Breta of their even letters, and reads leans. The three-mark centre enters every row of §15.4 but the banana, row four included, through the contraction of H at its single mark. The row with two marks doubled has only pairs and one mark at its leaves; its masters are $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$, $\Gamma(\frac{1}{2})^4$ and $P_R(1)$ (Theorem 17.12).

17 The masters of $\binom{[5]}{2}$; two marks doubled, closed

Both lists are read at a mark (Prop. 19.19, p. 144): the period is the Breta of the even letters of the two children, its share side is a Mellin–Barnes integral over the split d with the kernel of Lemma 5.16,

p. 34, and the closing (Thm. 4.5, p. 26) reads a values input and a slope input on the lattice of the split. For $\binom{[5]}{2}$ the children are $\binom{[4]}{2}$ and $\binom{[4]}{1}$ (§16.3). For the triangle with two marks doubled the mark is taken in a doubled class; the children are then those of the triangle H with one class doubled, opened at its single mark: the three-mark banana $H/1$ and a pair with a coloop $H \setminus 1$, and the class enters through its Mellin read. In both lists the lattice values of the input are leans of the three-mark banana. This entry finds the law of each input along its lines, the boundaries of its slope input, and the masters. Throughout, $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ is the three-mark centre and $P_{\binom{[4]}{1}}(\mathbf{1}; 1, 1)$ the four-mark centre.

17.1 The input of $\binom{[5]}{2}$ along its lines

For $s \in \{1, 2, \dots\}$ put $\Phi_s(d) = \Phi(d, s-d) = \int X_{/}^{-d} X_{\backslash}^{-(s-d)}\, dy$, the family of Proposition 16.5 on the line s; Φ_1 is the Φ of Proposition 16.6. By Proposition 16.7 every lattice value is a finite sum of leans of the three-mark banana,

$$\Phi_s(-m) = r_s(m) + c_s(m)\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1), \qquad r_s(m),\, c_s(m) \in \mathbb{Q} \quad (m \in \mathbb{N}),$$

and c_s is the *three-mark coordinate* of the values input.

Proposition 17.1 (The closing at every integer strength)**.** [derived] *For $s \in \{1, 2, \dots\}$ let $V_s(m) = \big((m+1)(m+2)\cdots(m+s-1)\big)^2$. Then*

$$\Gamma(s)^2 P_{\binom{[5]}{2}}(s\mathbf{1}; s, s) = \sum_{m \geq 0} \Big(V_s(m)\, \Phi_s'(-m) - V_s'(m)\, \Phi_s(-m) \Big),$$

and $x_s = \Gamma(s)^{-2} \sum_{m \geq 0} V_s(m)\, \Phi_s(-m)$ is the lowered setting, the values half of the closing.

Proof. Proposition 16.5 and the left closing of Proposition 16.6 at the strength s. By the reflection (Prop. 1.3(v), p. 11), $\Gamma(d)^2\Gamma(s-d)^2 = V_s(-d)\,\Gamma(d)^2\Gamma(1-d)^2$, with double poles at $d = -m$ and no simple parts; the residue at $d = -m$ is $\big(V_s(-d)\Phi_s(d)\big)'$ there. □

At $s = 1$ this is Proposition 16.6.

Proposition 17.2 (The law on a line)**.** [derived] *in Proposition 17.15 On the line $s = 1$,*

$$\sum_{i=0}^{4} c_i(m)\, \Phi_1(-m-i) = 0,$$

$$\begin{aligned}
c_0 &= -8(m+1)^3(m+2)(m+3)(7m+16),\\
c_1 &= (m+2)(m+3)(630m^4 + 4583m^3 + 12351m^2 + 14584m + 6336),\\
c_2 &= -(m+3)(2m+5)(1295m^4 + 11241m^3 + 35869m^2 + 49386m + 24408),\\
c_3 &= 3(2m+5)(2m+7)(385m^4 + 3855m^3 + 13913m^2 + 21153m + 11286),\\
c_4 &= -54(m+2)(m+4)(2m+5)(2m+7)(2m+9)(7m+9),
\end{aligned}$$

and r_1, c_1 obey it separately. Its symbol is

$$216\rho^4 - 330\rho^3 + 185\rho^2 - 45\rho + 4 = (2\rho-1)(9\rho-4)(3\rho-1)(4\rho-1).$$

The lines $s = 2, 3$ have laws of order four and degree eight with the same symbol.

Receipts [computed]. Found by exact linear algebra on the values $\Phi_s(-m)$, $m \le 80$, of Proposition 16.7 (at $s = 1$: 138 equations, 35 unknowns, one solution). At $m = -1$, outside the data, c_0 has a triple zero and the law holds exactly, $\sum_{i\ge1} c_i(-1)\Phi_1(1-i) = 0$ with $(c_1,\dots,c_4)(-1) = (300, -5670, 25920, -34020)$. It was read off first; Proposition 17.15 derives it from the relations along a ray.

Proposition 17.3 (The roots are the native values of the ratio of the strides)**.** [derived] *On the chart $X_/ / X_\backslash = e_1e_3/e_2^2$. At the table's gauge its value is $\frac{4}{9}$, the maximum. On the pattern $(1,1,x,x)$ its native equation is $(1-x)(1+x^2) = 0$, and at $(1,1,i,i)$ the value is $\frac{1}{2}$. Where one mark goes to zero it becomes e_1e_3/e_2^2 of three marks, of maximum $\frac{1}{3}$; where one mark goes to zero and another to infinity it becomes $\alpha_1\alpha_2/(\alpha_1+\alpha_2)^2$, the reciprocal stride of the two-mark banana, of maximum $\frac{1}{4}$. These four values are the roots of the symbol of Proposition 17.2.*

The values input decays like $(\frac{4}{9})^m$; its coordinates r_1 and c_1 decay like $(\frac{1}{2})^m$, and

$$P_{\binom{[3]}{1}}(\mathbf{1};1,1) = -\lim_{m\to\infty}\frac{r_1(m)}{c_1(m)} \quad \text{[computed]}:$$

the three-mark centre is the multiplier that removes the component along the native value $\frac{1}{2}$ of the complex gauge.

Proposition 17.4 (Lines tied by a first-order relation)**.** [derived] *for $s \le 4$ in Proposition 17.15 With $F(m,n) = \Phi(-m,n)$,*

$$\begin{aligned}&4(m+1)(3m+2n+4)\,F(m,n+1) - 12(n+1)(2n+3)\,F(m+1,n+2) - (m+1)(2n-m)\,F(m,n)\\ &+ (6n^2+4mn-3m^2+16n-4m+5)\,F(m+1,n+1) - 6(n+1)(2n+3)\,F(m+2,n+2) = 0.\end{aligned}$$

On the lines $n = m+s$ it ties the line $s+1$ to the line s, so with the law every line is read off the line 1. Receipts [computed]: found at $s \le 4$, the coefficients polynomial in s; the lines 2 to 5 read off the line 1 agree with Proposition 16.7 at every $m \le 80$.

17.2 The slope input and its boundaries

The law holds as a function of the split (Proposition 17.15), so its derivative is the recurrence of the slope input,

$$\sum_{i=0}^{4} c_i(m)\,\Phi_1'(-m-i) = \sum_{i=0}^{4} c_i'(m)\,\Phi_1(-m-i),$$

as in Proposition 15.21. Its four initial slopes $\Phi_1'(0),\dots,\Phi_1'(-3)$ are tied by the boundaries of the law.

Proposition 17.5 (The cut)**.** [derived] *from Proposition 17.2 At $m = -1$, where the four-mark centre $\Phi_1(1) = P_{\binom{[4]}{1}}(\mathbf{1};1,1)$ sits, c_0 has a triple zero; neither $\Phi_1(1)$ nor its slope enters, and*

$$300\,\Phi_1'(0) - 5670\,\Phi_1'(-1) + 25920\,\Phi_1'(-2) - 34020\,\Phi_1'(-3) = -\frac{21380}{63} + \frac{8866}{63}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1).$$

The lines 2 and 3 give $\frac{1}{2}$ and $-\frac{5}{9}$ times the same relation.

Proposition 17.6 (The wall at $d=2$)**.** *residue* [derived]*; relation* [computed] Φ_1 *has a simple pole at* $d=2$*, the wall* $2-d$ *of the two-element* S *(Proposition 16.6), with residue* -6*: each of the six two-element* S *collapses (Cor. 8.8, p. 56) to a product of two two-mark centres,* $P_{\binom{[2]}{1}}(\mathbf{1};1,1)^2=1$*; the law at* $m=-2$ *returns the same* -6*. There the law meets the slopes only through*

$$c_2(-2)\,\Phi_1'(0)+c_3(-2)\,\Phi_1'(-1)=96\,\Phi_1'(0)-432\,\Phi_1'(-1),$$

and

$$6\,\Phi_1'(0)-27\,\Phi_1'(-1)=7\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)-22$$

(PSLQ, 140 *digits), which is the same as the finite part*

$$\Phi_1(d)=-\frac{6}{d-2}+\frac{21}{2}+\frac{3}{4}\,P_{\binom{[4]}{1}}(\mathbf{1};1,1)+O(d-2).$$

Proposition 17.7 (One free slope, along the three-mark coordinate)**.** [computed] *A third relation,* $1680\,\Phi_1'(0)-24300\,\Phi_1'(-2)=2308\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)-7207$ *(PSLQ), with Propositions 17.5 and 17.6, says that* $\Phi_1'(-m)-\Phi_1'(0)\,c_1(m)\in\mathbb{Q}+\mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)$ *for* $m\le 3$*. By the slope recurrence and Proposition 17.4 it holds for every* m *and every line:*

$$\Phi_s'(-m)=\rho_s(m)+\lambda_s(m)\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)+c_s(m)\,\Phi_1'(0),\qquad \rho_s(m),\,\lambda_s(m)\in\mathbb{Q}.$$

The one free direction of the slope input is the three-mark coordinate of the values input. Minimality fixes it: the slope input has no component along the native value $\frac{1}{2}$*.*

Receipts [computed]. With $\Phi_1'(0)$ from the masters below, $\Phi_1'(-m)/\Phi_1(-m)=0.81342$ at $m=600$, tending to $\log\frac{9}{4}=0.81093$, where a component along $\frac{1}{2}$ would grow like $(\frac{9}{8})^m$. The free slope is

$$\Phi_1'(0)=\int\log\frac{X_\backslash}{X_/}\,\frac{dy}{X_\backslash}=2.7943139847582708461546135530266971416 39\ldots,$$

$$\Phi_1'(0)=2\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)-\partial_\sigma P_{\binom{[3]}{1}}(\sigma\mathbf{1};\sigma,\sigma)\big|_{\sigma=1}-\int\log X_/\,\frac{dy}{X_\backslash},$$

$$\int\log X_/\,\frac{dy}{X_\backslash}=9.9686527561522909453181622662\ldots,$$

by three-dimensional quadrature 2.7943139847583; and by the same quadrature

$$\Phi_1'(-1)=0.8280938236176\quad\text{against}\quad 0.82809382361938566084\ldots.$$

There is no rational relation among 1, $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, its slope, $P_{\binom{[4]}{1}}(\mathbf{1};1,1)$, the four-mark slope and $\Phi_1'(0)$ with coefficients below 10^{15}.

17.3 The masters of $\binom{[5]}{2}$

Theorem 17.8 (The masters of $\binom{[5]}{2}$)**.** *(i)* [derived] *from Propositions 17.1 and 17.7; (ii)–(iv)* [computed]

(i) Put

$$A_s=\frac{\sum_m\big(V_s\rho_s-V_s'r_s\big)(m)}{\Gamma(s)^2},\quad B_s=\frac{\sum_m\big(V_s\lambda_s-V_s'c_s\big)(m)}{\Gamma(s)^2},$$

$$C_s=\frac{\sum_m V_s(m)c_s(m)}{\Gamma(s)^2},\quad R_s=\frac{\sum_m V_s(m)r_s(m)}{\Gamma(s)^2},$$

each convergent, the coordinates decaying like $(\frac12)^m$*. Then*

$$P_{\binom{[5]}{2}}(s\mathbf{1};s,s)=A_s+B_s\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)+C_s\,\Phi_1'(0),\qquad x_s=R_s+C_s\,P_{\binom{[3]}{1}}(\mathbf{1};1,1):$$

the same C_s *weighs the three-mark centre in the lowered setting and the free slope in the period.*

(ii) $C_1=\dfrac{\sqrt3}{80}\Gamma\!\begin{bmatrix}\frac{1}{20},&\frac{3}{20},&\frac{7}{20},&\frac{9}{20}\\ \frac12,&\frac12,&\frac12,&\frac12\end{bmatrix}$ *(200 digits).*

(iii) The ray law of §15.4 and its corner are the boundary at the cut. As functions of the four initial slopes,

$$\sum_{i=1}^{4}c_i^{\binom{[5]}{2}}(0)\,P(i)-1800=-\tfrac{55}{27}\,\beta,\qquad\sum_{i=0}^{4}c_i^{\binom{[5]}{2}}(1)\,P(1+i)=-\tfrac{898}{189}\,\beta,$$

with β *the left side minus the right side of Proposition 17.5,* $P(i)=P_{\binom{[5]}{2}}(i\mathbf{1};i,i)$*, and* $1800=60\cdot30$ *the corner times* $-[\sigma^4]c_0$*. The lowered setting obeys the ray law with no corner.*

(iv) A second corner lands on the three-mark centre:

$$20\,P(1)-1672\,P(2)+5120\,P(3)=384-135\,P_{\binom{[3]}{1}}(\mathbf{1};1,1).$$

It is the ray law at $\sigma=-1$*:* $(c_2,c_3,c_4)(-1)=40\cdot(20,-1672,5120)$ *and* $\mathrm{CT}_{\sigma=-1}\big(c_0(\sigma)P(\sigma)+c_1(\sigma)P(\sigma+1)\big)=5400\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)-15360$*. In the coordinates of (i) it reads* $5C_1-418C_2+1280C_3=0$*, so the free slope drops out.*

Over $\mathbb{Q}+\mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)$ *the masters of* $\binom{[5]}{2}$ *are* $P(1)$ *and* $P(2)$*; the third master of §15.4 is the three-mark centre.*

Proof. (i) Insert Proposition 17.7 in Proposition 17.1; every coordinate is a combination of solutions of the law with the native values $\frac12,\frac49,\frac13,\frac14$, so each sum converges. (ii)–(iv) are identities among the sums of (i), checked to 140 to 200 digits. □

	$s=1$	$s=2$
A_s	1.261755008105184772419029227146394 2 …	−0.317137091035644599774019003293805 5 …
B_s	−0.406246694015285373669023942108240 1 …	−0.053025588162281981225369425786174 5 …
C_s	1.331333468498450081182459214601857 4 …	0.160794738034738416114870435505479 2 …
R_s	0.352748081018928506473428236341875 6 …	−0.266846421599537631024703106560763 9 …
$P(s)$	4.029714170711049607320695734428406 9 …	0.007886834201028778645599151768532 4 …
x_s	3.473270234941990339750353356341627 8 …	0.110041528823260999643597127118453 0 …

Receipts [computed]. The four initial slopes solved from $P(1),P(2),P(3)$ and Proposition 17.5 give $P(4)$ and $P(5)$ to 154 digits against Procedure 15.6, and the relations of Propositions 17.6 and 17.7 then come out of PSLQ with coefficients below 10^5. Apart from (iv) there is no rational relation among 1, $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, $\Phi_1'(0)$, $P(1)$, $P(2)$, x_1, x_2 with coefficients below 10^8; R_1, C_2, R_2, C_3 are not in $\mathbb{Q}\,aC_1+\mathbb{Q}\,b\,\Gamma(\frac12)^2+\mathbb{Q}\,c\,\Gamma(\frac12)^4/C_1+\mathbb{Q}+\mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)$ for any $a,b,c\in\{1,\sqrt3,\sqrt5,\sqrt{15}\}$, coefficients below 10^{12}.

17.4 Two marks doubled, closed

Let H be the triangle with the class 2 doubled. Opened at its mark 1 (Prop. 19.19), its children on the chart of $\{2a, 2b, 3\}$ are the three-mark banana $H/1 = \binom{[3]}{1}$, of stride $X_{H/1} = (B+\alpha_3)(B'+\alpha_3^{-1})$, and $H \setminus 1$, the pair $\{2a, 2b\}$ with 3 a coloop, of stride $X_{H\setminus 1} = BB'$, where $B = \alpha_{2a} + \alpha_{2b}$ and $B' = \alpha_{2a}^{-1} + \alpha_{2b}^{-1}$. As $X_{H/1} = BB' + B\alpha_3^{-1} + B'\alpha_3 + 1$, the deletion lies under the contraction. Put $\Psi(a,b) = \int X_{H/1}^{-a} X_{H\setminus 1}^{-b}\, dy$ and $\Psi_s(d) = \Psi(d, s-d)$.

Proposition 17.9 (Opened at a mark of a doubled class)**.** [derived] *For the triangle R with two marks doubled, $0 < c < \sigma$,*

$$\Gamma(\sigma)^2 P_R(\sigma\mathbf{1}; \sigma, \sigma) = \frac{1}{2\pi i} \int_{(c)} \Gamma(d)^2 \Gamma(\sigma - d)^2\, P_{\binom{[2]}{1}}(d\mathbf{1}; d, d)\, \Psi_\sigma(d)\, dd,$$

and for $s \in \{1, 2, \dots\}$, with $W_s(x) = \prod_{j=1}^{s-1}(x-j)^2$ as in Proposition 15.22,

$$\Gamma(s)^2 P_R(s\mathbf{1}; s, s) = \sum_{x \geq s} \big(W_s\, f_s - W_s'\, q_s\big)(x), \qquad q_s(x) = P_{\binom{[2]}{1}}(x\mathbf{1}; x, x)\, \Psi_s(x), \quad f_s = -q_s'.$$

At the lattice points, with $k = x - s$,

$$q_s(x) = \frac{1}{\Gamma(2x)} \sum_{u=-k}^{k} \binom{2k}{k+u} \int_0^\infty y^{x-1-|u|}\, \mathsf{G}_{|u|;(1,-1)}(1,y)^2\, \mathsf{G}_{0;(1,-1)}(1,y)\, dy \;\in\; \mathbb{Q} + \mathbb{Q}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1).$$

Proof. R is H with its mark 1 replaced by a class of two. In the opening of H at 1 the mark contributes $\Gamma(d)^2$; the class contributes the Mellin read of the product of its two Grammas, $\Gamma(d)^2 P_{\binom{[2]}{1}}(d\mathbf{1}; d, d)$ (Thm. 22.13(ii), p. 188, as in the proof of Proposition 16.3). Ψ_σ is holomorphic on $\operatorname{Re} d > 0$, its one wall being the coloop, and $X_{H\setminus 1}/X_{H/1} < 1$ while $P_{\binom{[2]}{1}}$ decays like 4^{-d}: close to the right as in Proposition 15.22. At $d = x$ the insertion $(BB')^k = \sum_u \binom{2k}{k+u}(\alpha_{2a}/\alpha_{2b})^u$ moves the masses (Lemma 35.2(iii), p. 308), and the share side of the three-mark banana at a lean (proof of Proposition 16.7) has the indices $(-u, u, 0)$: $\mathsf{G}_{-u}\mathsf{G}_u\mathsf{G}_0 = y^{-|u|}\mathsf{G}_{|u|}^2\mathsf{G}_0$. □

Proposition 17.10 (The input's law)**.** [derived] *in Proposition 17.15 On the line* 1*,*

$$\begin{aligned} x^2(2x-1)^2(60x^2+137x+65)\, q_1(x) - (2x+1)(1680x^5+4316x^4+2782x^3+231x^2-223x-56)\, q_1(x+1) \\ + (2x+1)(2x+3)(2940x^4+9113x^3+7350x^2-89x-1080)\, q_1(x+2) \\ - 18(x+2)(2x+1)(2x+3)(2x+5)(60x^2+17x-12)\, q_1(x+3) = 0, \end{aligned}$$

with the symbol $36\rho^3 - 49\rho^2 + 14\rho - 1 = (\rho-1)(4\rho-1)(9\rho-1)$, and the lines are tied by

$$\begin{aligned} -4(2x-2s-1)^2\, q_{s+1}(x) + 2(2x+1)(16x-1)\, q_{s+1}(x+1) \\ + \big(4x^2 - (8s+1)x + 2s(2s+1)\big)\, q_s(x) - 2(2x+1)(4x-1)\, q_s(x+1) = 0. \end{aligned}$$

The roots are the native value $\frac{1}{4}$ of the pair's period times the native values $4, 1, \frac{4}{9}$ of $X_{H\setminus 1}/X_{H/1}$: 4 at the sign $(1,1,-1)$ of the three-mark banana, where $BB' = 4$ and $X_{H/1} = 1$; 1 where $\alpha_3 \to 0$ and the pair splits; $\frac{4}{9}$ at the table's gauge [derived].

Receipts [computed]: found on $x \leq 91$ (176 equations, 28 unknowns) and at $s \leq 4$; the lines 2 to 5 read off the line 1 agree with Proposition 17.9. The native value 1 sits on the symbol, as it does for the five-mark law (Ledger, O1): the coordinates of q_1 decay only like powers, $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) = -\lim r/c$ again, and the closing cannot be summed coordinate by coordinate. It is summed whole, with the minimality below.

Proposition 17.11 (Zero shares)**.** *Laurent data and boundary* [derived]*; second relation* [computed] *At* $d = 0$*,*

$$P_{\binom{[2]}{1}}(d\mathbf{1}; d, d) = \frac{2}{d} - \frac{\Gamma(\frac{1}{2})^4}{3}\, d + O(d^3), \qquad \Psi_1(d) = \frac{2}{d} + 2 + O(d).$$

The pole of Ψ_1 *is the coloop's wall: each end of* α_3 *gives* $\int X_{H\setminus 1}^{-1} dy = P_{\binom{[2]}{1}}(\mathbf{1}; 1, 1) = 1$*, and the finite part is* $\int \log X_{H\setminus 1}\, X_{H\setminus 1}^{-1} dy = 2$ *on the chart of the pair. So* $q_1(d) = 4/d^2 + 4/d + O(1)$ *where* c_0 *has a double zero: the law at* $x = 0$ *returns the* 4*, and its derivative is the boundary*

$$56\, f_1(1) - 3240\, f_1(2) + 6480\, f_1(3) = -\frac{1726}{3} - \frac{16}{15}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1).$$

The next relation carries $\Gamma(\frac{1}{2})^4$*, the class's term of order* d*:*

$$3\, f_1(1) - 27\, f_1(2) = 11 + 9\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) - 2\,\Gamma(\tfrac{1}{2})^4 \qquad (PSLQ,\ 130\ digits).$$

Hence $f_1(x) - f_1(1)\, c_1(x) \in \mathbb{Q} + \mathbb{Q}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) + \mathbb{Q}\,\Gamma(\frac{1}{2})^4$ *for every* x *and, by the lines, on every line: as for* $\binom{[5]}{2}$ *the one free slope lies along the three-mark coordinate. Minimality fixes it: no component along the native value* 1*.*

The free slope is

$$f_1(1) = 2\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) - \partial_\sigma P_{\binom{[3]}{1}}(\sigma\mathbf{1}; \sigma, \sigma)\big|_{\sigma=1} - \int \log X_{H\setminus 1}\, \frac{dy}{X_{H/1}},$$

$$f_1(1) = 7.458823870410082651014428317985859793 05\ldots,$$

$$\int \log X_{H\setminus 1}\, \frac{dy}{X_{H/1}} = 5.304142870500479140458347501226070 8\ldots,$$

by two-dimensional quadrature 5.3041428705004785 [computed].

Theorem 17.12 (The masters of two marks doubled)**.** [computed]

(i) $P_R(\mathbf{1}; 1, 1) - 8\, P_R(2\mathbf{1}; 2, 2) = 6\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) - 2 - \frac{4}{9}\,\Gamma(\frac{1}{2})^4$*.*

(ii) $521\, P_R(\mathbf{1}; 1, 1) - 12288\, P_R(2\mathbf{1}; 2, 2) + 28672\, P_R(3\mathbf{1}; 3, 3) = 648 + 1350\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$*.*

(iii) Through the closing, the corner relation of §15.4 is the minimality alone, with the value $4432269600 = -[\sigma^4]c_0 \cdot 44$*; the second corner is* $-\frac{1741824}{13}$ *times the boundary of Proposition 17.11 plus the minimality, with the value* $-80640 \cdot 2503$*. The second corner is the boundary of the input's law at zero shares, where the class and the coloop put the double pole* $4/d^2 + 4/d$*.*

(iv) The lowered settings $x_s = \Gamma(s)^{-2} \sum_x W_s(x) q_s(x)$ *obey the ray law with no corner; at* $\sigma = -1$ *they have the rational second corner* $80640 \cdot 432$*, and* $239 - 148\, x_1 + 5244\, x_2 - 14336\, x_3 = 0$*.*

With the corner and the second corner of §15.4, every $P_R(s\mathbf{1}; s, s)$ *lies in* $\mathbb{Q} + \mathbb{Q}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) + \mathbb{Q}\,\Gamma(\frac{1}{2})^4 + \mathbb{Q}\, P_R(\mathbf{1}; 1, 1)$*: one master over* $\mathbb{Q} + \mathbb{Q}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) + \mathbb{Q}\,\Gamma(\frac{1}{2})^4$*, and the* $3^\dagger$ *of §15.4 are* $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$*,* $\Gamma(\frac{1}{2})^4$ *and* $P_R(\mathbf{1}; 1, 1)$*.*

Receipts [computed]. The three initial slopes solved from the minimality at $x = 640$ and $P_R(1), P_R(2)$ give $P_R(3), P_R(4), P_R(5)$ to 155 digits against Procedure 15.6; the slope at $x = 548$ is then $1.7 \cdot 10^{-336}$, of the size 4^{-x}. The values:

$$P_R(\mathbf{1}; 1, 1) = 7.865837368364021592421974697446 1682\ldots,$$

$$x_1 = 2.509422383492932056627701120329708 8\ldots.$$

There is no rational relation among 1, $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, $\Gamma(\frac{1}{2})^4$, $f_1(1)$ and $P_R(\mathbf{1};1,1)$, nor with $\partial_\sigma P_{\binom{[3]}{1}}$ and $P_{\binom{[4]}{1}}(\mathbf{1};1,1)$ adjoined, below 10^{10}.

Remark 17.13 (What the closing found). (i) In both lists the lattice values are three-mark leans, and the slope input has one free direction, the three-mark coordinate of the values; the other initial slopes are fixed by the boundaries of the law, at the cut and the wall for $\binom{[5]}{2}$, at zero shares for the triangle, and the free one by minimality along the leading native value: $\frac{1}{2}$ at the complex gauge $(1,1,i,i)$ for $\binom{[5]}{2}$, 1 at the sign of the three-mark banana for the triangle. The words of the masters, V_s with $s \le 3$ and W_s with $s \le 2$, have degree below what the adjoint of the law adds (six for $\binom{[5]}{2}$; five for the triangle, whose symbol vanishes at $\rho = 1$), so summation by parts in the sense of Proposition 15.23 has nothing to reduce; the content is in the boundaries.

(ii) The corners of the ray laws are those boundaries. For $\binom{[5]}{2}$ the corner and the law itself are the boundary at the cut; for the triangle the corner is the minimality and the second corner is the boundary at zero shares.

(iii) Over $\mathbb{Q}$ the masters are $\{P_{\binom{[3]}{1}}(\mathbf{1};1,1), P(1), P(2)\}$ for $\binom{[5]}{2}$ and $\{P_{\binom{[3]}{1}}(\mathbf{1};1,1), \Gamma(\frac{1}{2})^4, P_R(1)\}$ for the triangle with two marks doubled. The three-mark centre is a master of both; in the triangle it enters through the contraction $H/1$.

(iv) The weight C_1 of the three-mark centre in the lowered setting of $\binom{[5]}{2}$ is a stacked Gamma at twentieths, and the same number weighs the free slope in the period.

Remark 17.14 (The leaf and the closing). In each of the four rays met so far, the three-mark ray, the four-mark ray, the input of $\binom{[5]}{2}$ along its lines and the input of the triangle with two marks doubled, the lattice values are $r + c\,\ell$ and the lattice slopes are $\rho + \lambda\,\ell + c\,\ell'$: r, c, ρ, λ rational, the same c in both, and ℓ the centre of the smallest piece, $\ell = P_{\binom{[3]}{1}}(\mathbf{1};1,1)$ for the three-mark ray and the two inputs and $\ell = P_{\binom{[4]}{1}}(\mathbf{1};1,1)$ for the four-mark ray. For the two banana rays this is [derived]: the cut relation and its derivative (the corner and its vanishing term of order σ, Proposition 15.3) have the same homogeneous part, so the one new slope $\ell' = \partial_\sigma P(\sigma\mathbf{1};\sigma,\sigma)|_{\sigma=1}$ follows c through the recursion (exactly, for every lattice point checked). For the two inputs it is Propositions 17.7 and 17.11.

In every case the coordinates r and c carry the leading native value of the law, and that native value sits at a gauge off the positive chart: the sign $(1,1,-1)$ of three marks (value 1), the sign $(1,1,1,-1)$ of four marks (value $\frac{1}{4}$), the complex gauge $(1,1,i,i)$ of $X_/ /X_\backslash$ (value $\frac{1}{2}$), the sign of the three-mark banana inside the triangle (value 1). The leaf's centre is the multiplier that removes it,

$$\ell = -\lim \frac{r}{c} \qquad \text{[computed]},$$

and the true values decay like the extreme the positive chart reaches: $\frac{1}{9}$ and $\frac{1}{16}$ at the gauges of the bananas, $\frac{4}{9}$ at the table's gauge for $\binom{[5]}{2}$, and $\frac{1}{4}$ for the triangle, from the stratum where $\alpha_3 \to 0$ and the pair splits.

So the closing pairs two kinds of number: Trailmoments of rational coordinate sequences against the closing's words, and the leaf's centre with its one slope. For $\binom{[5]}{2}$ this is Theorem 17.8(i). The stacked Gammas sit differently in the two prime lists. For the banana the whole lowered setting $x^{(1)}(1) = R + C\,P_{\binom{[4]}{1}}(\mathbf{1};1,1)$ is the stacked Gamma at fifteenths (Proposition 15.27), while the weight $C = 1.07051077985786214296521 09212947808\ldots$ of the four-mark centre is not: there is no relation with that stacked Gamma, with $\Gamma(\frac{1}{2})^8$ over it, with 1 and $P_{\binom{[4]}{1}}(\mathbf{1};1,1)$, times $1, \sqrt{3}, \sqrt{5}, \sqrt{15}$, below 10^{12}. For $\binom{[5]}{2}$ the weight C_1 of the three-mark centre is the stacked Gamma at twentieths and x_1 is not [computed].

17.5 The laws from the relations along a ray

The laws of §§17.1–17.4 were read off exact lattice values. They follow from the two relations along a ray (Thm. 35.3, p. 308), integration by parts in every mark and the shift by the stride. The relation space has finite codimension (Thm. 35.19, p. 313), so every insertion has a law that follows from it (Cor. 35.6, p. 308). Finding it is linear algebra, and what the linear algebra returns is an insertion that certifies the law.

Proposition 17.15 (Certificates)**.** [derived]*; the certificates* [computed] *On the line s of $\binom{[5]}{2}$ put $\langle g\rangle_m = \int g\, t^m\, X_\backslash^{-s}\, dy$ with the stride $t = X_/ / X_\backslash = e_1e_3/e_2^2$, so that $\langle t^i\rangle_m = \Phi_s(-m-i)$ and $\langle t^i X_\backslash^{-1}\rangle_m = \Phi_{s+1}(-m-i)$. For each law and each line relation $\sum_j p_j(m)\langle g_j\rangle_m = 0$ used in §§17.1–17.3 (the laws at $s \le 3$; the first-order relations and the explicit relations of the lines at $s \le 4$) there is an insertion*

$$H = \frac{P}{e_2^{\,d}},$$

P symmetric in three marks, of degree $2d$, with every exponent at most d and coefficients in $\mathbb{Q}[m]$, with H_e its image under the exchange of the first mark with e, such that

$$\sum_j p_j(m)\, g_j = \sum_{e\in\mathcal{E}} \Big(\alpha_e\partial_{\alpha_e} H_e + H_e\big(m\,\alpha_e\partial_{\alpha_e}\log t - s\,\alpha_e\partial_{\alpha_e}\log X_\backslash\big)\Big).$$

For the triangle with two marks doubled the same holds on the chart $\alpha_{2a}=1$ of $\{2a,2b,3\}$ for the laws of Ψ_s (the law of Proposition 17.10 divided by the shift of $P_{\binom{[2]}{1}}$), with the stride $X_{H\backslash 1}/X_{H/1}$, the origin form $X_{H\backslash 1}^{-s}dy$, and $H_e = P_e/(L_1^{d_1}L_2^{d_2}L_4^{d_4})$, where L_1, L_2 are the letters of the three-mark banana, $L_4 = \alpha_{2a}+\alpha_{2b}$, and P_e lies inside the Newton polygon of the denominator.

	law	*first-order relation*	*explicit relation*
$\binom{[5]}{2}$	$d=7$, *degree* 5 ($s=1$); $d=9$, *degree* 7 ($s=2,3$)	$d=3$, *degree* 5	$d=5$, *degree* 5
triangle	$(d_1,d_2,d_4)=(3,3,2)$, *degree* 6	$(2,2,1)$, *degree* 5	$(2,2,1)$, *degree* 5

The degrees are in the line coordinate.

Proof. The identity integrates to the law by Thm. 35.3, once $H_e\, t^m X_\backslash^{-s}$ tends to zero at the boundary of the chart. The hypothesis of Thm. 35.3 fails here. Along the line the contraction's exponent goes down as the deletion's goes up, and the direction has $\kappa_{1,S} = (d+d^*)_{\binom{[4]}{1}}(S) - (d+d^*)_{\binom{[4]}{2}}(S)$, which is 0 for $|S|$ odd and 1 for $|S|=2$; on the odd walls the stride does not decay but tends to the ratio of the face. The origin form carries the decay instead. Along a direction w with $w_1\ge w_2\ge w_3\ge w_4$,

$$\operatorname{ord}_w t = w_3 - w_2 \le 0, \qquad \operatorname{ord}_w X_\backslash^{-s} = s(w_3+w_4-w_1-w_2) < 0,$$
$$\operatorname{ord}_w H_e \le d(w_1+w_2) - d(w_1+w_2) = 0,$$

the last because no exponent of P exceeds d. So the boundary terms vanish for $m \ge 0$. Both sides of the law are holomorphic in m where the values converge ($\operatorname{Re} m > -2$ on the line 1), so the law holds identically in the split. For the triangle, $\operatorname{ord}_w H_e \le 0$ by the Newton polygon, and $(X_{H\backslash 1}/X_{H/1})^x X_{H\backslash 1}^{-s}$ decays in every direction for $x>0$. □

Receipts [computed]. Each certificate comes from exact linear algebra in under a second; the symmetric ansatz brings the law of the line 1 of $\binom{[5]}{2}$ down to 444 unknowns, and every equation is checked exactly. The certificates of the two line-1 laws were checked again by direct evaluation at

random rational points, with the line coordinate negative and non-integral. The two banana rays, whose strides decay in every direction, have certificates of the same kind (Thm. 35.3 as stated gives the relations for large k; down to $k \to 0^+$ it needs bounds on the pieces, which §19, Step 5, supplies for three marks), with the denominators $e_1^2 e_2$ (three marks) and $e_1^2 e_3^2$ (four marks).

Corollary 17.16 (What is now derived)**.** [derived] *The laws of Propositions 17.2 and 17.10, the line relations of Propositions 17.4 and 17.10 for $s \le 4$, the slope recurrences (their derivatives in the split), and the boundaries at the cut (Proposition 17.5) and at zero shares (Proposition 17.11, with its Laurent data) are derived. Still computed: the relations among the initial slopes found by PSLQ (Propositions 17.6 and 17.7, and the relation carrying $\Gamma(\frac{1}{2})^4$ in Proposition 17.11), the identities of the closings in Theorems 17.8(iii) and 17.12(iii), and the relations among the masters.*

Remark 17.17 (The centre read off the law)*.* With the laws derived, the exchange rate of Remark 17.14 is an exact identity at every lattice point, not only in the limit. On the three-mark centre ray $p(\sigma) = P_{\binom{[3]}{1}}(\sigma\mathbf{1}; \sigma, \sigma)$ the three-mark law (its certificate is among the receipts of Proposition 17.15) and the first two values,

$$\sigma^2 p(\sigma) - (10\sigma^2 + 10\sigma + 3)\, p(\sigma+1) + 9(\sigma+1)^2\, p(\sigma+2) = 0,$$
$$p(1) = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1), \qquad p(2) = -\tfrac{2}{3} + \tfrac{1}{3}\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1),$$

give $p(q) = r(q) + c(q)\, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ with r, c rational, and $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) = -r(q)/c(q) + p(q)/c(q)$ for every q. The ratios $-r/c$ are

$$0, \quad 2, \quad \tfrac{23}{10}, \quad \tfrac{145}{62}, \quad \tfrac{1331}{568}, \quad \tfrac{242353}{103400}, \quad \tfrac{5495507}{2344600}, \quad \tfrac{46422941}{19805800}, \quad \ldots,$$

all below $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ at every q, since $c(q) > 0$ at every q (§19, Step 8), the error shrinking by 0.1112 per step at $q = 30$, towards $\frac{1}{9}$, the native value of the gauge over that of the sign $(1, 1, -1)$ [computed]: about a digit of the three-mark centre per step of exact arithmetic. On the line 1 of $\binom{[5]}{2}$, $\Phi_1(-m) = \int t^m X_{\setminus}^{-1}\, dy > 0$ and $\Phi_1(0) = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$, so $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) = -r_1(m)/c_1(m) + \Phi_1(-m)/c_1(m)$ [derived]; the exact $c_1(m)$ is positive for $m \le 80$, so these ratios are lower bounds as well, with error $6.7 \cdot 10^{-4}$ at $m = 80$, shrinking per step towards $\frac{8}{9} = \frac{4}{9}/\frac{1}{2}$ [computed]. With $V_1 = 1$, Theorem 17.8(ii) is a series of the rationals that the law of Proposition 17.2 produces:

$$\sum_{m \ge 0} c_1(m) = 1 + \frac{2}{9} + \frac{28}{405} + \frac{208}{8505} + \frac{304}{32805} + \frac{16672}{4546773} + \cdots = \frac{\sqrt{3}}{80}\, {}_4\Gamma_4\!\left[\begin{matrix} \frac{1}{20}, & \frac{3}{20}, & \frac{7}{20}, & \frac{9}{20} \\ \frac{1}{2}, & \frac{1}{2}, & \frac{1}{2}, & \frac{1}{2} \end{matrix}\right].$$

18 The reading of a ray through the origin

Remark 17.17 read the three-mark centre off its law, and Remark 17.14 found the same exchange rate on four rays. This entry states that reading for every ray through the origin of every family, as derived statements, and runs it where §§15–17 had not looked: off the centre, at a lean of three marks, and on the one connected list on four marks that had not been read. The receipts are in the script ray_reading_s18.py (python-flint, exact arithmetic and ball arithmetic, no quadrature in the procedure).

Throughout, a *ray through the origin* is a ray of Def. 35.1, p. 307, with origin 0 and an integral direction $(\mathbf{m}_1; \mathbf{s}_1, \mathbf{t}_1)$ on the balance wall, $s_{1j}, t_{1j} \in \mathbb{N}$, all of whose proper walls are positive, $\kappa_{1,S} > 0$ for every nonempty proper S. Its values and its stride are

$$P(k) = P_F(k\mathbf{m}_1; k\mathbf{s}_1, k\mathbf{t}_1) = \int_{y_{e_0}=0} f^k\, dy \quad (k \ge 1), \qquad f = \alpha^{\mathbf{m}_1} \prod_j K_j^{-s_{1j}} \Psi_j^{-t_{1j}},$$

with $\alpha^\star$ the native gauge of the direction on the orthant, where f takes its largest value $f(\alpha^\star)$ there (Thm. 35.10(i), p. 309). The centre ray of Def. 35.9, p. 309, is the case $\mathbf{m}_1 = \sigma\mathbf{1}$, $\mathbf{s}_1 = \mathbf{t}_1 = \boldsymbol{\sigma}$.

18.1 One pencil, two chains

Proposition 18.1 (The closings of a direction)**.** [derived] *Put* $x_j = [\alpha^0]\,(1/f)^j$ *for* $j \in \mathbb{N}$*, the constant terms of the powers of the Laurent polynomial* $1/f = \alpha^{-\mathbf{m}_1}\prod_j K_j^{s_{1j}}\Psi_j^{t_{1j}}$*, and*

$$\mathrm{Cor} = \sum_{\pi}\prod_{i=2}^{n}\kappa_{1,U_i}^{-1},$$

the sum over the orders π *of* $\mathcal{E}$*, with tails* $U_i = \{\pi(i),\dots,\pi(n)\}$*. Then* P *continues meromorphically in* k*, with its poles at negative rational* k*, and*

$$\lim_{k\to -j}(k+j)^{n-1}P(k) = \mathrm{Cor}\cdot x_j \qquad (j\in\mathbb{N}).$$

At the centre this is Proposition 15.4: there $1/f = \prod_j (K_j\Psi_j/\alpha^{\mathcal{E}})^{\sigma_j}$ *and* $\kappa_{1,S} = \sum_j \sigma_j c_j(S)$*.*

Proof. In the sector coordinates of Thm. 8.7(ii), p. 55, $\alpha_{\pi(1)} = \rho$, $\alpha_{\pi(i)} = \rho\, q_2\cdots q_i$, one has $K_j = \rho^{r_j}\prod_i q_i^{d_j(U_i)}k_{j,\pi}$, $\Psi_j = \rho^{n-r_j}\prod_i q_i^{d_j^*(U_i)}\psi_{j,\pi}$ and $\alpha^{\mathbf{m}_1} = \rho^{m_{1,\mathcal{E}}}\prod_i q_i^{m_1(U_i)}$, so by the balance of the direction

$$f = \prod_{i=2}^{n} q_i^{\kappa_{1,U_i}}\,\varphi_\pi(q), \qquad \varphi_\pi = \prod_j k_{j,\pi}^{-s_{1j}}\psi_{j,\pi}^{-t_{1j}},$$

and by Thm. 8.10, p. 56, $P(k) = \sum_\pi \int_{[0,1]^{n-1}}\prod_i q_i^{k\kappa_{1,U_i}-1}\varphi_\pi(q)^k\,dq$. The sector letters have non-negative coefficients and positive constant terms (Lemma 8.5, p. 55), so $(1/\varphi_\pi)^j = \sum_v c_{\pi,v}q^v$ is a polynomial. At $k = -j+\varepsilon$ write $\varphi_\pi^k = (1/\varphi_\pi)^j\varphi_\pi^\varepsilon$. By Lemma 8.6, p. 55, the term q^v has a pole in ε of order at most the number of i with $v_i = j\kappa_{1,U_i}$, so the pole of order $n-1$ comes from $v = j\kappa_1$ alone, with the leading coefficient $c_{\pi,j\kappa_1}\prod_i \kappa_{1,U_i}^{-1}$. Finally $c_{\pi,j\kappa_1} = [q^0]\big(\prod_i q_i^{-\kappa_{1,U_i}}/\varphi_\pi\big)^j$ is the constant term of $(1/f)^j$ in the sector coordinates, and $\alpha^v \mapsto \prod_i q_i^{v(U_i)}$ is injective on exponents of degree zero, so it is x_j for every π. □

Corollary 18.2 (The law and its closings)**.** [derived] *If* $\sum_{i=0}^{h} c_i(k)P(k+i) = 0$ *identically, then* $\sum_{i=0}^{\min(h,j)} c_i(-j)\,x_{j-i} = 0$ *for every* $j\in\mathbb{N}$*, and* c_0 *vanishes to order at least* $n-1$ *at* 0*, with*

$$[k^{n-1}]c_0\cdot\mathrm{Cor} + \sum_{i\ge 1} c_i(0)\,P(i) = 0.$$

Proof. As Corollary 15.5, with Proposition 18.1 in place of Proposition 15.4; the second statement is the case $j=0$, $x_0 = 1$. □

So a law of the values is the recurrence of the integers x_j read backwards, and reading it is exact linear algebra on integers. The converse, that the least recurrence of the x_j, reflected, is a law of P, is what a certificate of Thm. 35.3, p. 308, settles, as in Proposition 17.15.

Remark 18.3 (The torus and the chart)*.* [derived] The two sequences are one integrand on two chains. For $|q| < f(\alpha^\star)^{-1}$, $\sum_{k\ge 1}P(k)q^{k-1} = \int_{y_{e_0}=0} f/(1-qf)\,dy$ (Thm. 35.10(ii) with $k_0 = 1$). On the torus

$T = \{|\alpha_e| = 1\}$, where $|1/f| \le (1/f)(\mathbf{1})$ because $1/f$ has nonnegative coefficients, expanding in $1/(qf)$ gives for $|q| > f(\mathbf{1})^{-1}$

$$\sum_{j\ge 0} x_j\, q^{-j-1} = -\frac{1}{(2\pi i)^{n-1}} \oint_T \frac{f}{1-qf} \prod_{e\neq e_0} \frac{d\alpha_e}{\alpha_e}.$$

The pencil $\prod_j K_j^{s_{1j}} \Psi_j^{t_{1j}} - q\,\alpha^{\mathbf{m}_1}$ of Thm. 35.10 carries both: the positive chart gives its solution at $q = 0$, whose coefficients are the periods, and the torus its solution at $q = \infty$, whose coefficients are integers for a list. Corollary 18.2 is the one equation, inhomogeneous by its boundary terms, read at its two ends, and Cor is the ratio of the two normalizations.

18.2 The slit

Proposition 18.4 (The ray lives on the slit plane)**.** [derived] *$G(q) = \int_{y_{e_0}=0} f/(1-qf)\,dy$ is holomorphic on $\mathbb{C} \setminus [f(\alpha^\star)^{-1}, \infty)$, and it continues $\sum_{k\ge 1} P(k) q^{k-1}$.*

Proof. On the chart $0 < f \le f(\alpha^\star)$ (Thm. 35.10(i)). For q off the slit the segment $q\cdot(0, f(\alpha^\star)]$ misses 1, so $|1-qf|$ is bounded below by the distance from 1 to that segment, locally uniformly in q, and $\int f\,dy = P(1) < \infty$; dominated convergence gives holomorphy. On $|q| < f(\alpha^\star)^{-1}$ the integral is the series. □

Theorem 18.5 (Positive selection)**.** *(i)–(iii)* [derived]*; the count in (iii)* [computed] *on every ray below. Let $\sum_{i=0}^{h} c_i(k) P(k+i) = 0$ be a law of a ray through the origin, $c_0 c_h \neq 0$, let D be the largest degree of the c_i, and $\chi(\lambda) = \sum_i [k^D] c_i\, \lambda^i$ its symbol, with roots λ_c; it vanishes at the native value of every nondegenerate native gauge (Cor. 35.6, p. 308), and a root need not be a native value.*

(i) For every solution $(x_k)_{k\ge 1}$ of the law, $X(q) = \sum_k x_k q^{k-1}$ satisfies a linear differential equation in q with polynomial coefficients, inhomogeneous by a polynomial that depends linearly on $x_1, \dots, x_h$; in $\mathbb{C}^\times$ its singular points are the $q_c = 1/\lambda_c$. So X continues along every path in $\mathbb{C}^\times$ that avoids them.

(ii) The values are the solution whose X is holomorphic on $\mathbb{C} \setminus [f(\alpha^\star)^{-1}, \infty)$ (Proposition 18.4): holomorphic at every q_c off the slit, that is at every root $\lambda_c \notin (0, f(\alpha^\star)]$.

(iii) At a singular point q_c, holomorphy of X is ν_c linear conditions on $(x_1, \dots, x_h)$, ν_c the number of independent local solutions of the equation that are not holomorphic there, at most the multiplicity of λ_c as a root of χ. At a simple root that is the native value of a nondegenerate native gauge it is one: the branch of Thm. 35.11, p. 310, read at that gauge in place of $\alpha^\star$, of exponent $\frac{n-3}{2}$, with a logarithm for odd n. With the corner of Corollary 18.2, if $1 + \sum_{q_c \text{ off the slit}} \nu_c = h$ and the conditions are independent, the law, the corner and positivity determine the values.

(iv) Suppose further that every root $\lambda_c \neq f(\alpha^\star)$ that carries a local solution not holomorphic at q_c has $|\lambda_c| > f(\alpha^\star)$; a root in $(-f(\alpha^\star), 0)$, or a complex one in the disc $|\lambda| < f(\alpha^\star)$, carries a solution smaller than P, and the limits below diverge. Then, for a basis $x^{(0)}, \dots, x^{(h-1)}$ of rational solutions fixed by the corner, $P = x^{(0)} + \sum_{i\ge 1} M_i x^{(i)}$, and the M_i are the limits as $K \to \infty$ of the solutions of $x^{(0)}_{K+l} + \sum_i M_i x^{(i)}_{K+l} = 0$, $0 \le l < h-1$, with error $O\big((f(\alpha^\star)/\min_{c\neq\star} |\lambda_c|)^K\big)$ up to powers of K.

Proof. (i) Multiply the law by q^{k-1} and sum over $k \ge 1$: k acts as $q\partial_q + 1$, a shift by i as q^{-i} after removing the first i terms, which leaves a polynomial. The coefficient of the top power of $q\partial_q$ is $\sum_i [k^D] c_i\, q^{-i}$, which vanishes in $\mathbb{C}^\times$ exactly where $\chi(1/q) = 0$. (ii) Proposition 18.4. (iii) The inhomogeneity is holomorphic at q_c, so the singular part of X there lies in the span of the

local homogeneous solutions that are not holomorphic, with coefficients linear in $(x_1, \dots, x_h)$. At a nondegenerate native gauge the value over a steepest-descent disc (Thm. 35.5, p. 308) gives one branch of exponent $\frac{n-1}{2} - 1$; Thm. 35.11 is this computation at $\alpha^\star$, and it is local. (iv) The coefficients of X grow at the rate $|\lambda_c|$ of the singular point nearest 0 at which X is singular. The rational solutions are singular at the q_c off the slit and P is not, so the combination annihilating their $h-1$ largest rates is P, and its own size $f(\alpha^\star)^K$ against the smallest of the others is the error. □

Procedure 18.6 (The reading of a ray)**.** (1) From the family and the direction write $1/f$ and the walls $\kappa_{1,S}$, and Cor (Proposition 18.1). (2) The closings $x_j = [\alpha^0](1/f)^j$, exactly. (3) Their least recurrence by exact linear algebra, reflected (Corollary 18.2): the law; its certificate by Thm. 35.3 as in Proposition 17.15. (4) The native values, the roots of the symbol, and the native gauges, the critical points of $\log f$: the signs and what a lean moves them to, and families. (5) The seeds: the corner, and a second corner wherever c_0 and c_1 share a zero (§15.4). (6) Positivity, Theorem 18.5(iii)–(iv): the masters as limits of rational solutions, at the rate $f(\alpha^\star)/\min_{c\neq\star}|\lambda_c|$. (7) Integer relations among the masters and against the members already in use; what is left over $\mathbb{Q}$ are the masters of the ray.

Remark 18.7 (What sits on the slit). Positivity says nothing at a native value in $(0, f(\alpha^\star))$. For a ray strictly inside its walls f has one critical point on the orthant (Thm. 8.18(ii)–(iii), p. 58), so such a value would sit at a native gauge off the orthant; on every ray strictly inside its walls read in §§15–18 there is none. Example 20.12 has one, a conjugate of an irrational $f(\alpha^\star)$; the gauge end reads that ray all the same (§21). On a ray along walls the slit carries the faces: on the line 1 of $\binom{[5]}{2}$ (§17.1) the native values $\frac{1}{3}$ and $\frac{1}{4}$ of Proposition 17.3 lie below $\frac{4}{9}$, they come from strata at the boundary of the chart, and the constants there are the faces' own, the three-mark centre among them (Theorem 17.8). That is Thm. 35.15, p. 312, read on the slit.

18.3 Off the centre: a lean of three marks

The centre is the fixed point of Cremona (Thm. 8.13, p. 57), where both letters take one exponent; a lean gives them two. For $\binom{\mathcal{E}}{1}$, $K = \sum_e \alpha_e$ and $\Psi = \alpha^{\mathcal{E}} \sum_e \alpha_e^{-1}$, and at uniform masses $m\mathbf{1}$ the balance is $nm = s + (n-1)t$.

Proposition 18.8 (The lean is the index of the Gramma)**.** [derived]*; receipt* [computed] *For masses* $\mathbf{m}$ *on the balance wall and* $\operatorname{Re} s, \operatorname{Re} t > 0$*,*

$$\Gamma(s)\Gamma(t)\, P_{\binom{\mathcal{E}}{1}}(\mathbf{m}; s, t) = \int_0^\infty y^{t-1} \prod_{e\in\mathcal{E}} \mathsf{G}_{m_e - t;(1,-1)}(1, y)\, dy, \qquad \sum_e (m_e - t) = s - t.$$

At the centre this is Proposition 15.2; the lean and the offsets move the index slots of the marks' Grammas. In the direction $(2\mathbf{1}; 4, 1)$ *of three marks,* $P_{\binom{[3]}{1}}(2\mathbf{1}; 4, 1) = \frac{1}{6}\int_0^\infty \mathsf{G}_{1;(1,-1)}(1, y)^3\, dy =$ $0.030080945282262304399884274898349847\ldots$ *in ball arithmetic.*

Proof. On the share side (Proposition 15.1) the mark e carries the share a_e of the basis $\{e\}$ in K and the share b_e of the cobasis $\mathcal{E} \setminus e$ in Ψ, and the pin at e reads $a_e + \sum_{e'\neq e} b_{e'} = m_e$, that is $a_e = b_e + m_e - t$. So each mark carries $\Gamma(b_e + m_e - t)\Gamma(b_e)$, the share kernel of $\mathsf{G}_{m_e - t;(1,-1)}(1, y)$ (Prop. 4.2, p. 25), and the Mellin read of the product pins $\sum_e b_e = t$, as in Proposition 15.2. □

Proposition 18.9 (The native gauges of a lean of $\binom{\mathcal{E}}{1}$)**.** [derived] *In the direction* $(m\mathbf{1}; s, t)$*,* $s \neq t$*, put* $u_e = \alpha_e / K(\alpha)$*,* $v_e = \alpha_e^{-1}/K(\alpha^{-1})$ *and* $\lambda = \big(K(\alpha)K(\alpha^{-1})\big)^{-1}$*. At a native gauge* $su_e - tv_e = \frac{s-t}{n}$

and $u_e v_e = \lambda$ for every e, so every u_e is a root of $su^2 - \frac{s-t}{n}u - t\lambda = 0$; with j marks at one root u_- and $n-j$ at the other u_+,

$$(n-2j)\,u_+ = 1 - j\,\frac{s-t}{ns}, \qquad f = \Big(\prod_e u_e\Big)^{\frac{s-t}{n}} \lambda^t.$$

At $s = t$ the two roots are $\pm u_+$ and this is Proposition 15.7. For three marks in the direction $(\mathbf{21}; 4, 1)$: $j = 0$ is the table's gauge, $f = \frac{1}{243}$; $j = 1$ is $\alpha \propto (3, 3, -2)$ and its images, $f = -\frac{27}{64}$, the sign $(1, 1, -1)$ moved along the real line.

Proof. Nativity is $m = sR_e(\alpha) + tR_e^*(\alpha)$ with $R_e = u_e$ and $R_e^* = 1 - v_e$ (Thm. 7.14, p. 50), and $u_e v_e = 1/\big(K(\alpha)K(\alpha^{-1})\big)$ for every e. The u_e sum to 1, which fixes u_+. For the value, $\alpha^{\mathbf{1}} = K(\alpha)^n \prod_e u_e$ and $K(\alpha^{-1}) = 1/(K(\alpha)\lambda)$, and the powers of $K(\alpha)$ cancel by the balance. □

Receipts [computed]. The walls of $(\mathbf{21}; 4, 1)$ are 2 on marks and 3 on pairs, so $\mathrm{Cor} = 6/(2 \cdot 3) = 1$. The closings are $x_j = (4j)!\, j! \sum_{b_1+b_2+b_3=j} \prod_f \big(b_f!\,(b_f + j)!\big)^{-1} = 1, 36, 4620, 762300, 140216076, \ldots$; sixty of them have a least recurrence of order 2 and degree 6, and reflected it is the law

$$\begin{gathered} -3k^2(3k+1)(3k+2)(265k^2+636k+383)\,P(k) \\ +(1721705k^6+7575502k^5+13047276k^4+11068046k^3+4766659k^2+965100k+75600)\,P(k+1) \\ +972(k+1)^2(4k+5)(4k+7)(265k^2+106k+12)\,P(k+2) = 0, \end{gathered}$$

$P(k) = P_{\binom{[3]}{1}}(2k\mathbf{1}; 4k, k)$, with the symbol $265(243\lambda - 1)(64\lambda + 27)$: the two native values of Proposition 18.9, which were read off the list before the closings were computed. The corner gives $12600\,P(1) + 68040\,P(2) = 383$; positivity gives

$$P_{\binom{[3]}{1}}(\mathbf{21}; 4, 1) = 0.03008094528226230439988427489834984673367385 11\ldots,$$

about two digits per step (the rate $\frac{64}{6561}$), against 0.95 for the centre, and Proposition 18.8 confirms it. Over $\mathbb{Q}$,

$$54\,P_{\binom{[3]}{1}}(\mathbf{21}; 4, 1) = 11 - 4\,P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) \qquad (380 \text{ digits}).$$

The Cremona partner $(3\mathbf{1}; 1, 4)$ has the same closings, by $\alpha \mapsto \alpha^{-1}$, and the same law. So the lean is a new coordinate system on the content of the centre: a different law, different native gauges, a different corner, the same master.

The rational coordinates of this ray do not give a gap (Thm. 34.3, p. 300): on $(1, P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1))$ their denominators beyond d_k^2 grow like $e^{8.7k}$, with primes of the apparent factor $265k^2 + \cdots$ among them, against the decay $243^{-k} = e^{-5.49k}$ [computed]; at the centre the excess is $3^{1.88k}$, against 9^{-k}.

18.4 The fourth list on four marks

On four marks the connected lists are $\binom{[4]}{1}$, $\binom{[4]}{2}$, $\binom{[4]}{3}$ and the triangle H with one mark doubled; the first two were read in §15.1, and the third is the dual of the first (Prop. 8.14, p. 57). H is the list of §17.4, where it was a step toward the triangle with two marks doubled; its own centre was not read there. With the doubled class $\{a, b\}$ and the marks 2, 3 its bases are the pairs other than $\{a, b\}$, and with $A = \alpha_a + \alpha_b$, $B = \alpha_2 + \alpha_3$

$$K = AB + \alpha_2\alpha_3, \qquad \Psi = AB + \alpha_a\alpha_b :$$

the letters exchange under $\{a, b\} \leftrightarrow \{2, 3\}$, so H is self-dual up to that relabeling. The widths are 1 on marks, on $\{a, b\}$, on $\{2, 3\}$ and on triples, and 2 on the four other pairs (Def. 7.10, p. 49), so at the centre $\mathrm{Cor} = 8 \cdot 1 + 16 \cdot \frac{1}{2} = 16$.

Proposition 18.10 (The native gauges of the centre of H)**.** [derived]*; that there are no others* [computed] *The table's gauge, of native value* $\frac{1}{25}$*; the sign* $(-1,-1,1,1)$*, of native value* $\frac{1}{9}$*; and, of native value* 1*, the three curves*

$$\{A=0,\ \alpha_a^2=\alpha_2\alpha_3\},\qquad \{B=0,\ \alpha_2^2=\alpha_a\alpha_b\},\qquad \{A=B=0\},$$

the second the image of the first under the self-duality; on them the class letters vanish and K*,* Ψ *collapse to monomials, the case of Proposition 15.15(ii) with one class. Every sign of value* 1 *lies on them.*

Proof. At the centre nativity is $R_e(\alpha)=R_e(\alpha^{-1})$ (§15.2). On $A=0$ also $A(\alpha^{-1})=A/(\alpha_a\alpha_b)=0$, so $K(\alpha)=\alpha_2\alpha_3$, $K(\alpha^{-1})=1/(\alpha_2\alpha_3)$ and $R_2=R_3=1$ at α and at α^{-1}, while $R_a(\alpha)=\alpha_aB/(\alpha_2\alpha_3)$ and $R_a(\alpha^{-1})=B/\alpha_a$ agree exactly when $B=0$ or $\alpha_a^2=\alpha_2\alpha_3$, and likewise at b; there $K(\alpha)K(\alpha^{-1})=1$. The second curve follows by the self-duality. At a sign $K(\eta)=(\eta_a+\eta_b)(\eta_2+\eta_3)+\eta_2\eta_3\in\{\pm5,\pm3,\pm1\}$ (Proposition 15.14), and $K(\eta)=\pm1$ forces $\eta_a=-\eta_b$ or $\eta_2=-\eta_3$, which puts the sign on one of the curves. Newton's method from several thousand starts in $(\mathbb{C}^\times)^3$ finds no critical point of $\log f$ off these. □

Receipts [computed]. By Proposition 15.10 the closings are

$$x_k=\sum_{i+j+l=k}\Big(\frac{k!}{i!\,j!\,l!}\Big)^2\binom{2i+2j}{i+j}=1,5,53,761,12661,229705,4410665,\dots,$$

and the least recurrence of 160 of them has order 4 and degree 7 (156 equations, 40 unknowns, one solution). Reflected, the law of the centre of H, $P(k)=P_H(k\mathbf{1};k,k)$, is

$$\begin{gathered}
k^3(k+1)(144k^3+864k^2+1693k+1075)\,P(k)\\
-2(k+1)(2592k^6+19440k^5+56322k^4+80296k^3+60004k^2+23017k+3580)\,P(k+1)\\
+2(21168k^7+211680k^6+884919k^5+2003456k^4\\
+2646984k^3+2031207k^2+830703k+137260)\,P(k+2)\\
-2(k+2)^2(34848k^5+261360k^4+741842k^3+984642k^2+598948k+127215)\,P(k+3)\\
+225(k+2)^2(k+3)^2(144k^3+432k^2+397k+102)\,P(k+4)=0,
\end{gathered}$$

with the symbol $144(25\lambda-1)(9\lambda-1)(\lambda-1)^2$: the curves double the root 1, an integral direction in the exceptional set of Thm. 35.19(i), p. 313. (A search by total size first returns a recurrence of order 5 and degree 5: this law composed with a factor of order one, whose symbol has the extra root $\frac{131003}{4771109}$.) The corner relation is $17200-7160P(1)+274520P(2)-1017720P(3)+826200P(4)=0$, and c_0, c_1 share the zero $k=-1$; the second corner (§15.4) is rational,

$$57\,P(1)-807\,P(2)+1050\,P(3)=196\qquad(530\text{ digits}),$$

the constant term at $k=-1$ being 1176. Positivity (Theorem 18.5: three conditions, one at $\frac{1}{9}$ and two at the double root) gives

$$\begin{aligned}
P_H(\mathbf{1};1,1)&=4.0962369983378828500555532283512686376346708 96\dots,\\
P_H(2\mathbf{1};2,2)&=0.0477164986915131102974120808906636666177626293 6\dots.
\end{aligned}$$

Proposition 18.11 (The centre of H on its share side)**.** [derived]*; receipt* [computed] *H is a series root over the parallel cell of $\{a,b\}$ and the two marks left in place (Def. 22.17(5), p. 191), so, as in Proposition 16.3,*

$$\Gamma(\sigma)^2 P_H(\sigma\mathbf{1};\sigma,\sigma)=\frac{1}{2\pi i}\int_{(c)}\Gamma(d)^2\,\Gamma(\sigma-d)^2\,P_{\binom{[2]}{1}}(d\mathbf{1};d,d)^2\,dd,\qquad P_{\binom{[2]}{1}}(d\mathbf{1};d,d)=\Gamma_2\begin{bmatrix}d,\ d\\ 2d\end{bmatrix},$$

$0<c<\sigma$*. At $\sigma=1$, on $c=\frac12$, the integral is* $4.096236998337882850055553228351268637634670 9\ldots$ *in ball arithmetic: the value of the positive selection, to* 44 *digits.*

Proof. A series node multiplies the transforms of its cells (Thm. 22.13, p. 188): the two marks give $\mathsf{G}_{0;(1,-1)}(1,y)^2$, whose Mellin read is $\Gamma(z)^2P_{\binom{[2]}{1}}(z\mathbf{1};z,z)$ (Proposition 15.2), and the parallel cell gives the transform whose Mellin read is $\Gamma(w)^2P_{\binom{[2]}{1}}((\sigma-w)\mathbf{1};\sigma-w,\sigma-w)$ (Proposition 16.3); the Mellin read of the product pins the sum of the shares, and $d=\sigma-w$. □

The integrand is $\Gamma(d)^6\Gamma(\sigma-d)^2/\Gamma(2d)^2$. Its poles at $-\mathbb{N}$ have order four and residues that grow like 16^m, and to the right the double poles of $\Gamma(\sigma-d)^2$ meet a factor that grows faster than every power: neither closing of Thm. 4.5, p. 26, converges. The centre of H is a fibre integral of stacked Gammas that does not land on a lattice, as the five-mark centre does not (Remark 15.20).
The lean. The direction $(2\mathbf{1};3,1)$ has the walls 2 on marks, 3 on $\{a,b\}$, 1 on $\{2,3\}$, 4 on the other pairs and 2 on triples, so $\mathrm{Cor}=\frac13+1+1=\frac73$. On the slice $\alpha_a=\alpha_b=x$, $\alpha_2=\alpha_3=1$ the stride is $x^3/\big((4x+1)^3(x+4)\big)$, whose critical points are the roots of $2x^2-x-6=(x-2)(2x+3)$ [derived]: the native gauge of the direction on the orthant is $(2,2,1,1)$, where $K=9$, $\Psi=12$ and $f=\frac{4}{2187}$, and the sign has moved to $(-\frac32,-\frac32,1,1)$, $f=\frac{27}{2500}$; the curves move to curves of value $-\frac{16}{27}$ [computed]. The closings $x_j=[\alpha^{2j\mathbf{1}}]K^{3j}\Psi^j=1,48,10240,3131184,\ldots$ have a least recurrence of order 4 and degree 25 (196 equations, 104 unknowns, one solution), with the symbol a multiple of $(2187\lambda-4)(2500\lambda-27)(27\lambda+16)^2$; positivity converges at the rate $\frac{10000}{59049}$, about 0.77 digits per step, and the lean's own second corner is rational: $413667900\,P'(1)+6131684550\,P'(2)+12031031250\,P'(3)=24789199$, $P'(k)=P_H(2k\mathbf{1};3k,k)$.

Theorem 18.12 (One content, two rays)**.** [computed]

$$28\,P_H(2\mathbf{1};3,1)=14-3\,P_H(\mathbf{1};1,1)-P_H(2\mathbf{1};2,2),$$
$$5670000\,P_H(4\mathbf{1};6,2)=126002\,P_H(\mathbf{1};1,1)+143398\,P_H(2\mathbf{1};2,2)-522781,$$
$$95681250000\,P_H(6\mathbf{1};9,3)=3048368669-731189098\,P_H(\mathbf{1};1,1)-1115791502\,P_H(2\mathbf{1};2,2),$$

to 530 *digits. The centre ray and the lean ray of H, with two laws, two corners and two positive selections, land on the one space $\mathbb{Q}+\mathbb{Q}\,P_H(\mathbf{1};1,1)+\mathbb{Q}\,P_H(2\mathbf{1};2,2)$.*

Remark 18.13 (What the masters of H are not). [computed] There is no rational relation among 1, $P_H(\mathbf{1};1,1)$, $P_H(2\mathbf{1};2,2)$ and the members in use, $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, $\Gamma(\frac12)^4$, $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\Gamma(\frac12)^6/\sqrt3$, $\sqrt3\,\Gamma(\frac12)^2$, $\sqrt3\,\Gamma(\frac12)^2\log3$, $\Gamma(\frac12)^4\log2$, $\Gamma(\frac12)^4\log3$, $\log2$, $\log3$, $P_{\binom{[3]}{1}}(\mathbf{1};1,1)\log2$, $P_{\binom{[3]}{1}}(\mathbf{1};1,1)\log3$, $\Gamma(\frac13)^6/\Gamma(\frac12)^4$ and $\Gamma(\frac12)^8/\Gamma(\frac13)^6$, with coefficients below 10^{24} (420 digits); nor with $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, the two masters of $\binom{[5]}{1}$, the stacked Gamma X at fifteenths of Proposition 15.27, $\Gamma(\frac12)^8/X$, $\Gamma(\frac12)^4$, $\sqrt3\,\Gamma(\frac12)^2$, $\sqrt{15}$, and $\sqrt{15}$ times X, $\Gamma(\frac12)^8/X$ and $\Gamma(\frac12)^4$, below 10^{28} (380 digits). In that framework, over $\mathbb{Q}$, the two masters of H are new numbers. H and $\binom{[5]}{1}$ have the same native values $\frac{1}{25},\frac19,1$ at their centres and different masters: the native values do not decide the numbers.

What the tree expects [not yet computed]. Opened at the mark 2 (Prop. 19.19, p. 144), H has the children $\binom{[3]}{1}$ and the pair $\{a, b\}$ with the coloop 3, the strides $X_{H/1}$ and $X_{H\setminus 1}$ of §17.4, and its input is Ψ_σ of §17.4 without the factor $P_{\binom{[2]}{1}}$. Its lattice values are three-mark leans, in $\mathbb{Q}+\mathbb{Q}\,P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, and its slopes should have one free direction (Proposition 17.11). As for $\binom{[5]}{2}$ (Theorem 17.8), the masters over the field of the weights of that closing should be $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$ and one free slope, the weights carrying what Remark 18.13 sees as new.

18.5 Two observations

(i) *The exact pairing belongs to the banana.* Propositions 15.13 and 15.11 give

$$\mathrm{Cor}\cdot\det H = -\frac{\lambda^\star\chi'(\lambda^\star)}{[k^{n-1}]c_0} = \prod_{c\neq\star}\Big(1-\frac{\lambda^\star}{\lambda_c}\Big)$$

whenever the closings solve the adjoint of the law and c_0 has degree $n-1$; here H in $\det H$ is the Hessian of Proposition 15.11. The laws of $\binom{[n]}{1}$ for $n=2,3,4,5$ (Theorem 15.25, §15.7, Proposition 15.13) are self-adjoint up to sign, $c_i(-k-i)=\pm c_i(k)$ [computed], and the identity is

$$n!\cdot\frac{2^{n-1}}{n^n} = \prod_{j=1}^{\lceil n/2\rceil-1}\Big(1-\frac{(n-2j)^2}{n^2}\Big) = \prod_{j=1}^{\lceil n/2\rceil-1}\frac{4j(n-j)}{n^2} \quad \text{[derived]},$$

which holds for every n: the low end of strength, the corner (Thm. 38.5(i), p. 330), and the high end, the Gaussian at the native gauge (Thm. 38.5(ii)), tied through the native values. On H, whose law is not self-adjoint, and on $\binom{[5]}{2}$ (§15.5) the closings solve no adjoint and the identity fails.

(ii) *The content belongs to the list.* On $\binom{[3]}{1}$ and on H, two rays each, the masters of the ray are those of the list. The letter and mark relations (Thm. 8.3, p. 54) keep integer data on the balance wall with rational coefficients, so every integer period of a list lies in the span they leave, and each ray samples that span. A ray with the symmetries of the list may sample only their invariant part, which offsets would break. Hypothesis: every ray through the origin strictly inside its walls, without the list's symmetries, spans the whole content. [not yet computed]

19 The three-mark centre, set as an examination

Remark 17.17 read the three-mark centre off its law as a list of ratios, and Procedure 18.6 states the reading of every ray through the origin. This entry runs the reading once, on that one ray, written as an examination question with its answer key and its marking scheme. A key has to say what a complete solution shows at every step and what each omission costs. So every hypothesis that §§15–18 checked once in passing is checked here, at the step that uses it, and the places where a solution can look right and be wrong are named, with the credit they earn. The ratios of Remark 17.17 become a certified bracket. The receipts are in the script exam_s19.py (python-flint: exact arithmetic for the solution, ball arithmetic only for the independent check of §19.5).

19.1 The question

Question (40 points). Let $F = \binom{[3]}{1}$ be the list on the marks $\{1,2,3\}$ whose bases are the singletons, with unit weights, so that its letters are

$$K = \alpha_1 + \alpha_2 + \alpha_3, \qquad \Psi = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3,$$

and let $\ell = P_F(\mathbf{1}; 1, 1)$ be the period of its centre. By the reading of its centre ray (Procedure 18.6), that is, the law of the ray certified by the relations along a ray (Thm. 35.3, p. 308), its corner, and positive selection (Theorem 18.5), find a rational number q with

$$0 < \ell - q < 10^{-6},$$

and prove both inequalities.

You may cite the handbook and §§15–18 of this notebook. When you cite a statement, check its hypotheses in the case at hand. Exact computation (integers, rationals, polynomials, linear algebra over $\mathbb{Q}$) may be done by machine; report it so that it can be repeated. Floating-point arithmetic, numerical integration, integer-relation searches and comparison with a known decimal value of ℓ may be used as checks. They earn no credit.

19.2 The answer and the scheme

Answer. $q = a_9 = \dfrac{2447622221}{1044248800} = 2.3439071426\ldots$, with

$$\frac{2447622221}{1044248800} < \ell < \frac{6414407843}{2736630400}, \qquad b_9 - a_9 = \frac{6561}{59511099776} < 1.11 \cdot 10^{-7}.$$

Any rational q with a complete proof of $0 < \ell - q < 10^{-6}$ earns full credit, whatever bounds the proof uses (other consequences of positivity, such as the determinants of Thm. 35.10(iii), are admissible), and the points of the scheme then go to the corresponding steps of that proof; §19.4, items 1, 5 and 6, lists the alternatives a grader will meet most often. What the key proves is this.

Theorem 19.1 (The three-mark centre, bracketed)**.** [derived]*; the table* [computed] *exactly. Let c and r be the solutions of*

$$k^2 x(k) - (10k^2 + 10k + 3)\, x(k+1) + 9(k+1)^2\, x(k+2) = 0 \qquad (k \geq 1)$$

with $c(1) = 1$, $c(2) = \frac{1}{3}$ and $r(1) = 0$, $r(2) = -\frac{2}{3}$. For every integer $K \geq 2$, $c(K) > 0$ and $9c(K) > c(K-1)$, and

$$a_K = -\frac{r(K)}{c(K)} \;<\; P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) \;<\; b_K = \frac{r(K-1) - 9\, r(K)}{9\, c(K) - c(K-1)},$$

$$b_K - a_K = \frac{2 \cdot 9^{2-K}}{3(K-1)^2\, c(K)\big(9c(K) - c(K-1)\big)}.$$

Step	What is shown	Points
1	The ray: the chart and its measure, the stride, the values, the walls; $\ell = P(1)$	3
2	The stride: $f \le \frac{1}{9}$; $\frac{1}{9}e^{-\varphi} \le f \le e^{-\varphi}$; $\int e^{-k\varphi}dy = 6/k^2$; what follows at every real $k > 0$	5
3	The closings, their recurrence, the reflected candidate law	3
4	A certificate for Thm. 35.3 and its exact verification	5
5	The law at every real $k > 0$: the integration by parts, with the bounds of Step 2	5
6	The native values; what positivity can select	2
7	The corner: $k^2P(k) \to 6$, so $9P(2) = 3\ell - 6$	5
8	The rational solutions, their signs, the exact table	4
9	The bracket $a_K < \ell < b_K$, the choice of K, the answer	8
		40

19.3 The key

Throughout, e_1, e_2, e_3 are the elementary symmetric polynomials of $\alpha_1, \alpha_2, \alpha_3$, so that $K = e_1$ and $\Psi = e_2$, and $D_e = \alpha_e\, \partial/\partial\alpha_e$.

Step 1 (3 points). The ray. Take the chart $\alpha_3 = 1$ of the mark $e_0 = 3$, with $\alpha_1 = e^{y_1}$ and $\alpha_2 = e^{y_2}$, so that

$$dy = dy_1\, dy_2 = \frac{d\alpha_1\, d\alpha_2}{\alpha_1\alpha_2}.$$

The centre ray at the strength $\sigma = 1$ (Def. 35.9, p. 309) has origin 0 and direction $(\mathbf{1}; 1, 1)$. Its stride and its values are

$$f = \frac{\alpha^{\mathbf{1}}}{K\Psi} = \frac{e_3}{e_1e_2}, \qquad P(k) = P_F(k\mathbf{1}; k, k) = \int_{\mathbb{R}^2} f^k\, dy \qquad (k = 1, 2, \dots),$$

so that

$$\ell = P(1) = \int_0^\infty\int_0^\infty \frac{d\alpha_1\, d\alpha_2}{(1 + \alpha_1 + \alpha_2)(\alpha_1 + \alpha_2 + \alpha_1\alpha_2)}.$$

For real $k > 0$ the same integral defines $P(k)$; Step 2 shows that it converges. The walls of the direction are $\kappa_{1,S} = |S| - d(S) - d^*(S)$ (Def. 8.1, p. 53), the width $c(S)$ of Def. 7.10, p. 49. Every nonempty S has rank $r(S) = 1$, so for S nonempty and proper $d(S) = |S| - r^*(S) = r - r(\mathcal{E} \setminus S) = 0$ and $d^*(S) = |S| - r(S) = |S| - 1$, and

$$\kappa_{1,S} = 1 \qquad (\varnothing \ne S \subsetneq \mathcal{E}).$$

All proper walls are positive, the hypothesis of Thm. 35.3 and of Thm. 35.10, p. 309.

Marking. 1 point for the stride with the chart and its measure; 1 for $\ell = P(1)$ with the definition of the values cited; 1 for the walls. The measure $d\alpha_1 d\alpha_2$ in place of dy makes the integral diverge (the integrand is $\sim 1/(\alpha_1 + \alpha_2)$ at infinity): -1.

Step 2 (5 points). The stride. (a) *The gauge.* The identity

$$e_1e_2 - 9e_3 = \alpha_1(\alpha_2 - \alpha_3)^2 + \alpha_2(\alpha_3 - \alpha_1)^2 + \alpha_3(\alpha_1 - \alpha_2)^2$$

holds because both sides equal $\sum_{i\ne j} \alpha_i^2\alpha_j - 6e_3$, the left since $e_1e_2 = \sum_{i\ne j} \alpha_i^2\alpha_j + 3e_3$. So $0 < f \le \frac{1}{9}$ on the orthant, with equality exactly on the diagonal $\alpha_1 = \alpha_2 = \alpha_3$, which on the chart is the single point $y = 0$. This is Thm. 35.10(i) here, with $\alpha^\star = \mathbf{1}$ and $f(\alpha^\star) = \frac{1}{9}$.

(b) *The decay.* Order the coordinates, $\alpha_{(1)} \ge \alpha_{(2)} \ge \alpha_{(3)}$. Then $\alpha_{(1)} \le e_1 \le 3\alpha_{(1)}$, $\alpha_{(1)}\alpha_{(2)} \le e_2 \le 3\alpha_{(1)}\alpha_{(2)}$ and $e_3 = \alpha_{(1)}\alpha_{(2)}\alpha_{(3)}$, so

$$\frac{1}{9}\,\frac{\alpha_{(3)}}{\alpha_{(1)}} \le f \le \frac{\alpha_{(3)}}{\alpha_{(1)}}.$$

On the chart $\alpha_{(3)}/\alpha_{(1)} = e^{-\varphi(y)}$ with

$$\varphi(y) = \max(y_1, y_2, 0) - \min(y_1, y_2, 0),$$

and since 0 and y_i are both among the three entries, $\varphi(y) \ge |y_i|$: $\varphi(y) \ge \|y\|_\infty$.

(c) *The volume.* For real $k > 0$,

$$\int_{\mathbb{R}^2} e^{-k\varphi(y)}\,dy = \frac{6}{k^2}.$$

The lines $y_1 = y_2$, $y_1 = 0$, $y_2 = 0$ cut the plane into six cones, one for each order of the entries $(y_1, y_2, 0)$, and on each cone φ is the largest entry less the smallest:

$$\int_{y_1 \ge y_2 \ge 0} e^{-ky_1} dy = \int_0^\infty y_1 e^{-ky_1} dy_1 = \frac{1}{k^2}, \qquad \int_{y_1 \ge 0 \ge y_2} e^{-k(y_1 - y_2)} dy = \frac{1}{k}\cdot\frac{1}{k},$$
$$\int_{0 \ge y_1 \ge y_2} e^{ky_2} dy = \int_0^\infty u e^{-ku} du = \frac{1}{k^2},$$

and the three cones with y_1 and y_2 exchanged give the same.

(d) *Consequences*, for real $k > 0$:

$$\frac{6 \cdot 9^{-k}}{k^2} \le P(k) \le \frac{6}{k^2}, \qquad P(k+1) < \frac{1}{9}\,P(k),$$

the second from (a), strict because $f < \frac{1}{9}$ off the single point $y = 0$. And P is continuous on $(0, \infty)$: on $[k_1, \infty)$ the integrands f^k are dominated by f^{k_1}, because $0 < f < 1$, and f^{k_1} is integrable.

Marking. (a) 2 points; the identity must be shown or verified by expansion. The two inequalities of the arithmetic and geometric means, $e_1 \ge 3e_3^{1/3}$ and $e_2 \ge 3e_3^{2/3}$, are an accepted proof of $f \le \frac{1}{9}$. The strict inequalities of Step 9 need only $f < \frac{1}{9}$ on a set of positive measure; without any strictness the bracket is closed at the top, which suffices for the answer. (b) 1 point for the upper side, the decay; citing Thm. 35.10(i) for $f \le C'e^{-c'\|y\|}$ on the chart earns it, and gives convergence at every real $k > 0$. 1 point for the lower side of (b) and the volume (c) together; they serve only the corner, and a solution that obtains the corner by a correct citation of Proposition 15.3 or 18.1 in Step 7 earns this point there. (d) 1 point: convergence, positivity, continuity and $P(k+1) < \frac{1}{9}P(k)$. Step 2(d) credits the proof of the last inequality and Step 9(b) its use.

Step 3 (3 points). The candidate law. Here $1/f = e_1e_2/e_3 = (\alpha_1+\alpha_2+\alpha_3)(\alpha_1^{-1}+\alpha_2^{-1}+\alpha_3^{-1})$, and the closings of the direction (Proposition 18.1) are

$$x_j = [\alpha^0]\Big(\frac{1}{f}\Big)^j = \sum_{a_1+a_2+a_3=j} \Big(\frac{j!}{a_1!\,a_2!\,a_3!}\Big)^2 = 1,\ 3,\ 15,\ 93,\ 639,\ 4653,\ 35169,\ 272835,\ 2157759,\ \ldots,$$

since the constant term pairs the monomial α^a of $(\sum_e \alpha_e)^j$ with the monomial α^{-a} of $(\sum_e \alpha_e^{-1})^j$, and both have the coefficient $j!/(a_1!a_2!a_3!)$. Exact linear algebra with the ansatz $\sum_{i=0}^{2} p_i(j)\,x_{j+i} = 0$, $\deg p_i \le 2$, on $x_0, \ldots, x_{15}$ (14 equations, 9 unknowns) has a one-dimensional space of solutions,

$$(j+2)^2 x_{j+2} - (10j^2 + 30j + 23)\,x_{j+1} + 9(j+1)^2 x_j = 0.$$

Reflected as in Corollary 18.2, $c_i(k) = p_{2-i}(-k-2)$:

$$k^2P(k) - (10k^2+10k+3)\,P(k+1) + 9(k+1)^2\,P(k+2) = 0. \tag{L}$$

At this point (L) is a candidate. Corollary 18.2 runs the other way, from a law of P to the recurrence of the closings, and sixteen closings do not prove their own recurrence.

Marking. 1 point each for the closings, the recurrence (with the size of the system, or any other exact derivation) and the reflection. Any route to (L) earns the 3 points: the closings are how the law is found, not part of its proof. A solution that takes (L) as proved here keeps these 3 points and earns nothing in Steps 4 and 5.

Step 4 (5 points). The certificate. The origin is 0, so the defects of the origin vanish, and the defect of the direction is (Lemma 35.2(ii), p. 308)

$$c_e = D_e \log f = 1 - R_e - R_e^*, \qquad R_e = \frac{\alpha_e}{e_1}, \qquad R_e^* = \frac{\alpha_e(e_1-\alpha_e)}{e_2},$$

the occupation of e under the basis law and under the cobasis law. Since $\sum_e R_e = 1$ and $\sum_e R_e^* = 2$, $\sum_e c_e = 0$. A *certificate* of (L) is a choice of insertions g_1, g_2, g_3 (Def. 35.1: regular functions on the letter complement; here rational functions of degree 0 whose denominators are products of e_1 and e_2, with coefficients polynomial in k) such that

$$k^2 - (10k^2+10k+3)\,f + 9(k+1)^2 f^2 = \sum_{e=1}^{3} \big(D_e g_e + k\,c_e\,g_e\big). \tag{C}$$

Ansatz: $g_e = N_e/(e_1^2 e_2)$, with N_e homogeneous of degree 4 with coefficients in $\mathbb{Q}+\mathbb{Q}k$, g_e the image of g_1 under the exchange of the marks 1 and e, and N_1 symmetric in α_2, α_3. Multiplied by $e_1^3e_2^2$, (C) becomes a polynomial identity, of degree 7 in α and 2 in k, through

$$e_1^3e_2^2\big(D_e g_e + k\,c_e g_e\big) = e_1e_2\,D_eN_e - N_e\big(2\alpha_e e_2 + \alpha_e(e_1-\alpha_e)e_1\big) + k\,N_e\big(e_1e_2 - \alpha_e e_2 - \alpha_e(e_1-\alpha_e)e_1\big),$$

and it is linear in the 18 coefficients of N_1: 99 equations of rank 10. One solution is

$$g_1 = -\frac{\alpha_1^2\big(k\,\alpha_1(\alpha_2+\alpha_3) + (2+3k)(\alpha_2^2+\alpha_3^2) - (1-k)\,\alpha_2\alpha_3\big)}{e_1^2e_2}, \tag{G}$$

and within the ansatz the others differ from it by one insertion h added at every mark, h with a symmetric numerator (an 8-dimensional space). Such an h changes nothing, because $\sum_e D_e h = 0$ (Euler's identity in degree 0) and $\sum_e c_e h = 0$.

Verification. Write $g_e = g_e^{(0)} + k\,g_e^{(1)}$. The three powers of k in (C) read

$$\begin{aligned} k^2: &\quad \textstyle\sum_e c_e g_e^{(1)} = 1 - 10f + 9f^2 = (1-f)(1-9f),\\ k^1: &\quad \textstyle\sum_e \big(D_e g_e^{(1)} + c_e g_e^{(0)}\big) = -10f + 18f^2,\\ k^0: &\quad \textstyle\sum_e D_e g_e^{(0)} = -3f + 9f^2. \end{aligned}$$

Times $e_1^3e_2^2$ each sum over the marks is a symmetric polynomial of degree 7, and in the basis of products of e_1, e_2, e_3 exact expansion gives

$$\textstyle\sum_e e_1^3e_2^2\,c_e g_e^{(1)} = e_1^3e_2^2 - 10\,e_1^2e_2e_3 + 9\,e_1e_3^2,$$

$$\sum_e e_1^3e_2^2\,D_eg_e^{(1)} = -2\,e_1e_2^3 - 3\,e_1^2e_2e_3 + 9\,e_1e_3^2, \qquad \sum_e e_1^3e_2^2\,c_eg_e^{(0)} = 2\,e_1e_2^3 - 7\,e_1^2e_2e_3 + 9\,e_1e_3^2,$$
$$\sum_e e_1^3e_2^2\,D_eg_e^{(0)} = -3\,e_1^2e_2e_3 + 9\,e_1e_3^2,$$

which, with $f = e_3/(e_1e_2)$, are the three right sides times $e_1^3e_2^2$. So (C) holds identically in α and k. Proposition 17.15 reported a certificate of this law with the same denominator.

Marking. 1 point for the defects c_e, in any equivalent form. 2 points for a certificate. Any certificate is accepted; it need not be symmetric, and any denominator is accepted for which Step 5 can be carried out. 2 points for the verification: by exact expansion, by the symmetric-function form above, or by exact evaluation of the cleared identity on a grid large enough to decide it. After clearing, both sides are polynomials of degree at most 7 in each α_i and 2 in k, homogeneous in α, so agreement at $\alpha_3 = 1$ on 8 values of α_1, 8 of α_2 and 3 of k (all 192 points) proves it. Agreement at a few random points is a test, not a proof. To test a certificate other than (G), a grader evaluates both sides of (C) at $\alpha = (1, 2, 4)$, $k = \frac{1}{3}$, where the left side is $-\frac{2809}{7203}$ (there $f = \frac{4}{49}$, and (G) gives $g_1, g_2, g_3 = -\frac{85}{1029}, -\frac{310}{1029}, -\frac{424}{1029}$); a certificate written on the chart, with $g_3 = 0$ and D_1, D_2 read as $\partial_{y_1}, \partial_{y_2}$, is tested at the same point $(\alpha_1, \alpha_2) = (\frac{1}{4}, \frac{1}{2})$. A certificate with floating-point coefficients, or verified only in floating point, earns 0 of the 4: the identity has to hold exactly. How the certificate was found does not matter: rational coefficients found by a numerical search and then verified exactly earn full credit.

Step 5 (5 points). The law at every real $k > 0$. Thm. 35.3 states the relations for all *large* k. The law is needed at $k = 1, \dots, K-2$ (Step 8) and as $k \to 0^+$ (Step 7), so the integration by parts is carried out here at every real $k > 0$, with the bounds of Step 2. Fix a real $k > 0$.

(a) $D_e(f^k) = k\,c_ef^k$, so the right side of (C), times f^k, is $\sum_e D_e(g_ef^k)$.

(b) On the chart $D_1 = \partial_{y_1}$ and $D_2 = \partial_{y_2}$, and for every function h of degree 0, $D_3h = -D_1h - D_2h$ (Euler's identity). So

$$\sum_{e=1}^{3} D_e(g_ef^k) = \partial_{y_1}H_1 + \partial_{y_2}H_2, \qquad H_i = (g_i - g_3)f^k.$$

(c) *Bounds.* Every monomial of every N_e occurs in the expansion of $e_1^2e_2$: for N_1, $\alpha_1^3\alpha_2 = \alpha_1^2 \cdot \alpha_1\alpha_2$, $\alpha_1^2\alpha_2^2 = \alpha_1\alpha_2 \cdot \alpha_1\alpha_2$ and $\alpha_1^2\alpha_2\alpha_3 = \alpha_1^2 \cdot \alpha_2\alpha_3$, with the images under $2 \leftrightarrow 3$, and the other N_e are images of N_1. The coefficients of $e_1^2e_2$ are positive, so $\alpha^v \le e_1^2e_2$ on the orthant for each such monomial, and $|g_e| \le \|N_e\|_1$, the sum of the absolute values of the coefficients. For every mark i,

$$D_ig_e = \frac{D_iN_e}{e_1^2e_2} - g_e\left(2R_i + R_i^*\right),$$

where D_iN_e has the monomials of N_e or fewer and $0 \le R_i, R_i^* \le 1$; and $|c_i| \le 1$. With Step 2(b) there is, for fixed k, a constant C with

$$|H_i| \le C\,e^{-k\varphi}, \qquad |\partial_{y_j}H_i| \le C\,e^{-k\varphi}.$$

(d) If H is C^1 on $\mathbb{R}^2$, tends to 0 at infinity and $\partial_{y_1}H$ is integrable, then $\int \partial_{y_1}H\,dy = 0$: for every y_2, $\int_{-R}^{R} \partial_{y_1}H\,dy_1 = H(R, y_2) - H(-R, y_2) \to 0$, and Fubini applies. By (c) and Step 2, H_1 and H_2 qualify ($e^{-k\varphi} \le e^{-k\|y\|_\infty} \to 0$, and $e^{-k\varphi}$ is integrable), so the right side of (C), times f^k, integrates to 0.

(e) The left side of (C), times f^k, integrates to the left side of (L), all three integrals convergent by Step 2. Hence (L) holds at every real $k > 0$.

Marking. 1 point for (a)–(b), including the mark 3 on its own chart. 2 points for the bounds (c). 1 point for (d). 1 point for the conclusion at every real $k > 0$, with the reason it is needed. Citing Thm. 35.3 at $k = 1, 2, \dots$ and as $k \to 0^+$ without (c): 2 of 5. Any complete extension earns full credit; for instance (C) integrated at large integers by Thm. 35.3 as stated (its hypothesis checked in Step 1), then carried to every k with $\operatorname{Re} k > 0$ by Carlson's theorem: the left side of (L) is holomorphic there, it is bounded by a polynomial in $|k|$ on $\operatorname{Re} k \geq 1$ because $|P(k)| \leq P(\operatorname{Re} k) \leq 6/(\operatorname{Re} k)^2$ (Step 2), and Carlson's theorem applied to $z \mapsto F(N+z)$, with F the left side of (L) and $N \geq 1$ an integer from which on (L) holds at every integer, makes F vanish identically. Integrating $D_3(g_3 f^k)$ by parts "in y_3" on the chart $\alpha_3 = 1$, where y_3 is not a coordinate: -1, unless repaired by Euler's identity or by a certificate with $g_3 = 0$.

Step 6 (2 points). The native values; what positivity can select. The symbol of (L) (Cor. 35.6, p. 308) is

$$\chi(\lambda) = 1 - 10\lambda + 9\lambda^2 = (1-\lambda)(1-9\lambda).$$

The native gauges of the centre ray are the signs η with $\eta_3 = 1$ (Proposition 15.7, Thm. 35.12, p. 310), of native value $K(\eta)^{-2}$: the table's gauge $\eta = (1,1,1) = \alpha^\star$, with $K(\eta) = 3$ and the value $\frac{1}{9} = f(\alpha^\star)$; and $\eta = (-1,1,1), (1,-1,1), (-1,-1,1)$, with $K(\eta) = \pm 1$ and the value 1. They are nondegenerate. By Thm. 35.12(ii) the Hessian of $-\log f$ at η is $2\big(\kappa_\eta(X_e, X_{e'})\big)_{e,e' \neq 3}$, the covariances of the indicators under the signed measure $\mathbb{P}_\eta(\{e\}) = \eta_e / K(\eta)$. Its determinant is $\frac{4}{27}$ at $(1,1,1)$ and -4 at each of the other three signs. So Cor. 35.6, with $g = 1$, makes both native values roots of χ, and since χ has degree 2 they are all its roots. (Cor. 35.6 runs from native values to roots; a root need not be a native value, as the extra root $\frac{131003}{4771109}$ in §18.4 shows.)

In Theorem 18.5 the singular points are $q = 9$, where the slit $[9, \infty)$ begins, and $q = 1$, off the slit. At $q = 1$, a simple root and the value of nondegenerate gauges, positivity imposes $\nu = 1$ condition, and with the corner $1 + \nu = 2 = h$. The condition is independent, and here that is seen directly: by Step 8, $c(K) \geq 3^{1-K}$ and $9c(K) - c(K-1) \geq 2c(K-1) \geq 2 \cdot 3^{2-K}$, so the width of Step 9(d) is at most $1/(K-1)^2$ and the bracket closes on one number. So the law, the corner and positivity determine P (Theorem 18.5(iii)). The rational solutions of (L) have the exponential rate 1 of the signs and P the rate $\frac{1}{9}$ of the gauge, so the bracket shrinks by about $\frac{1}{9}$ per step, $\log_{10} 9 = 0.954$ digits: about seven steps for 10^{-6}, before constants. The constants decide in Step 9.

Marking. 1 point for the native values with their nondegeneracy, matched to the roots of the symbol in the direction Cor. 35.6 gives. "The roots are native values by Cor. 35.6" reverses the corollary and earns 0 for this point unless the roots are matched to gauges directly, as above. 1 point for the count $1 + \nu = h$ with the independence, or any equivalent statement that the corner supplies the condition positivity does not. The step is not used in the proofs of the bounds; a complete solution without it loses these 2 points only.

Step 7 (5 points). The corner. (a) By Step 2(d), $6 \cdot 9^{-k} \leq k^2 P(k) \leq 6$ for real $k > 0$, so

$$\lim_{k \to 0^+} k^2 P(k) = 6.$$

(b) Take (L) at real $k \in (0,1)$ (Step 5) and let $k \to 0^+$. By the continuity of Step 2(d), $P(k+1) \to P(1) = \ell$ and $P(k+2) \to P(2)$, while $10k^2 + 10k + 3 \to 3$ and $9(k+1)^2 \to 9$. So $6 - 3\ell + 9P(2) = 0$:

$$P(2) = \frac{\ell - 2}{3}. \tag{S}$$

(c) This is Corollary 18.2 at $j = 0$,

$$[k^2]c_0 \cdot \mathrm{Cor} + c_1(0)P(1) + c_2(0)P(2) = 0, \qquad \mathrm{Cor} = \sum_{\pi} \prod_{i=2}^{3} \kappa_{1,U_i}^{-1} = 3! \cdot 1 = 6$$

(Proposition 18.1; for $\binom{\mathcal{E}}{1}$ the corner is $n!$, Proposition 15.3). The six cones of Step 2(c) are the six orders π, each contributing $\prod_i \kappa_{1,U_i}^{-1} = 1$.

Marking. 2 points for (a), 2 for (b), 1 for (c). A correct citation of Proposition 15.3 or 18.1 in place of (a), with the walls of Step 1 and the sum over the 3! orders, earns the 2 points of (a). Errors:

- Reading (L) at $k = 0$ as $0 \cdot P(0) - 3P(1) + 9P(2) = 0$. Here $P(0) = \int dy = \infty$, and the product $0 \cdot \infty$ is the corner, 6, not 0. The reading gives $P(2) = \ell/3$, which is false ($P(2) = 0.1146\ldots$, $\ell/3 = 0.78\ldots$) and contradicts Step 2(d) at once: $P(2) < \frac{1}{9}P(1)$ would say $\ell/3 < \ell/9$. After it every $P(K)$ is ℓ times a positive rational, positivity fixes only ratios, and the bracket is empty. 0 for Step 7, and in Steps 8–9 only the points that do not depend on (S): the signs and their use, 2 points.
- Passing the limit under the integral, $k^2 f^k \to 0$ pointwise, so "$k^2P(k) \to 0$": 0 for (a). There is no domination, since $\int dy = \infty$.
- The wrong power ($kP(k)$, the corner of a ray on two marks) or Cor = 1 (one order): 0 for (a); (b) is marked as a consequence.
- Citing Corollary 18.2 for (S). Its hypothesis is the law *identically*, as meromorphic functions of k; Step 5 gives it at real $k > 0$, and the continuation of Proposition 18.1 extends it. Citing the corollary with the law proved only at the integers leaves a gap: -1.

Step 8 (4 points). The rational solutions and their signs. For integers $k \geq 1$, $9(k+1)^2 \neq 0$, and (L) gives

$$P(k+2) = \frac{(10k^2 + 10k + 3)\,P(k+1) - k^2 P(k)}{9(k+1)^2}.$$

Let c and r be the solutions of (L) with $c(1) = 1$, $c(2) = \frac{1}{3}$ and $r(1) = 0$, $r(2) = -\frac{2}{3}$. From $P(1) = \ell$, (S) and induction,

$$P(K) = c(K)\,\ell + r(K) \qquad (K \geq 1).$$

Signs, at every K. Put $x'_j = 9^j c(j+1)$. Then (L) at $k = j+1$ reads $(j+2)^2 x'_{j+2} = (10j^2 + 30j + 23)\,x'_{j+1} - 9(j+1)^2 x'_j$, with $x'_0 = 1$, $x'_1 = 3$. If $x'_{j+1} \geq 3x'_j > 0$, then

$$(j+2)^2 x'_{j+2} \geq \big(10j^2 + 30j + 23 - 3(j+1)^2\big)x'_{j+1} = (7j^2 + 24j + 20)\,x'_{j+1} \geq 3(j+2)^2 x'_{j+1},$$

since $7j^2 + 24j + 20 - 3(j+2)^2 = 4(j+1)(j+2) \geq 0$. So $x'_{j+1} \geq 3x'_j > 0$ for every j, hence $x'_j \geq 3^j$, that is,

$$c(K) \geq 3^{1-K} > 0, \qquad 9c(K) \geq 3c(K-1) > c(K-1) \qquad (K \geq 2).$$

The x'_j obey the recurrence of the closings with the same first two terms, so $x'_j = x_j$ and $c(K) = 9^{1-K} x_{K-1}$ wherever that recurrence holds: in the table below by inspection, and for every j by Corollary 18.2 once (L) holds identically. The proof of the signs does not use this.

K	$c(K)$	$r(K)$	$a_K = -r(K)/c(K)$	
1	1	0	0	
2	1/3	−2/3	2	
3	5/27	−23/54	23/10	2.3
4	31/243	−145/486	145/62	2.33870967…
5	71/729	−1331/5832	1331/568	2.34330985…
6	517/6561	−242353/1312200	242353/103400	2.34383945…
7	11723/177147	−5495507/35429400	5495507/2344600	2.34389959…
8	10105/177147	−46422941/347208120	46422941/19805800	2.34390638…
9	26639/531441	−2447622221/20832487200	2447622221/1044248800	2.34390714…
10	5773279/129140163	−530455590781/5062294389600	530455590781/226312536800	2.34390722…

Two checks a grader can run on a student's table: $9^{K-1}c(K)$ is the closing x_{K-1} $(1, 3, 15, 93, \dots)$, and the frame of the two solutions (Prop. 35.8, p. 309: $C_{k+1} = \frac{c_0(k)}{c_2(k)}C_k = \frac{k^2}{9(k+1)^2}C_k$, $C_1 = -\frac{2}{3}$) is

$$c(k)\,r(k+1) - c(k+1)\,r(k) = -\frac{2}{3}\cdot\frac{9^{1-k}}{k^2}.$$

Marking. 2 points for $P(K) = c(K)\ell + r(K)$ with the initial values from (S). 1 point for the signs $c(K) > 0$ and $9c(K) - c(K-1) > 0$, at every K as above or at the K used from the exact table. 1 point for the table, exact, through the K used. A decimal table: 0 for the last point.

Step 9 (8 points). The bracket and the answer. (a) *Below, by positivity.* $P(K) = c(K)\ell + r(K) > 0$, divided by $c(K)$, gives

$$\ell > a_K = -\frac{r(K)}{c(K)}.$$

(b) *Above, by the gauge.* By Step 2(d), $P(K) < \frac{1}{9}P(K-1)$, that is, $c(K)\ell + r(K) < \frac{1}{9}\big(c(K-1)\ell + r(K-1)\big)$, or $\big(9c(K) - c(K-1)\big)\ell < r(K-1) - 9r(K)$, and divided by $9c(K) - c(K-1)$,

$$\ell < b_K = \frac{r(K-1) - 9\,r(K)}{9\,c(K) - c(K-1)}.$$

(c) *The signs.* Both divisions keep the direction of the inequality: $c(K) > 0$ and $9c(K) - c(K-1) > 0$ by Step 8.

At $K = 2$ this is the first bracket: $P(2) = \frac{\ell-2}{3} > 0$ gives $\ell > 2$, and $P(2) < \frac{1}{9}P(1)$ gives $\ell < 3$.

(d) *The width and the answer.* With the frame of Step 8,

$$b_K - a_K = \frac{r(K-1)\,c(K) - r(K)\,c(K-1)}{c(K)\big(9c(K) - c(K-1)\big)} = \frac{2\cdot 9^{2-K}}{3(K-1)^2\,c(K)\big(9c(K) - c(K-1)\big)},$$

and in exact arithmetic:

K	b_K	$b_K - a_K$	width with $P(K) < 9^{1-K}\ell$ instead
2	3	1	1
3	19/8	$7.50\cdot 10^{-2}$	$1.64\cdot 10^{-1}$
4	61/26	$7.44\cdot 10^{-3}$	$2.54\cdot 10^{-2}$
5	3413/1456	$7.84\cdot 10^{-4}$	$3.67\cdot 10^{-3}$
6	104539/44600	$8.43\cdot 10^{-5}$	$5.04\cdot 10^{-4}$
7	149014/63575	$9.17\cdot 10^{-6}$	$6.66\cdot 10^{-5}$
8	909876251/388187800	$1.0036\cdot 10^{-6}$	$8.59\cdot 10^{-6}$
9	6414407843/2736630400	$1.1025\cdot 10^{-7}$	$1.0863\cdot 10^{-6}$
10	232184895407/99058909600	$1.21\cdot 10^{-8}$	$1.35\cdot 10^{-7}$

At $K = 8$ the certified width is $1.0036 \cdot 10^{-6} > 10^{-6}$; at $K = 9$ it is $\frac{6561}{59511099776} < 1.11 \cdot 10^{-7}$. So

$$\frac{2447622221}{1044248800} = a_9 < \ell < b_9 = \frac{6414407843}{2736630400},$$

and $q = a_9$ satisfies $0 < \ell - q < b_9 - a_9 < 1.11 \cdot 10^{-7} < 10^{-6}$.

Marking. (a) 2 points. (b) 2 points: 1 for using $P(K) < \frac{1}{9}P(K-1)$ (its proof is credited in Step 2(d)), 1 for b_K. (c) 1 point for invoking the two signs at the divisions (their proof is credited in Step 8; a sign proved there and used here silently loses nothing). (d) 3 points: 1 for a K at which the width certified by (a)–(c) is below 10^{-6}, 1 for the exact comparison, 1 for a stated q whose inequalities $0 < \ell - q < 10^{-6}$ follow from the bracket.

19.4 Where a solution that looks right is wrong

1. *The answer one step early.* $a_8 = \frac{46422941}{19805800}$ is a correct answer: $\ell - a_8 = 8.58 \cdot 10^{-7}$. But the bracket at $K = 8$ has width $1.0036 \cdot 10^{-6}$, so a solution that certifies a_8 from $K = 8$ alone has an error in it. The upper bound at $K = 9$ certifies it: $\ell - a_8 < b_9 - a_8 = 8.72 \cdot 10^{-7}$. Full credit with that; with the bound claimed at $K = 8$, 5 of the 8 points of Step 9. The true error falls below the tolerance one step before the certified one does.

2. *The observed error.* Comparing a_K with a decimal value of ℓ (from the share side, from a quadrature, or from a closed form recalled elsewhere) proves nothing about $\ell - a_K$: 0 for Step 9(d).

3. *The rate as a bound.* Theorem 18.5(iv) gives the error $O(9^{-K})$ up to powers of K, with no constant: 0 for Step 9(d). Nor is the last step a bound: $a_9 - a_8 = 7.62 \cdot 10^{-7}$ is smaller than $\ell - a_8 = 8.58 \cdot 10^{-7}$, because the a_K increase to ℓ with ratio about $\frac{1}{9}$, so the remaining error is about $\frac{9}{8}$ of the next step.

4. *An unproved upper bound.* "$P(K) \le 9^{-K}\ell$, since $f \le \frac{1}{9}$". The gauge gives $P(K) < \frac{1}{9}P(K-1)$, and iterated $P(K) < 9^{1-K}P(1)$; the factor 9^{-K} does not follow from $f \le \frac{1}{9}$, it would need $\int dy < \infty$. (The inequality is in fact true for $K \ge 8$ and false for $K \le 7$: $9^K P(K)$ decreases by Step 2(d), and $9^7 P(7)/\ell = 1.03$, $9^8 P(8)/\ell = 0.90$. Nothing in such a solution proves it, and a grader who checks it numerically should not be misled.) With it the width at $K = 9$ is $1.21 \cdot 10^{-7}$, while the simple bound actually proved gives $1.09 \cdot 10^{-6}$ there, so the solution certifies nothing below 10^{-6}. The stated $q = a_9$ is right and the proof is not: 1 of 2 in Step 9(b), 0 in 9(d), 4 of the 8 points of Step 9.

5. *The simple bound, done right.* $P(K) < 9^{1-K}P(1) = 9^{1-K}\ell$ gives $\big(c(K) - 9^{1-K}\big)\ell < -r(K)$, and $c(K) - 9^{1-K} = 9^{1-K}(x'_{K-1} - 1) > 0$ for $K \ge 2$ by Step 8, so $\ell < -r(K)/\big(c(K) - 9^{1-K}\big)$: valid and cruder (the last column of the table). It needs $K = 10$, with $q = a_{10} = \frac{530455590781}{226312536800}$. Full credit, with the sign stated.

6. *Decimals.* $q = 2.343907$ is accepted, with the exact comparisons $\frac{2343907}{10^6} < a_9$ and $b_9 - \frac{2343907}{10^6} = 2.53 \cdot 10^{-7}$. A decimal above a_9, such as 2.3439072, is a correct answer ($\ell - 2.3439072 = 3.9 \cdot 10^{-8}$) but needs a new lower bound: $a_{10} = 2.34390722\ldots$ supplies it. A decimal rounded above the lower bound in hand: -1 in Step 9(d), its last point.

7. *The signs.* Signs of $c(K)$ and of $9c(K) - c(K-1)$ neither proved nor checked: -2 in all, the sign point of Step 8 and Step 9(c). They hold here at every K (Step 8), but where a rational solution changes sign the inequality reverses; on the line 1 of $\binom{[5]}{2}$, Remark 17.17 checked the sign only for $m \le 80$.

8. *The corner.* Reading the law at $k = 0$ with $0 \cdot P(0) = 0$, or passing the limit $k \to 0^+$ under the integral: the approximation is lost (Step 7). The first gives $P(2) = \ell/3$, every $P(K)$ is then ℓ times a positive rational, positivity gives only $\ell > 0$, and the gauge bound at $K = 2$, $\ell/3 < \ell/9$, gives $\ell < 0$: the student's own bracket is empty, which the grader can point to.
9. *The theorem at small k.* Thm. 35.3 cited at $k = 1, 2, \ldots$ or as $k \to 0^+$ without the decay of Step 2: Step 5.
10. *A certificate checked at sample points only*: Step 4.
11. *The measure* $d\alpha_1 d\alpha_2$ on the chart: Step 1.
12. *A quadrature for* ℓ *or for* $P(2)$ *used in the solution*: no credit at the step where it is used. It is a check (§19.5).

19.5 Receipts and the independent check

The script exam_s19.py prints every number of this entry. Exact: the sum of squares; the closings $x_0, \ldots, x_{15}$ and their recurrence (14 equations, 9 unknowns, nullity 1); the reflection; the certificate (G) and the identity (C) after clearing $e_1^3 e_2^2$; its three powers of k in the basis of products of e_1, e_2, e_3; the rank 10 of the symmetric ansatz in 18 unknowns; the monomial check of Step 5(c); the spot value $-\frac{2809}{7203}$; the symbol and the signs; the walls and Cor = 6; the table and the brackets to $K = 10$; and to $K = 16$ the signs with $c(K) \geq 3^{1-K}$, the identity $9^{K-1} c(K) = x_{K-1}$, the frame, and the two forms of the width with its bound $1/(K-1)^2$.
Check [computed], not part of the solution, in ball arithmetic, with the integrals cut where their tails are below 10^{-70}. On the chart, the integral over α_2 is elementary, $\int_0^\infty \frac{d\alpha_2}{(\alpha_2+A)(A\alpha_2+B)} = \frac{\log(A^2/B)}{A^2-B}$ with $A = 1 + \alpha_1$, $B = \alpha_1$, so

$$\ell = \int_0^\infty \frac{\log\big((1+x)^2/x\big)}{1+x+x^2}\, dx = 2.343907238689458890601562288872277069095\ldots.$$

On the share side (Proposition 15.2), $\Gamma(k)^2 P(k) = \int_0^\infty y^{k-1} \mathsf{G}_{0;(1,-1)}(1,y)^3\, dy$ gives the same ℓ to 40 digits, $P(2) - \frac{\ell-2}{3} = 0$ to 47 digits, and $P(9) = 4.8152090557463543143 \cdot 10^{-9} = c(9)\ell + r(9)$. With this value,

$$\ell - a_8 = 8.5773 \cdot 10^{-7}, \qquad \ell - a_9 = 9.6062 \cdot 10^{-8}, \qquad b_9 - \ell = 1.4186 \cdot 10^{-8}.$$

The upper side is the sharper. Its slack is $\frac{1}{9}P(K-1) - P(K)$, which is about $P(K)/(K-1)$ because $P(k)$ decays like $9^{-k}/k$ (for three marks the branch of Thm. 35.11, p. 310, is logarithmic), so $b_K - \ell$ is about $\frac{9}{8(K-1)}(\ell - a_K)$.

19.6 What the example shows

The reading needed three inputs, and each was elementary once it was named. The analytic input is one inequality, $e_1 e_2 \geq 9 e_3$, sharpened to $f \asymp e^{-\varphi}$ by the tropical stride φ: it gives the convergence at every real $k > 0$, the corner, and the upper side of the bracket. The algebraic input is one certificate: it gives the law at every real $k > 0$, not only at large k. The arithmetic is exact. Positivity gives the lower side and the gauge the upper side. On every ray through the origin the first two inputs exist [derived]: $f \leq f(\alpha^\star)$ (Thm. 35.10(i)), and the tropical stride, whose volume is Cor (the six cones of Step 2(c) are the six orders of Proposition 18.1). A certificate exists for every law that follows from the relations (Thm. 35.19, p. 313), but Step 5 needs more than existence: bounds on the g_e and their derivatives against the decay of f^k down to $k \to 0^+$. Here they came from the

numerators lying among the monomials of $e_1^2 e_2$, as the exponent condition did in Proposition 17.15; that a certificate with such bounds exists on every ray is [not yet computed]. With $h-1$ masters, positivity of $P(K),\dots,P(K+h-2)$ and the gauge bounds $P(K+i) < f(\alpha^\star)P(K+i-1)$ cut out a polytope in the coordinates of the masters, and Theorem 18.5(iv) is the rate at which it shrinks [not yet computed]; with all the conditions of the stride's law the rate is Theorem 20.9. The width of the bracket is a ratio of integers from the other end of the pencil (Remark 18.3):

$$b_K - a_K = \frac{2\cdot 9^{K-1}}{3(K-1)^2\, x_{K-1}\,(x_{K-1}-x_{K-2})} \qquad \text{[derived]},$$

from $c(K) = 9^{1-K}x_{K-1}$, which holds at every K by Corollary 18.2: by Step 5 and the continuation of Proposition 18.1, (L) holds identically.

20 The law of the stride: eight questions after the examination

The examination of §19 read one ray, the three-mark centre, with three inputs: a law certified by the relations along a ray, the corner, and positivity. It closed on the period with a certified bracket. The eight questions below ask what that reading is when it is pointed elsewhere: at every ray, at the corner alone, at the slopes, at the walls, at the other solutions, at the stride itself, at lists that no graph or field realizes, and at the arithmetic. They turn out to be questions about one measure, the law of the stride (§20.0): positivity is that it is a measure, the corner is its mass near 0, the slopes are its logarithmic moments, a wall is a singular point of its density inside its support, the other solutions are the other branches of its density, and the rates are the geometry of its support against the native values. §20.0 sets it up, with three objects that every ray carries: the tropical stride, the two ends of the stride's law, and the two ends of a certificate. §§20.1–20.8 state each question as it was posed and answer it.

Throughout, a ray is a ray through the origin in the sense of §18: origin 0, an integral direction $(\mathbf{m}_1;\mathbf{s}_1,\mathbf{t}_1)$ on the balance wall, every proper wall positive, values $P(k) = \int_{y_{e_0}=0} f^k\,dy$, stride f (Def. 35.1, p. 307), native gauge $\alpha^\star$ of the direction on the orthant and $f^\star = f(\alpha^\star)$, the largest value of f there (Thm. 35.10(i), p. 309). The *slit* is $(0, f^\star)$, read in the plane of native values. The laws below that are not those of §§15–19 are least recurrences of closings, reflected (Corollary 18.2); the converse, that such a recurrence is a law of the values, is what a certificate of Thm. 35.3, p. 308, settles, and none of them is certified yet. Every reading made with one of them is checked against an independent value.

20.0 The stride's law

Definition 20.1 (The stride's law)**.** The *stride's law* of a ray is the image of the Lebesgue input of the chart under the stride, $\mu = f_*\mathbb{I}_{\mathbb{R}^{\mathcal{E}-e_0}}$ (the law of a measure under a map, Def. 1.1(vii), p. 10; the Lebesgue input, Def. 5.4, p. 29), a measure on $(0,\infty)$; ϱ is its density, $\mu(dt) = \varrho(t)\,dt$. With the reciprocal stride $X = 1/f$, the image $X_*\mathbb{I}_{\mathbb{R}^{\mathcal{E}-e_0}}$ is the same measure read at $x = 1/t$.

Proposition 20.2 (A ray is one Trailmoment on a line, and one Gramma)**.** [derived]

(i) μ is carried by $(0, f^\star]$, it has a density, and for every real $k > 0$

$$P(k) = \int_0^{f^\star} t^k\,\mu(dt) = \big(\mathsf{T}_{k;(1)}[x]\,X_*\mathbb{I}_{\mathbb{R}^{\mathcal{E}-e_0}}\big)\big((0,\infty)\big)\,:$$

the values of the ray are the Trailmoment on a line of index k, one unit exponent and coefficient function x, of the transported Lebesgue input (Thm. 5.9, p. 31).

(ii) Write $X = \sum_i w_i\,\alpha^{t_i}$ on the chart, a Laurent polynomial with positive coefficients w_i and exponents t_i of degree 0. The Laplace transform of $X_*\mathbb{I}_{\mathbb{R}^{\mathcal{E}-e_0}}$ beyond the cut 0 (Def. 5.4) is the Gramma of the reciprocal stride at index 0,

$$\int e^{-ux}\,X_*\mathbb{I}_{\mathbb{R}^{\mathcal{E}-e_0}}(dx) = \int_{y_{e_0}=0} e^{-uX(e^y)}\,dy = \mathsf{G}_{\mathbf{0};t}(u\,\mathbf{w}),$$

and the Laplace side (Thm. 5.11, p. 31) reads the whole ray off it:

$$\Gamma(k)P(k) = \int_0^\infty u^{k-1}\mathsf{G}_{\mathbf{0};t}(u\,\mathbf{w})\,du.$$

(iii) For $U \sim \mathrm{Breta}^\circ(F; k\mathbf{m}_1; k\mathbf{s}_1, k\mathbf{t}_1)$ (Thm. 17.1(ii), p. 107), the law of $f(U)$ is $t^k\mu(dt)/P(k)$. So the ray is the exponential family of Thm. 17.4(ii), p. 108, restricted to the direction: sufficient statistic $\log f$, base measure μ, log-partition $\log P$, and

$$\frac{d}{dk}\log P(k) = \mathbb{E}\big[\log f(U)\big], \qquad \frac{d^2}{dk^2}\log P(k) = \mathrm{Var}\,\big[\log f(U)\big] > 0.$$

Proof. (i) $0 < f \le f^\star$ on the orthant with equality only on the ray of $\alpha^\star$ (Thm. 35.10(i)), and f has no critical point on the chart other than $\log\alpha^\star$ there, so the coarea formula gives a density. The integral of f^k is the integral of t^k against the image measure, which is Thm. 5.9 with the coefficient function $x \mapsto x$ and the kernel $\mathsf{Z}_{k;(1)}(x) = x^{-k}$ (Def. 5.2). (ii) $\int_0^\infty u^{k-1}e^{-uX}\,du = \Gamma(k)X^{-k}$ and Fubini, all terms positive; on the chart $x^{\nu-\mathbf{1}}dx = dy$ at $\nu = \mathbf{0}$, so the integral of e^{-uX} is the Gramma of Def. 2.1, p. 13, with the exponents t_i and coefficients uw_i. It converges because the direction's walls are positive, which puts $\mathbf{0}$ in the interior of the cone of the t_i (Thm. 2.2). (iii) On the balance wall the density of Breta$^\circ$ in y is proportional to the integrand of the period (Thm. 17.1(i)–(ii)), here f^k on the chart. The two derivatives are those of a log-partition (Thm. 17.4(ii)). □

At the centre of a list, $X = K(\alpha)K(\alpha^{-1})$, and for the banana Prop. 15.2 adds a third reading: since $\int_0^\infty y^{\sigma-1}\mathsf{G}_{0;(1,-1)}(1,y)\,dy = \Gamma(\sigma)^2$,

$$\mathsf{G}_{0;(1,-1)}(1,y)^n = \int_0^{f^\star} \mathsf{G}_{0;(1,-1)}(1,y/t)\,\mu(dt) \qquad \text{[derived]}:$$

the n-th power of the Gramma of one mark is a μ-mixture of its own dilations. Every question below is a question about μ.

Proposition 20.3 (The corner is the volume of the tropical stride)**.** [derived]*; sampled* [computed] *Let a ray through the origin of a family with positive weights have every proper wall positive. On the chart put*

$$\varphi(y) = \sum_j \Big(s_{1j}\max_{T\in\mathcal{B}_j}\langle\mathbf{1}_T, y\rangle + t_{1j}\max_{T\in\mathcal{B}_j}\langle\mathbf{1}_{\mathcal{E}\setminus T}, y\rangle\Big) - \langle\mathbf{m}_1, y\rangle,$$

the tropical stride*: $-\log f$ with every letter replaced by its largest monomial and the weights dropped.*

(i) On the cone of an order π of $\mathcal{E}$, $y_{\pi(1)} \ge \cdots \ge y_{\pi(n)}$, φ is linear, and on the generator $-\mathbf{1}_{U_i}$ of the cone, U_i the tails of π, it takes the value κ_{1,U_i}.

(ii) On the chart

$$C_-e^{-\varphi} \le f \le C_+e^{-\varphi}, \qquad C_- = \prod_j\Big(\sum_T \theta_{j,T}\Big)^{-s_{1j}-t_{1j}}, \qquad C_+ = \prod_j\big(\min_T \theta_{j,T}\big)^{-s_{1j}-t_{1j}}.$$

(iii) For real $k > 0$,

$$\int e^{-k\varphi}dy = \frac{\mathrm{Cor}}{k^{n-1}}, \qquad \mathrm{Cor} = \sum_{\pi}\prod_{i=2}^{n}\kappa_{1,U_i}^{-1} = (n-1)!\,\mathrm{vol}\{\varphi \le 1\},$$

so $C_-^k\,\mathrm{Cor} \le k^{n-1}P(k) \le C_+^k\,\mathrm{Cor}$ and $k^{n-1}P(k) \to \mathrm{Cor}$.

(iv) At the centre of a list, φ is the support function of $\mathcal{P}-\mathcal{P}$, $\mathcal{P}$ the convex hull of the $\mathbf{1}_T$, $T \in \mathcal{B}$; so $\{\varphi \le 1\}$ is the polar body of $\mathcal{P}-\mathcal{P}$ and

$$\mathrm{Cor} = (n-1)!\,\mathrm{vol}\big((\mathcal{P}-\mathcal{P})^\circ\big).$$

Proof. (i) Along π the largest of the $\langle \mathbf{1}_T, y\rangle$ is taken at the basis built greedily in the order π (the exchange property), the same basis on the whole cone, so each maximum is linear there. On $-\mathbf{1}_U$, $\max_T(-|T\cap U|) = -d_j(U)$ and $\max_T(-|U\setminus T|) = -d_j^*(U)$ (Def. 7.10, p. 49), so $\varphi(-\mathbf{1}_U) = m_1(U) - \sum_j \big(s_{1j}d_j(U) + t_{1j}d_j^*(U)\big) = \kappa_{1,U}$ (Def. 8.1, p. 53). (ii) Each letter lies between its largest monomial times the least weight and the same monomial times the sum of the weights. (iii) Modulo $\mathbf{1}$ the cone of π is generated by $-\mathbf{1}_{U_2}, \dots, -\mathbf{1}_{U_n}$, a basis of the lattice of the chart, so $\int_{\mathrm{cone}} e^{-k\varphi}dy = \prod_i (k\kappa_{1,U_i})^{-1}$, and homogeneity gives the volume; the squeeze is that of §19, Steps 2 and 7. (iv) At the centre $1/f = K(\alpha)K(\alpha^{-1})$ (Def. 35.9, p. 309), whose monomials are the $\alpha^{\mathbf{1}_T - \mathbf{1}_{T'}}$. □

So the corner of Proposition 18.1 at $j=0$ and of Proposition 15.3 is the volume of one polytope, read cone by cone, and the squeeze of §19 is general. For $\binom{[3]}{1}$, $\mathcal{P}-\mathcal{P}$ is a hexagon whose polar has area 3, and $\mathrm{Cor} = 2!\cdot 3 = 6$. Sampling $\{\varphi \le 1\}$ gives 12, 16 and 24 for $\binom{[4]}{2}$, H and $\binom{[4]}{1}$, the corners of §15 and §18.4.

Proposition 20.4 (The two ends of the stride's law)**.** [derived] *Let μ be the stride's law (Definition 20.1). It lives on $(0, f(\alpha^\star)]$, and for real $k > 0$ and $q \notin [f(\alpha^\star)^{-1}, \infty)$*

$$P(k) = \int t^k\,\mu(dt), \qquad G(q) = \int \frac{t}{1-qt}\,\mu(dt),$$
$$\mu\big([t, f(\alpha^\star)]\big) \sim \frac{\mathrm{Cor}}{(n-1)!}\Big(\log\frac{1}{t}\Big)^{n-1} \quad (t \to 0^+).$$

At the centre of a list the level sets $\{f = t\}$ are the hypersurfaces

$$K(\alpha)\,\Psi(\alpha) = t^{-1}\alpha^{\mathbf{1}}$$

of degree n in the projective chart, of dimension $n-2$, the same at every strength.

Proof. The first two are the image of dy under f and Thm. 35.10(ii), p. 309. For the third, $\{f \ge t\}$ lies between $\{\varphi \le \log(C_-/t)\}$ and $\{\varphi \le \log(C_+/t)\}$ by Proposition 20.3, and $\mathrm{vol}\{\varphi \le s\} = s^{n-1}\mathrm{vol}\{\varphi \le 1\}$. At the centre $\deg K + \deg\Psi = r + (n-r) = n = \deg\alpha^{\mathbf{1}}$: the balance of the centre puts the two sides in one degree. At strength σ the stride is f^σ, with the same level sets. □

So the corner is the mass of μ near 0, the native value of the table's gauge is the end of its support, and the branch of Thm. 35.11, p. 310, is its shape at that end. On two marks $\mu = 2\,dt/\big(t\sqrt{1-4t}\big)$ on $(0, \frac{1}{4}]$, whose moments are $\Gamma_{\substack{2\\1}}\begin{bmatrix} k, & k \\ & 2k \end{bmatrix}$.

Proposition 20.5 (The two ends of a certificate)**.** [derived] *Let* $\sum_{i=0}^{h} c_i(k) f^i = \sum_e \big(D_e g_e + k\, c_e g_e\big)$ *certify a law of a ray through the origin,* c_e *the defects of the direction,* D *the largest degree of the* c_i *and* $g_e = \sum_{d<D} k^d g_e^{[d]}$*. Then*

$$\chi(f) = \sum_e c_e\, g_e^{[D-1]}, \qquad \sum_{i\geq 1} c_i(0)\, f^i = \sum_e D_e\, g_e^{[0]}.$$

The first puts the symbol, read at the stride, in the ideal of the defects: at a native gauge every c_e *vanishes, which is Cor. 35.6, p. 308, read off a certificate. The second is a divergence. When the* $g_e^{[0]}$ *are bounded, integrating it over* $\{\varphi \leq R\}$ *on the chart (with the mark* e_0 *handled as in §19, Step 5(b)) and letting* $R \to \infty$ *gives*

$$\sum_{i\geq 1} c_i(0)\, P(i) = \lim_{R\to\infty} \big(\textit{flux of the field of the } g_e^{[0]} \textit{ through } \partial\{\varphi \leq R\}\big) = -[k^{n-1}]c_0 \cdot \mathrm{Cor},$$

the last by Corollary 18.2: the corner is the flux at infinity of the bottom piece of a certificate.

Proof. Compare the coefficients of k^D and of k^0; $c_0(0) = 0$ by Corollary 18.2. The left side of the second identity is integrable, so its integral over the chart is the limit of its integrals over $\{\varphi \leq R\}$, and on these the divergence theorem applies. □

For the three-mark law of §19 the top piece is $(1-f)(1-9f)$, the bottom piece $-3f+9f^2$, and the flux $-3\ell + 9P(2) = -6$.

Proposition 20.6 (The corner of two marks is the law of a logarithm)**.** [derived]*; checked* [computed] *For* $\binom{[2]}{1}$*,* $P(k) = \Gamma \begin{bmatrix} k,\ k \\ 2k \end{bmatrix}_1^2$*, the corner is* 2*, and with* X *of the Gamma law of index* 1*,*

$$\log\Big(\frac{k}{2}\, P(k)\Big) = 2\log \mathbb{E}[X^k] - \log \mathbb{E}[X^{2k}] = \sum_{m\geq 2} (-1)^m\, (2 - 2^m)\, \frac{\kappa_m}{m!}\, k^m,$$

where $\kappa_m = (m-1)!\, (\mathsf{T}_{m;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$*,* $m \geq 2$*, are the cumulants of* $-\log X$*, the law that opens Chapter G.*

Proof. $\frac{k}{2} P(k) = \Gamma(1+k)^2/\Gamma(1+2k)$ and $\mathbb{E}[X^k] = \Gamma(1+k)$; the cumulant generating function of $-\log X$ is $\log \Gamma(1-k)$, whose coefficients of order $m \geq 2$ are $(\mathsf{T}_{m;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))/m$. The terms of order one cancel, as Proposition 15.3 says they do for every list. □

20.1 Is every ray examinable?

Question 20.7 (Is every ray examinable?)**.** On a ray through the origin with every proper wall positive, do the corners (the Laurent data of Corollary 18.2 at $k = 0, -1, \dots$) and positivity always determine the values, with the masters held in brackets (polytopes, for several masters) whose widths shrink at the rate of Theorem 18.5(iv)?

The only thing that can fail is a native value on the slit, where positivity says nothing. Positivity is a moment problem, and the exam used only its first two conditions.

Hypothesis, as posed: positivity, all of it, examines a ray unless a native value sits on the slit; and a native value sits there only for a reason that can be named.

Proposition 20.8 (Positivity is the moment problem of the stride's law)**.** [derived] *For every polynomial $p \geq 0$ on $[0, f^\star]$ and every $K \geq 1$,*

$$\int_0^{f^\star} t^K p(t)\, \mu(dt) > 0,$$

and in the coordinates of the masters of Theorem 18.5(iv) each such condition is one linear inequality. The conditions with $p = q^2$, $p = t\,q^2$, $p = (f^\star - t)q^2$ and $p = t(f^\star - t)q^2$ generate all of them (alias, last). The examination used two: $p = 1$, giving $P(K) > 0$, and $p = f^\star - t$, giving $P(K) < f^\star P(K-1)$.

Proof. $\mu \geq 0$ is carried by $[0, f^\star]$ (Proposition 20.2(i)); a nonnegative polynomial times t^K integrates to a positive number unless it vanishes. The values are affine in the masters. □

Theorem 20.9 (The rate of positivity)**.** *Upper bound* [derived] *from the expansion of Thm. 35.5; attained* [computed]*. Let every root $\lambda_c \neq f^\star$ of the symbol that carries a local solution not holomorphic at $q_c = 1/\lambda_c$ lie off the segment $[0, f^\star]$. Put $w = 2\lambda/f^\star - 1$ and $e^{g(\lambda)} = |w + \sqrt{w^2 - 1}|$, the branch with $e^g \geq 1$. Then the brackets cut out by the conditions of Proposition 20.8 of degree at most N shrink like $e^{-N \min_c g(\lambda_c)}$ up to powers of N. For a real $\lambda_c > f^\star$, with $\rho = f^\star/\lambda_c$,*

$$e^{-g(\lambda_c)} = \frac{1 - \sqrt{1-\rho}}{1 + \sqrt{1-\rho}} = \frac{\rho}{4} + O(\rho^2),$$

against ρ for the two conditions of §19.

Proof of the upper bound. Take one master, $P = x^{(0)} + M\, x^{(1)}$, and $p = q(t)^2$ with $q(t) = T_m(2t/f^\star - 1)$, T_m the polynomial of degree m with $T_m(\cos\vartheta) = \cos m\vartheta$, so that $0 \leq p \leq 1$ on $[0, f^\star]$. The condition $\int t^K p\, \mu > 0$ reads $A + BM > 0$ with $B = \sum_j p_j\, x^{(1)}(K + j)$, and its error in M is $\int t^K p\, \mu / |B| \leq P(K)/|B|$. By Thm. 35.5(i), p. 308, $x^{(1)}(k) = C\lambda_c^k k^{-(n-1)/2}(1 + O(1/k))$ plus terms of smaller rate, $n = |\mathcal{E}|$, so $B = C\lambda_c^K \int p\, d\nu + \dots$ for a positive measure ν on $[0, \lambda_c]$ with mass near λ_c, and $|B| \gtrsim |T_m(w(\lambda_c))|^2 \lambda_c^K / \mathrm{poly}(m)$. As $|T_m(w)| \sim \frac{1}{2} e^{mg(\lambda)}$ and $2m = N$, the error is e^{-Ng} times powers of N. The side $p = (f^\star - t)q^2$ bounds M from the other side in the same way. With several masters the same polynomials are used one direction at a time. □

Receipts [computed], on the ray of §19: the full conditions of degree at most N, read on $P(1), \dots, P(N+1)$, against the two conditions of §19 at $K = N + 1$ (the same values). The interval is cut out exactly, its ends being the roots of Hankel determinants affine in ℓ:

N	two conditions, $b_K - a_K$	all conditions	ratio of consecutive widths
2	$7.50 \cdot 10^{-2}$	$4.73 \cdot 10^{-2}$	
4	$7.84 \cdot 10^{-4}$	$6.90 \cdot 10^{-5}$	0.0382
8	$1.10 \cdot 10^{-7}$	$9.52 \cdot 10^{-11}$	0.0335
12	$1.64 \cdot 10^{-11}$	$1.04 \cdot 10^{-16}$	0.0320
16	$2.46 \cdot 10^{-15}$	$1.03 \cdot 10^{-22}$	0.0314
22	$4.58 \cdot 10^{-21}$	$9.10 \cdot 10^{-32}$	0.0308

The ratio tends to $e^{-g(1)} = 17 - 12\sqrt{2} = 0.0294\dots$ ($\rho = \frac{1}{9}$), slowly, as the powers of N predict: 1.53 digits of the three-mark centre per value against 0.954. With the nine values of §19 the whole moment problem gives ℓ to 10^{-10} instead of 10^{-7}.

Proposition 20.10 (What positivity cannot see)**.** [derived] *If two solutions of the law with the same corner differ by $u(k) = \int_0^{f^\star} t^k \nu(dt)$ for a signed measure ν with $|\nu| \leq C\mu$, then every condition of Proposition 20.8 holds along $P + \varepsilon u$ for $|\varepsilon| \leq 1/C$: positivity cannot tell them apart. A root λ_c on*

the slit whose local solution is not holomorphic at q_c gives, by Theorem 18.5(iii), one more solution holomorphic off $[1/f^\star,\infty)$ than the count needs; its jump across the slit is its measure ν, singular at λ_c with the exponent $\frac{n-3}{2}$ of a nondegenerate gauge (Thm. 35.11, p. 310, read at that gauge; §20.6), so bounded by a multiple of ϱ near λ_c when $n \geq 4$.

Proof. $\mu \pm \nu/C \geq 0$. For the second statement: Theorem 18.5(ii)–(iii) with a root on the slit leaves the space of solutions holomorphic off the slit of dimension one more than the count; a solution there is a Stieltjes transform of its jump. □

So positivity examines exactly the rays with no seen root in $(0, f^\star)$, and off the segment it does so at the rate of Theorem 20.9, complex roots inside the disc $|\lambda| < f^\star$ included (the two conditions of §19 cannot see those). What remains is whether roots land on the slit.

Proposition 20.11 (Where the slit gets its native values)**.** [derived]

(i) At the centre of a list no native gauge on the torus $|\alpha_e| = 1$, the signs among them, has its native value on the slit: there $K(\alpha^{-1}) = \overline{K(\alpha)}$, so $f(\alpha) = |K(\alpha)|^{-2} \geq K(\mathbf{1})^{-2} = f^\star$, with equality only at $\alpha \propto \mathbf{1}$.

(ii) If $f^\star$ is irrational, each of its conjugates is a native value: the equations of nativity have rational coefficients, $\alpha^\star$ is algebraic, its conjugates are native gauges, and their values are the conjugates of $f^\star$ (so they are roots of every symbol, Prop. 35.7, p. 309). A real conjugate in $(0, f^\star)$ sits on the slit.

At the centre of a list $f^\star = |\mathcal{B}|^{-2}$ is rational; at a lean it need not be.

Survey [computed]: all native gauges (Newton's method in ball arithmetic from many complex starts) of

- the three-mark list $\binom{[3]}{1}$ at the 55 integral directions with entries at most 4 and positive walls: no native value in $|\lambda| \leq f^\star$ other than $f^\star$;
- $\binom{[4]}{1}$ (144 directions) and $\binom{[4]}{2}$ (243), with $s_1, t_1 \leq 3$: none;
- H, the triangle with one mark doubled (§18.4), at the 74 directions with $s_1, t_1 \leq 3$ up to its symmetries: 10 have a native value on the slit, every one of them at a gauge where the doubled class is flipped, $\alpha_a, \alpha_b < 0$, and every one the conjugate of an irrational $f^\star$ (Proposition 20.11(ii)).

Example 20.12 (A ray strictly inside its walls that positivity cannot examine)**.** [derived] where marked; the rest [computed]. Let H carry the class $\{a, b\}$ and the marks $2, 3$, with $A = \alpha_a + \alpha_b$, $B = \alpha_2 + \alpha_3$, $K = AB + \alpha_2\alpha_3$, $\Psi = AB + \alpha_a\alpha_b$, and take the direction $\mathbf{m}_1 = (1, 1, 3, 3)$ on $(a, b, 2, 3)$, $s_1 = 3$, $t_1 = 1$. Its walls are 1 on the marks and on $\{a, b\}$, 3 or 4 on the other pairs, 3 on the triples; $\mathrm{Cor} = \frac{160}{27}$.

(i) [derived] On the slice $\alpha = (x, x, 1, 1)$, fixed by the symmetries $a \leftrightarrow b$, $2 \leftrightarrow 3$, the stride is $f = x/\big((4x+1)^3(x+4)\big)$, critical at $3x^2 + 8x - 1 = 0$, so

$$f^\star = f\Big(\tfrac{\sqrt{19}-4}{3}\Big) = \frac{7643 + 1064\sqrt{19}}{1366875} = 0.0089846\ldots,$$
$$f\Big(\tfrac{-\sqrt{19}-4}{3}\Big) = \frac{7643 - 1064\sqrt{19}}{1366875} = 0.0021985\ldots,$$

the conjugate, on the slit, at the gauge $(-2.786, -2.786, 1, 1)$, the class flipped.

(ii) The closings, $x_j = [\alpha^{j(1,1,3,3)}]K^{3j}\Psi^j = \sum_{b=0}^{j} \binom{3j}{2j+b}\binom{j}{b}\binom{2j-2b}{j-b}^2$ (expand both letters in AB and read the two pairs), are $1, 13, 589, 37597, 2791949, \ldots$; their least recurrence has order 4 and

degree 25 (261 closings), and the law has the symbol $(1366875\lambda^2 - 15286\lambda + 27)(\lambda - 1)^2$: $f^\star$, its conjugate, and the double root 1 of the curves of §18.4. The count of Theorem 18.5(iii) is 1 (corner) $+2$ (at most, at $\lambda = 1$) < 4: law, corner and positivity leave one direction, the conjugate's. The procedure of Theorem 18.5(iv) diverges: at $K = 40, 80, 120$ it returns $P(1) \approx -1.6 \cdot 10^{22}, -4.5 \cdot 10^{46}, -1.3 \cdot 10^{71}$.

(iii) [derived] On its share side, splitting both letters (Cor. 4.4, p. 26) and integrating the two pairs (two Beta corners, the scale between them a pin),

$$\Gamma(3k)\Gamma(k)\,P(k) = \frac{1}{2\pi i}\int_{(c)} \frac{\Gamma(v)^6\,\Gamma(3k-v)\,\Gamma(k-v)}{\Gamma(2v)^2}\,dv, \qquad 0 < c < k,$$

the integrand of Prop. 18.11 for the centre of H with the strengths of the two letters pulled apart. In ball arithmetic [computed]:

$$P(1) = 0.48830319489526888388541464396010900874354031922042897524562\ldots,$$
$$P(2) = 0.00132119179498773222717641714721510065075969610885135498475\ldots,$$

with $P(3)$, $P(4)$, $P(5)$ to the same precision; the law holds at $k = 1$ to 60 digits.

(iv) The second corner. The coefficients c_0 and c_1 share the factor $k+1$, and the constant term at $k = -1$ is rational:

$$\operatorname*{CT}_{k=-1}\big(c_0(k)P(k) + c_1(k)P(k+1)\big) = 2077616120857361856,$$

that is $12610378582200\,P(1) - 133936231480200\,P(2) + 160181043750000\,P(3) = 5981712159277$ [computed] (30 digits). With it the count is $1 + 1 + 2 = 4$, and law, corner, second corner and positivity read the values: from $K = 20$ on they agree with (iii) to all 60 digits [computed].

So for the inputs of §19, law, corner and positivity, the answer is no, and the reason has a name: positivity loses one direction exactly when a seen root sits on the slit; the roots that can sit there are not at the torus, and the first ones met are the conjugates of an irrational $f^\star$. In the example that direction is fixed by a second corner (§20.4). The ray itself is examinable. From its gauge end, with the law and the Hessian at $\alpha^\star$ and neither corner nor positivity, it is read to 68 digits, both corners coming out as outputs (Theorem 21.1, Example 21.7). The "no" belongs to the inputs, not to the ray.

20.2 How much of a period does its corner know?

Question 20.13 (How much of a period does its corner know?)**.** Bound $P(1)$ on every ray above and below by multiples of Cor read off the law at $k = 0$ and $f(\alpha^\star)$, and decide how far the corner determines the first value.

The corner plus positivity already gives a two-sided bound: $\mathrm{Cor}/3 < \ell < \mathrm{Cor}/2$ *for three marks. At* $(\mathbf{1}; 0, 2)$ *on a graph,* $P(1)$ *is the period of the graph (the note, Corollary 5.5, p. 35), and the corner is the volume already observed to track that period (alias, last). The law explains why: it ties* $k \to 0$ *to* $k = 1$ *with rational coefficients.*

Proposition 20.14 (A one-master ray is its corner times a continued fraction of its law)**.** [derived] *Let the law have order* 2, $c_0(k) = a\,k^{n-1} + O(k^n)$ *(Corollary 18.2). Then*

$$P(1) = \mathrm{Cor} \cdot \frac{-a}{c_1(0) + c_2(0)\rho_1}, \qquad \rho_k = \frac{P(k+1)}{P(k)} = \frac{-c_0(k)}{c_1(k) + c_2(k)\,\rho_{k+1}},$$

the second the ratio of the minimal solution (Prop. 35.7), a continued fraction in the coefficients of the law. Since $0 < \rho_1 < f^\star$ *(Proposition 20.8 with* $p = 1$ *and* $p = f^\star - t$*),* $P(1)/\mathrm{Cor}$ *lies between* $-a/c_1(0)$ *and* $-a/\big(c_1(0) + f^\star c_2(0)\big)$ *whenever both are positive. For three marks these are* $\frac{1}{3}$ *and* $\frac{1}{2}$*.*

Proof. The corner relation (Corollary 18.2) is $a\,\mathrm{Cor} + c_1(0)P(1) + c_2(0)P(2) = 0$; divide by $P(1)$. The ratio ρ_k of the solution selected by positivity satisfies the law divided by $P(k+1)$. □

With more masters the corner fixes the scale and the law fixes the ratio: the selection of Theorem 18.5(iv) is homogeneous, so $P(1)/\mathrm{Cor}$ is a function of the law alone [derived]. How much of the period the corner knows is the first bracket; what the law adds is that ratio.

Theorem 20.15 (The fixture at $(\mathbf{1}; 0, 2)$, read off its ray)**.** *The law* [computed]*; the reading* [computed]*; the corner exact. Let the fixture be the six-mark list* $\mathrm{M}(K_4)$ *(Ledger, O19), at the direction* $(\mathbf{1}; 0, 2)$*:* $f = \alpha^{\mathbf{1}}/\Psi^2$*,* $P(1)$ *its full cobasis lean at unit masses.*

(i) Its walls are $\kappa_{1,S} = 2r(S) - |S| > 0$ *on every nonempty proper* S*, and* $\mathrm{Cor} = 84$*.*

(ii) The closings $x_j = [\alpha^{j\mathbf{1}}]\Psi^{2j}$ *are* $1, 12, 756, 78960, 10451700, 1587790512, 263964176784, \ldots$*, and on* 31 *of them (*29 *equations,* 18 *unknowns, one solution)*

$$(j+2)^5 x_{j+2} - 2(2j+3)(65j^4 + 390j^3 + 885j^2 + 900j + 346)\, x_{j+1} + 16(j+1)(2j+1)(2j+3)(4j+3)(4j+5)\, x_j = 0.$$

Reflected (Corollary 18.2), the law of the ray is

$$-k^5 P(k) + 2(2k+1)(65k^4 + 130k^3 + 105k^2 + 40k + 6)\, P(k+1) - 16(k+1)(2k+1)(2k+3)(4k+3)(4k+5)\, P(k+2) = 0,$$

of symbol $-(1-4\lambda)(1-256\lambda)$*:* $f^\star = \frac{1}{256}$ *and one other native value,* $\frac{1}{4}$*.*

(iii) The corner relation is $-84 + 12P(1) - 720P(2) = 0$*, so* $P(2) = \frac{P(1)-7}{60}$ *and every value lies in* $\mathbb{Q} + \mathbb{Q}P(1)$*. The first bracket of Proposition 20.14 is*

$$\frac{\mathrm{Cor}}{12} = 7 < P(1) < \frac{64}{7} = \frac{16\,\mathrm{Cor}}{147},$$

and the brackets of §19 shrink by $\frac{1}{64}$ *per step (by Theorem 20.9, by* 0.0039 *with all conditions). At* $K = 60$ *the reading gives*

$$P(1) = 7.212341418957565712398428969068699944589917754042993290753 63\ldots,$$

which is $6\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ *to all* 60 *digits* [recalled].

So at $(\mathbf{1}; 0, 2)$ on the fixture the corner, the volume of the tropical stride (Prop. 18.1), knows the period of the graph within the factor $\frac{64}{49}$ before positivity is used past its first step, and the law makes it exact. The ratios $P(1)/\mathrm{Cor}$ met in this entry:

ray	Cor	$P(1)$	$P(1)/\mathrm{Cor}$	law
three-mark centre	6	$2.343907238689\ldots$	0.3907	order 2
fixture, $(\mathbf{1}; 0, 2)$	84	$7.212341418958\ldots$	$0.0859 = \frac{1}{14}(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$	order 2
N, $(\mathbf{1}; 0, 2)$ (§20.7)	66	$4.756802563758\ldots$	0.0721	order 6
N, centre (§20.7)	63	$4.578912478882\ldots$	0.0727	order 11
V, $(\mathbf{1}; 0, 2)$ (§20.7)	300	7.12 ± 0.01 (check)	0.0237	[not yet computed]

20.3 What do the slopes carry?

Question 20.16 (What do the slopes carry?)**.** What numbers are the derivatives of a ray in its strength, the $P^{(j)}(i)$ at the integers and the coefficients of $k^{n-1}P(k)$ at the corner, and how do they enter the content of larger lists?

The law and the corner tie the derivatives together, $3P'(1) - 9P'(2) = -4\ell - 12$*, with* $P'(1) = -8.0751522635316\ldots$*, not in* $\mathbb{Q} + \mathbb{Q}\ell$ *below* 10^{18}*. In §17 slopes of a smaller list became masters of* $\binom{[5]}{2}$*.*

Throughout this subsection $P(k) = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ at $k\mathbf{1}$, $\ell = P(1)$, and $\ell^{(j)} = P^{(j)}(1)$.

Proposition 20.17 (The slopes are examinable by positivity, one corner per order)**.** [derived] *Write* $k^2P(k) = \sum_{j\geq 0} a_j k^j$ *near* $k = 0$*, the corners of every order (*$a_0 = \mathrm{Cor} = 6$*,* $a_1 = 0$ *by Prop. 15.3).*

(i) Differentiating the law (L) of §19 j *times and letting* $k \to 0$ *gives one relation among* $P^{(i)}(1)$*,* $P^{(i)}(2)$*,* $i \leq j$*, and* a_j*; at* $j = 1$ *it is the relation of the question, at* $j = 2$

$$\tfrac{3}{2}P''(1) - \tfrac{9}{2}P''(2) = a_2 - 10P(1) - 10P'(1) + 9P(2) + 18P'(2).$$

(ii) $(-1)^j P^{(j)}(K) = \int t^K \log^j(1/t)\,\mu(dt)$*: positive, and at most* $f^\star$ *times its value at* $K - 1$*. The* j*-th slopes obey the law with an inhomogeneity built from lower slopes, so* $P^{(j)}(K)$ *is a rational combination of* $\ell^{(j)}$ *and of lower data, with the coefficient* $c(K)$ *of* $\ell^{(j)}$ *the solution of §19 with* $c(1) = 1$*,* $c(2) = \frac{1}{3}$*.*

(iii) So with a_j *known, positivity examines* $\ell^{(j)}$ *exactly as it examined* ℓ*, at the same rate. Each order of slope costs one corner, and the first slope costs none. From the gauge end no order costs a corner (Proposition 21.9).*

Proof. (i) Taylor-expand (L) at $k = 0$ to order j. (ii) Differentiate $\int f^k\,dy$ under the integral, $0 < f \leq f^\star$. The homogeneous part of the differentiated law is (L), and $P^{(j)}(1)$ is the free initial value, the second being fixed by (i). (iii) As Step 9 of §19. □

Proposition 20.18 (The corners of the three-mark centre)**.** *Order* 2 [derived]*; orders* 3 *and* 4 [computed]*.*

$$k^2P(k) = 6 - \Gamma(\tfrac{1}{2})^4\,k^2 + 4\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))\,k^3 + \Big(\tfrac{7}{60}\Gamma(\tfrac{1}{2})^8 + \tfrac{3}{4}\,\ell^2\Big)k^4 + a_5k^5 + O(k^6),$$

with $a_5 = -40.660206658370993534908264126234905408687957 1\ldots$*. The term of order two is the sum over the three pairs of the term of order* d *of the two-mark centre,* $\Gamma\begin{bmatrix} d,\ d \\ 2d \end{bmatrix}_{1}^{2} = \frac{2}{d} - \frac{1}{3}\Gamma(\frac{1}{2})^4 d + O(d^3)$ *(Prop. 17.11).*

Proof of order two. In the sector coordinates of Prop. 18.1 every order π has $\kappa_{1,U_2} = \kappa_{1,U_3} = 1$ and the same sector letter, $f = q_2q_3\varphi$ with $\varphi = \big((1 + q_2 + q_2q_3)(1 + q_3 + q_2q_3)\big)^{-1}$, $\varphi(0) = 1$. Expand $\varphi^k = \sum_m k^m \log^m \varphi/m!$ and split each $\log^m \varphi$ into its value at 0, its restrictions to the two faces $q_3 = 0$ and $q_2 = 0$, and a remainder vanishing on both. In $\int_{[0,1]^2} q_2^{k-1}q_3^{k-1}(\cdot)\,dq$ the value at 0 gives k^{-2}, a face gives k^{-1} times $\int_0^1 \big(g(q) - g(0)\big)\,dq/q$, the remainder $O(1)$. Only $m = 1$ reaches k^0, through the faces, where $\varphi = (1 + q)^{-1}$: each contributes $-\int_0^1 \log(1 + q)\,dq/q$, which is $-\frac{1}{12}\Gamma(\frac{1}{2})^4$ (the finite part at $d = 0$ of the face's own two-mark letter, Def. 1.4, p. 11). Two faces, six orders: $a_2 = -\Gamma(\frac{1}{2})^4$. □

Receipts [computed]. The coefficients a_j by Cauchy's integral on $|k| = 0.42$ of the share side $k^2 P(k) = k^2\, 4^{2-k}\Gamma(k)^{-2} \int_0^\infty x^{2k-1} K_0(x)^3 dx$ (alias), the part on $(0,1)$ summed exactly from the expansion of the integrand at 0: $a_0 = 6$ and $a_1 = 0$ to 10^{-45}, a_2, a_3, a_4 as displayed to 44 digits, the relation for a_4 having coefficients $60, 7, 45$. With a_2 the second slope is examinable, and positivity gives

$$\ell' = -8.07515226353164401026965124146737650134735782164232219222\ldots,$$
$$\ell'' = 31.24865700624850564562564881961470385267 56\ldots,$$

ℓ' to 380 digits ($K = 400$) and ℓ'' to 39; a Cauchy integral of the share side around $k = 1$ (a check) agrees to 14 digits for both. There is no rational relation among ℓ' and 1, ℓ, $\ell \log 2$, $\ell \log 3$, $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\Gamma(\frac{1}{2})^4 \log 2$, $\Gamma(\frac{1}{2})^4 \log 3$, the cubes and mixed cubes of $\log 2, \log 3$, $\Gamma(\frac{1}{2})^6/\sqrt{3}$, $\frac{\Gamma(1/2)^2}{\sqrt{3}}$ times $\log^2 2, \log^2 3, \log 2 \log 3, \log 2, \log 3, 1$, $\Gamma(\frac{1}{2})^4$, $\log 2$, $\log 3$, $\log^2 2$, $\log 2 \log 3$, $\log^2 3$ (24 numbers) with coefficients below 10^{13} (350 digits). With the arc integrals S_3 and S_4 admitted, ℓ' and ℓ'' have relations of small height (Proposition 21.10).

Remark 20.19 (What the slopes carry). (i) New numbers, one per order: ℓ' is free of any corner datum, and for positivity ℓ'' needs a_2, which the two-mark faces supply; from the gauge end neither needs one, and the new numbers are the arc integrals S_3, S_4 (Proposition 21.10). (ii) The next mark: by Prop. 15.19 the four-mark centre is the closing of the three-mark slope input, $P_{\binom{[4]}{1}}(\mathbf{1}; 1, 1) = -\sum_{m \geq 1} P'(m)$, and the five-mark centre that of the four-mark one (Remark 15.20); slopes are how a list's numbers feed the list with one more mark. (iii) Along a hinge: the free slopes of §17, $\Phi_1'(0) = 2\ell - \ell' - \int \log X_{/}\, dy/X_{\backslash}$ and $f_1(1) = 2\ell - \ell' - \int \log X_{H\backslash 1}\, dy/X_{H/1}$, both carry ℓ', with a logarithmic integral of the children beside it; neither integral is a small rational combination of $1, \ell, \ell', (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)), \Gamma(\frac{1}{2})^4, \Gamma(\frac{1}{2})^4 \log 2, \Gamma(\frac{1}{2})^4 \log 3, \ell \log 2, \ell \log 3, \log 2, \log 3, \Gamma(\frac{1}{2})^2/\sqrt{3}$, four at a time, coefficients below 2000 (at the 27 to 37 digits known) [computed]. (iv) The corners carry the faces and, from order four, the masters: a_4 holds ℓ^2. A product of values written linearly in the slopes of order up to four is a relation of degree two, of the kind asked for in the Ledger, O16.

20.4 Are corners the channels of the tree?

Question 20.20 (Are the corners the channels of the tree?)**.** Along a wall, is the inhomogeneity that replaces the corner a value or a slope of the faces, so that every number a smaller list contributes to a larger one enters through a corner of a law; and are the roots of the symbol on the slit exactly the native values of the faces?

Along a wall the squeeze fails exactly in the face's directions, and the ray carries the faces (Thm. 35.15, p. 312). The question is whether every number a smaller list contributes enters as a corner. This turns "the tree in one equation" into a computation.

The corners are read here at every order: $a_j = [k^j]\, k^{n-1} P(k)$ at $k = 0$, and the same at the other poles $k \in -\mathbb{N}$ and, along a hinge, at the integer points of the split. The leading one is Cor (Prop. 18.1); a *second corner* is a constant term at a pole where c_0 and c_1 share a zero (§15.4).

Proposition 20.21 (The corners of a ray are its faces)**.** [derived] *In the sector coordinates of the proof of Prop. 18.1, $f = \prod_{i=2}^n q_i^{\kappa_i} \varphi_\pi(q)$ with $\kappa_i = \kappa_{1,U_i}$ and $\varphi_\pi(0) = 1$. For $A \subseteq \{2, \ldots, n\}$ let Δ_A restrict a function of q to the face $\{q_i = 0,\ i \notin A\}$ and subtract its restrictions to the faces of that face, so that $\Delta_A h$ vanishes wherever some $q_i = 0$, $i \in A$. Then*

$$a_j = \sum_{\pi} \sum_{A} \prod_{i \notin A} \kappa_i^{-1} \sum_{m + |\mathbf{b}| + |A| = j} \frac{1}{m!} \int_{[0,1]^A} \prod_{i \in A} \frac{(\kappa_i \log q_i)^{b_i}}{b_i!\, q_i}\ \Delta_A\big(\log^m \varphi_\pi\big)\, dq_A.$$

So the corner of order j is a sum, over the orders and over the faces of the sector cube of dimension at most j, of integrals of powers of the logarithm of the stride restricted to those faces. Up to order $n-1$ only proper faces enter, since $\Delta_A(\log^0\varphi_\pi)=0$ for $A\neq\varnothing$; the faces of the cube are the strata where a tail U_i is scaled to zero, on which φ_π is the product of the bottom letters of the bunch and its room (Lemma 35.14, p. 311). From order n on the whole cube enters as well.

Proof. Thm. 8.10, p. 56, as in the proof of Prop. 18.1: $P(k)=\sum_\pi\int_{[0,1]^{n-1}}\prod_i q_i^{k\kappa_i-1}\varphi_\pi^k\,dq$. Write $\varphi_\pi^k=\sum_m k^m\log^m\varphi_\pi/m!$, decompose every function of q as $\sum_A\Delta_A$ (inclusion–exclusion over the faces), and integrate: the coordinates outside A give $\prod_{i\notin A}(k\kappa_i)^{-1}$, those in A an integral analytic at $k=0$ whose Taylor coefficients are the logarithmic moments displayed. $\Delta_A(\log^0\varphi)=0$ for $A\neq\varnothing$ and $\log\varphi_\pi(0)=0$, which gives $a_1=0$ for unit weights (Prop. 15.3). □

At three marks this is the computation of Proposition 20.18: a_2 is the two-mark faces, $-\Gamma(\frac12)^4=3\cdot[d]\Gamma_{\binom21}\left[\begin{smallmatrix}d,\ d\\ 2d\end{smallmatrix}\right]$, and $a_3=4(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and a_4 involve the whole cube, the second through ℓ^2. So the corners carry the faces, and from order n on the list itself.

The evidence, three kinds of ray.

(a) *Strictly inside the walls, no root on the slit* (every centre ray of §§15–18). Law, corner and positivity determine the values (Theorem 18.5(iii)); the higher corners are then consequences, through the slope relations of Proposition 20.17. The faces' numbers reach the values as well: $P_{\binom{[3]}{1}}(\mathbf{1};1,1)$ and $\Gamma(\frac12)^4$ are among the masters of the triangle with two marks doubled (Theorem 17.12), selected by positivity, while its two corners in σ are rational (§15.4). On such a ray a face number is not confined to the corners.

(b) *Along walls* (the inputs of §17, rays in the split d between two children). The faces' native values sit on the slit (Prop. 17.3: $\frac13$ and $\frac14$ below $\frac49$), and the faces' numbers are exactly the boundary data of the input's law at the integer points of the split: $\Phi_1(0)=P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, the deletion's centre; $\Phi_1(1)=P_{\binom{[4]}{1}}(\mathbf{1};1,1)$, the contraction's; at the wall $d=2$ the residue $-6=-6\,P_{\binom{[2]}{1}}(\mathbf{1};1,1)^2$ and the finite part $\frac{21}{2}+\frac34 P_{\binom{[4]}{1}}(\mathbf{1};1,1)$ (Prop. 17.6); at zero shares the class's $\Gamma(\frac12)^4$ (Prop. 17.11). Each number a child contributes enters there as a corner of the input.

(c) *Strictly inside the walls, a root on the slit* (Example 20.12). The direction that positivity cannot squeeze is fixed by the second corner, and that corner is rational. From the gauge end no condition is missing, and the second corner is an output (Example 21.7).

Hypothesis 20.22 (Corners are the channels)**.** For every ray of a family, the numbers its faces contribute enter its law only through its corners, at the poles in the strength and, along a hinge, at the integer points of the split; positivity carries them from there into the masters.

What it turns into a computation, node by node of the Foreman tree, leaves first: (1) the law of the node's input in the split (or of its ray); (2) its corners, read from the children by Proposition 20.21; (3) positivity off the slit, or the gauge end (§21), which needs neither (3) nor (4); (4) the corners on the slit; (5) the closing (summation by parts, Prop. 15.23) up to the node's masters. Steps (1), (3), (5) are those of §§15.7 and 17; step (2) is derived above at the strength 0; at the other poles and at the walls of a split it is the same expansion around $q^{\kappa\cdot j}$ (Prop. 18.1) and around the wall's face (Thm. 35.15), carried out at three marks only. That the second corner of Example 20.12 is a sum of face numbers, and which: [not yet computed].

20.5 Is a ray's full set of solutions the coaction made visible?

Question 20.23 (Is the period matrix of a ray the face of the coaction?)**.** Are the solutions of a law selected at its other native values and at the torus, the rest of the period matrix of the ray, built from the open periods of the small blocks that the note's Theorem A attaches to the tree?

The law's other solutions, picked out at the other native values and at the torus, should be built from the small blocks of the note's Theorem A. The exam already found one: the dominant rational solution is the closings.

On the banana the other solutions have integral representations: the lowered settings of Remark 15.8, $x^{(j)}(s) = \int_0^\infty y^{s-1} g^{n-j} h^j \, dy$ with $g = \mathsf{G}_{0;(1,-1)}(1,\cdot)$ and $h(y) = \sum_m y^m/\Gamma(1+m)^2$, the sign classes with j minus signs, of native value $(n-2j)^{-2}$; divided by $\Gamma(s)^2$ they obey the ray law (Theorem 15.25), and their corners vanish (Remark 15.8, computed there at five marks; at three and four marks in the proof below).

Proposition 20.24 (The other solutions of the bananas)**.** *At $k = 1, 2$* [computed]*; at every k* [derived] *from the law and the zero corner. Let u_n be the rational solution of the ray law of $\binom{[n]}{1}$ with zero corner and $u_n(1) = 1$.*

(i) Three marks: $u_3(k) = 9^{1-k}[\alpha^0](K\Psi/\alpha^{\mathbf{1}})^{k-1}$, the closings (§19.3, Step 8), and

$$\frac{x^{(1)}(k)}{\Gamma(k)^2} = \frac{1}{3}\Gamma{}^{2}_{0}\big[\tfrac{1}{3},\ \tfrac{2}{3}\big]\ u_3(k).$$

(ii) Four marks: $u_4(2) = \frac{1}{16}$, and $x^{(1)}(k)/\Gamma(k)^2 = \frac{1}{4}\,\Gamma(\frac{1}{2})^4\, u_4(k)$.

(iii) Five marks: the two sign classes span the zero-corner solutions with the coefficients of Prop. 15.27, the stacked Gamma

$$X = \frac{1}{120}\Gamma{}^{8}_{4}\left[\begin{matrix}\frac{1}{15},\ \frac{2}{15},\ \frac{4}{15},\ \frac{8}{15},\ \frac{1}{5},\ \frac{2}{5},\ \frac{3}{5},\ \frac{4}{5}\\ \frac{1}{2},\ \frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\end{matrix}\right],$$

$\Gamma(\frac{1}{2})^8/X$ and $\sqrt{15}/\Gamma(\frac{1}{2})^2$.

Proof. At three and four marks the space of solutions with zero corner has dimension one (order 2, one corner relation), and $x^{(1)}/\Gamma^2$ lies in it; so it is $x^{(1)}(1)\,u_n$. The values at $k = 1, 2$, by quadrature as a check (18 and 30 digits): $x^{(1)}(1) = \frac{2\pi}{3\sqrt{3}}$ and $x^{(1)}(2) = \frac{1}{3}x^{(1)}(1)$ at three marks; $x^{(1)}(1) = \frac{\pi^2}{4}$ and $x^{(1)}(2) = \frac{\pi^2}{64}$ at four (aliases, last). □

So the frame of the chart and the sign class (Prop. 35.8, p. 309) is a stacked Gamma of a two-mark block times the frame of §19: at three marks

$$P(k)\,\frac{x^{(1)}(k+1)}{\Gamma(k+1)^2} - P(k+1)\,\frac{x^{(1)}(k)}{\Gamma(k)^2} = \frac{1}{3}\Gamma{}^{2}_{0}\big[\tfrac{1}{3},\ \tfrac{2}{3}\big]\cdot\frac{2}{3}\cdot\frac{9^{1-k}}{k^2} \qquad \text{[derived]}.$$

Remark 20.25 (The period matrix of a banana ray). The solutions split into the one that carries the corner, the chart, and those with zero corner. The latter are rational solutions times one number each, so their rows of the period matrix are a Gamma product times a rational row; the chart's row is the only one that mixes. That is the shape a coaction has: the pure part is read off the frames, and the mixed entries sit in one row.

Against Theorem A of the note: Theorem A is about left factors, periods of the same chain against other forms, and the other solutions are other chains. What can be compared is what the

frames are made of. At three and four marks they are Beta corners $\Gamma_{\,1}^{\,2}\left[\begin{smallmatrix} a,\ b \\ a+b \end{smallmatrix}\right]$ of a two-mark block at masses in $\frac{1}{3}\mathbb{Z}$ and $\frac{1}{2}\mathbb{Z}$; at five marks the frame is the stacked Gamma X at fifteenths, which is a product of Beta corners of two-mark blocks at fifteenths and halves (Proposition 20.26). So the hypothesis holds through five marks in the form "the other solutions are rational solutions times algebraic multiples of products of Beta corners of two-mark blocks and their inverses". At the fixture's full lean (Theorem 20.15) the zero-corner solution has $u(2) = \frac{1}{60}$ and the other native value is $\frac{1}{4}$; the number that multiplies u there, by analogy a stacked Gamma of weight three: [not yet computed].

Proposition 20.26 (The five-mark frame is a product of Beta corners)**.** [derived]

$$X = \frac{1}{120}\Gamma_{\,4}^{\,8}\left[\begin{smallmatrix} \frac{1}{15},\ \frac{2}{15},\ \frac{4}{15},\ \frac{8}{15},\ \frac{1}{5},\ \frac{2}{5},\ \frac{3}{5},\ \frac{4}{5} \\ \frac{1}{2},\ \frac{1}{2},\ \frac{1}{2},\ \frac{1}{2} \end{smallmatrix}\right]$$
$$= \frac{1}{15\sqrt{15}}\,\Gamma_{\,1}^{\,2}\left[\begin{smallmatrix} \frac{1}{2},\ \frac{1}{2} \\ 1 \end{smallmatrix}\right]\Gamma_{\,1}^{\,2}\left[\begin{smallmatrix} \frac{1}{15},\ \frac{4}{15} \\ \frac{1}{3} \end{smallmatrix}\right]\Gamma_{\,1}^{\,2}\left[\begin{smallmatrix} \frac{2}{15},\ \frac{8}{15} \\ \frac{2}{3} \end{smallmatrix}\right] = 8.57028044337446126 81\ldots$$

Proof. The reflection formula gives $\Gamma(\frac{1}{5})\Gamma(\frac{4}{5})\Gamma(\frac{2}{5})\Gamma(\frac{3}{5}) = \Gamma(\frac{1}{2})^4/(\sin\frac{\pi}{5}\sin\frac{2\pi}{5}) = 4\Gamma(\frac{1}{2})^4/\sqrt{5}$, so $X = \Gamma(\frac{1}{15})\Gamma(\frac{2}{15})\Gamma(\frac{4}{15})\Gamma(\frac{8}{15})/(30\sqrt{5})$. Also $\Gamma(\frac{1}{3})\Gamma(\frac{2}{3}) = 2\Gamma(\frac{1}{2})^2/\sqrt{3}$. So the right side is $\Gamma(\frac{1}{15})\Gamma(\frac{4}{15})\Gamma(\frac{2}{15})\Gamma(\frac{8}{15})\cdot\sqrt{3}/(2\cdot 15\sqrt{15})$, the same number. □

Remark 20.27 (Balance is not an invariant)*.* "Its top entries sum to 3, its bottom ones to 2" is a property of the written monomial. The reflection formula writes the unbalanced $\Gamma(a)\Gamma(1-a)$ (top 1, bottom 0) as an algebraic multiple of the balanced $\Gamma(\frac{1}{2})^2/\Gamma(1)$, and the proof above is three such moves. A Gamma monomial whose function $F(t) = \sum_a n_a\langle ta\rangle$ on $(\mathbb{Z}/30)^\times$ is constant is algebraic up to a power of $\Gamma(\frac{1}{2})^2$ [recalled]; the receipt finds $F_X = F$ of the product, identically, the same statement as the proposition. So the hypothesis above holds through five marks in its strong form: the coefficients of the zero-corner solutions are algebraic multiples of products of Beta corners of two-mark blocks and their inverses, for instance $\Gamma(\frac{1}{2})^8/X = 15\sqrt{15}\,\Gamma(\frac{1}{2})^6/\Gamma_{\,1}^{\,2}\left[\begin{smallmatrix} \frac{1}{15},\ \frac{4}{15} \\ \frac{1}{3} \end{smallmatrix}\right]\Gamma_{\,1}^{\,2}\left[\begin{smallmatrix} \frac{2}{15},\ \frac{8}{15} \\ \frac{2}{3} \end{smallmatrix}\right]$, at thirds (three marks), halves (four), and fifteenths with halves (five). The fifteenths are the orbit $\{1,2,4,8\}$ of doubling in $(\mathbb{Z}/15)^\times$, as in the top row of X. Which block of the note's Theorem A gives these corners: [not yet computed].

20.6 What is the law of the stride?

Question 20.28 (What is the law of the stride?)**.** Is μ a member of the family, with its density on $(0, f(\alpha^\star))$ a value of the serif family; and what decides its moments when the native values do not?

H and $\binom{[5]}{1}$ share native values but not numbers, and their level sets differ in dimension.

The law is μ of Definition 20.1; its density satisfies the law of the ray read on the other side of the Trailmoment.

Proposition 20.29 (The density's law)**.** [derived] *Let $\sum_{i=0}^{h} c_i(k)P(k+i) = 0$ hold identically, and $\theta = t\,d/dt$. Then on $(0, f^\star)$*

$$\Big(\sum_{i=0}^{h} t^i\, c_i(-1-i-\theta)\Big)\varrho = 0,$$

an equation of order $D = \max_i \deg c_i$ whose leading coefficient is $(-1)^D\chi(t)$, χ the symbol: its singular points in $\mathbb{C}^\times$ are the roots of the symbol, the native values. At $t = 0$ the exponents are those of $c_0(-1-$

θ), *among them* -1 *with multiplicity* $n-1$, *which is the corner,* $\varrho(t) \sim \operatorname{Cor}(-\log t)^{n-2}/\big((n-2)!\,t\big)$. *At a simple root carried by a nondegenerate native gauge the exponents are* $0, 1, \dots, D-2$ *and* $\frac{n-3}{2}$ *(with a logarithm when* n *is odd); a family of native gauges of dimension* d *(Remark 15.16) moves the last one to* $\frac{n-3-d}{2}$.

Proof. $k\,t^k = \theta\,t^k$, and integrating θ by parts against dt gives $-1-\theta$; $c_i(k)t^{k+i} = t^{k+i}c_i(\dots)$ moves t^i through, $\theta t^i = t^i(\theta+i)$. The law holds if and only if $\varrho\mathbf{1}_{(0,f^\star)}$ solves the equation in the sense of distributions (Theorem 21.1(ii)); in particular ϱ solves it on the open interval. The exponents at the native values are those of the branch of Thm. 35.11, read at each gauge; the families are the case of Remark 15.16. □

Receipts [computed]: the exponents, from the laws of §§15–19 and Theorem 20.15.

ray	n	native values	exponents of ϱ beyond $0, \dots, D-2$
$\binom{[3]}{1}$, centre	3	$\frac{1}{9}$, 1	0 with logarithm at both
$\binom{[4]}{1}$, centre	4	$\frac{1}{16}$, $\frac{1}{4}$	$\frac{1}{2}$ at both
$\binom{[5]}{1}$, centre	5	$\frac{1}{25}$, $\frac{1}{9}$, 1	1 with logarithm at all three
H, centre	4	$\frac{1}{25}$, $\frac{1}{9}$, 1 (double)	$\frac{1}{2}$ at $\frac{1}{25}$ and $\frac{1}{9}$; 0 thrice at 1
fixture, $(\mathbf{1}; 0, 2)$	6	$\frac{1}{256}$, $\frac{1}{4}$	$\frac{3}{2}$ at both

At the native value 1 of H the indicial polynomial is $s^3(s-1)(s-2)(s-3)(s-4)$: besides the holomorphic exponents, 0 twice more, the logarithms of the curves of Prop. 18.10.

Remark 20.30 (H against $\binom{[5]}{1}$). The two laws of the stride have the same singular points, $\frac{1}{25}, \frac{1}{9}, 1$, and different exponents at every one of them: $\frac{1}{2}$ against 1 with a logarithm at the gauge and at the sign, and at 1 a triple exponent 0 from the curves of native gauges against a nondegenerate sign class. "The level sets differ in dimension" is the exponent $\frac{n-3}{2}$, the level sets $\{f = t\}$ having dimension $n-2$, two for H and three for the banana; and the curves at 1 are the critical locus of dimension one that the banana does not have. The numbers of a ray are the connection data of this one equation between its ends, the corner at $t = 0$ and the branch at $f^\star$, so the native values alone do not decide them (Remark 18.13): the exponents do their part. Whether one law is a transform of the other is taken up in Remark 20.31 below (Ledger, O8); such a transform would have to turn the exponent $\frac{1}{2}$ at two singular points into 1 with a logarithm while turning a triple 0 at the third into a single one.

Remark 20.31 (What adding a mark accounts for, and what it does not). [derived] from Prop. 20.29 and the display after Prop. 20.2. At a nondegenerate native gauge the exponent beyond $0, \dots, D-2$ is $\frac{n-3}{2}$, with a logarithm when n is odd; at a family of native gauges of dimension d it is $\frac{n-3-d}{2}$. A transform from the law of H ($n = 4$) to that of $\binom{[5]}{1}$ ($n = 5$) adds a mark. At the gauges $\frac{1}{25}$ and $\frac{1}{9}$ it must move $\frac{1}{2}$ to 1 with a logarithm, which is exactly what adding a mark does there. These two mismatches of Remark 20.30 are the signature of such a transform; they rule out only the transforms that keep n, the rational changes of the unknown, which keep exponents modulo $\mathbb{Z}$ and logarithms. At $t = 1$ adding a mark moves the exponent of the curves of native gauges ($d = 1$) from 0 to $\frac{1}{2}$, while $\binom{[5]}{1}$ has 1 with a logarithm, the exponent of an isolated sign class. The curves would have to become a point as well. Adding a mark does not account for that, and this mismatch stands as a constraint on any transform. On bananas the mark-adding transform is at hand. The display after Prop. 20.2 makes $\mathsf{G}_{0;(1,-1)}(1,\cdot)^n$ a μ_n-mixture of dilations. Applying it once more, $\mu_{n+1} = \int \nu_t\,\mu_n(dt)$, with ν_t the stride's law of two marks at coefficients $(1, t)$: splitting the two letters of $\mathsf{G}_{0;(1,-1)}(1,y)\,\mathsf{G}_{0;(1,-1)}(1,y/t)$ at their sum gives $\int_0^1 \mathsf{G}_{0;(1,-1)}\big(1, y(\frac{1}{w} + \frac{1}{t(1-w)})\big)\frac{dw}{w(1-w)}$. Its density has the exponent $-\frac{1}{2}$ at its gauge end (Prop. 20.29 at $n = 2$): a convolution with a square root, which at a nondegenerate gauge turns a sign into a logarithm and a logarithm into a sign.

Whether H and $\binom{[5]}{1}$ are related by such a transform together with one that shrinks the curves at $t = 1$: [not yet computed].

So the answer to the sixth question: the law of the stride is the base measure of the ray as an exponential family (Proposition 20.2(iii)), the input of its Trailmoment on a line (Proposition 20.2(i)), whose Laplace transform is one Gramma (Proposition 20.2(ii)) and whose density obeys the law of the ray read backwards (Proposition 20.29); its singular points are the native values and its exponents the dimensions of the level sets and of the critical loci.

20.7 What does a list without a realization make?

Question 20.32 (What does a list without a realization make?)**.** What are the contents, at the centre and at the full lean $(\mathbf{1}; 0, 2)$, of prime lists that are not lists of spanning trees: the note's list of rank three on six marks whose bases are the triples other than $\{1, 2, 3\}$, and a list on eight marks that is realized over no field?

Two candidates: the note's prime rank-3 list on six marks, and an eight-mark list realizable over no field. The second is balanced, with every wall positive, at $(\mathbf{1}; 0, 2)$.

Nothing in §§15–19 or above uses a realization: the stride, the walls, the corner, the closings, the law and the selection are read off the bases. So both lists make rays, and the question is what the rays hold.

Definition 20.33 (The two lists)**.** N is the list of rank three on [6] whose bases are all triples but $a = \{1, 2, 3\}$ (the note, Remark 5.6, p. 37); put $b = \{4, 5, 6\}$. It is realized by six points of a plane, three of them on a line, and by no graph: contracting the mark 4 and deleting the mark 6 leaves $\binom{\{1,2,3,5\}}{2}$, a minor that the list of a graph never has [recalled]. V is the list of rank four on [8] with the pairs $P_1 = \{1, 2\}$, $P_2 = \{3, 4\}$, $P_3 = \{5, 6\}$, $P_4 = \{7, 8\}$, whose bases are the 65 four-sets other than the five unions $P_i \cup P_j$ with $\{i, j\} \neq \{3, 4\}$; it is realized over no field [recalled].

Proposition 20.34 (The letters and the closings of N)**.** [derived] *Let* $K_a = \alpha_1 + \alpha_2 + \alpha_3$ *and* $\Psi_a = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3$ *be the letters of* $\binom{a}{1}$, $\alpha^a = \alpha_1\alpha_2\alpha_3$, *and the same on* b.

(i) With $Q = \Psi_a K_b + K_a \Psi_b$,

$$K_N = Q + \alpha^b, \qquad \Psi_N = Q + \alpha^a.$$

(ii) For $i \equiv l \pmod 3$ *let* $N(i, l)$ *be the coefficient of* $\alpha^{D\mathbf{1}}$ *in* $\Psi^i K^l$ *on* $\binom{[3]}{1}$, $D = \frac{2i+l}{3}$*: the first closing of the three-mark direction* $(D\mathbf{1}; l, i)$. *Then*

$$N(i, l) = \sum_w \binom{i}{w} \binom{l}{w + \frac{l-i}{3}} \binom{2D - 2w}{D - w},$$

and the closings of N *are*

$$\text{centre:} \quad x_j = [\alpha^{j\mathbf{1}}](K_N \Psi_N)^j = \sum_{p,q=0}^{j} \binom{j}{p} \binom{j}{q} \binom{p+q}{2q-p} N(2q-p, 2p-q)\, N(2p-q, 2q-p),$$

$$(\mathbf{1}; 0, 2): \quad x_j = [\alpha^{j\mathbf{1}}]\Psi_N^{2j} = \sum_{p} \binom{2j}{p} \binom{p}{2p-3j} N(2p-3j, 3j-p)\, N(3j-p, 2p-3j),$$

$1, 19, 1567, 224821, 41485807, \ldots$ *and* $1, 18, 1494, 211860, 38486070, \ldots$.

Proof. (i) A triple of [6] other than a and b meets a in one or two marks, so its monomial is a mark of a times a pair of b or the reverse; b is a basis and a is not. The cobases are the complements of the bases, all triples but b. (ii) In $(K_N\Psi_N)^j$ take u_1,u_2,u_3 factors $\Psi_aK_b, K_a\Psi_b, \alpha^b$ from K_N^j and v_1,v_2,v_3 factors $K_a\Psi_b, \Psi_aK_b, \alpha^a$ from Ψ_N^j. The part on a is $\Psi_a^{u_1+v_2}K_a^{u_2+v_1}(\alpha^a)^{v_3}$ and the part on b is $\Psi_b^{u_2+v_1}K_b^{u_1+v_2}(\alpha^b)^{u_3}$, so the two triples separate and each gives an $N(i,l)$ with $i=u_1+v_2$, $l=u_2+v_1$; with $p=j-u_3$, $q=j-v_3$ the degrees force $i=2q-p$, $l=2p-q$, and the sum over u_1 is Vandermonde's. The lean is the same with one letter. For $N(i,l)$ write $\Psi=\alpha_1\alpha_2+\alpha_3(\alpha_1+\alpha_2)$, $K=\alpha_3+(\alpha_1+\alpha_2)$ and read the power of α_3, then the central binomial coefficient of $\alpha_1+\alpha_2$. □

So the closings of N are bilinear in first closings of the three-mark list along all its directions, one factor per triple: the classes of the lean, joined at the centre by $i\mapsto i+3$ (the note, Remark 5.6). From them the closings cost $O(j^3)$ operations, against the expansion of a power of a letter in five coordinates.

Theorem 20.35 (N at $(\mathbf{1};0,2)$)**.** *The law and the reading* [computed]*; the native gauge* [derived].

(i) Every wall is positive (the least is 1*) and* $\mathrm{Cor}=66$.

(ii) The closings obey a recurrence of order 6 *and degree* 24 *(found on* 220 *closings, verified on all* 221*). The law of the ray has order* 6, $c_0(k)=k^5(k+1)^3(k+2)(k+3)\,r(k)$ *with* r *irreducible of degree* 14*, and symbol*

$$\chi(\lambda)\propto(24\lambda-1)(10800\lambda^2+324\lambda-1)(1728\lambda^3-320\lambda^2+28\lambda+1).$$

(iii) [derived] *On the slice* $\alpha=(x,x,x,1,1,1)$*, the fixed locus of the permutations of* a *and of* b*,* $f=x/(x^2+9x+9)^2$*, critical at* $x^2+3x-3=0$*, so* $\alpha^\star=\big(\frac{\sqrt{21}-3}{2}\mathbf{1}_a,\mathbf{1}_b\big)$ *and*

$$f^\star=\frac{7\sqrt{21}-27}{1800}=0.0028211277\ldots,$$

irrational; its conjugate $-\frac{7\sqrt{21}+27}{1800}$ *is negative. The other native values are* $\frac{1}{24}$ *and the roots of the cubic,* -0.0265231 *and* $0.105854\pm0.103023\,i$ *(modulus* 0.147712*): none lies on* $[0,f^\star]$*, so positivity examines the ray (§20.1), and the selection converges at the rate* $\big(f^\star/0.0265231\big)^K$.

(iv) From $K=60$ *on the reading is stable; at* $K=200$ *it is good to* 189 *digits, the rate being* $10^{-0.973K}$ *(the gauge end gives* 500*, §21.4):*

$$P(1)=4.75680256375822944340803728605708480533517181028275669568732\ldots,$$

and $P(1)/\mathrm{Cor}=0.07207$*. Check: tropical sampling gives* 4.751 ± 0.010.

(v) The relations among $1,P(1),\ldots,P(6)$ *form a lattice of rank* 3*: the values lie in* $\mathbb{Q}+\mathbb{Q}P(1)+\mathbb{Q}P(2)+\mathbb{Q}P(3)$*, three masters against the fixture's one (Theorem 20.15), and the content holds the three-mark centre (Proposition 20.39(i)–(ii)). The constants searched against first, the* numbers of §20.7*, were* 1, $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\Gamma(\frac12)^4$, ℓ, $\Gamma(\frac12)^4\log2$, $\Gamma(\frac12)^4\log3$, $\ell\log2$, $\ell\log3$, Λ_{-7}, Λ_{-8}, Λ_{-24}, Λ_{-84}, $\Gamma(\frac12)^2/\sqrt{3}$, $\log2$, $\log3$*, where* $\ell=P_{\binom{[3]}{1}}(\mathbf{1};1,1)$ *and* $\Lambda_D=\sum_{a=1}^{|D|}\big(\frac{D}{a}\big)(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{a+|D|\mathbb{Z}})((0,\infty))$ *with* $\big(\frac{D}{a}\big)$ *the Kronecker sign, and the seven numbers* 1, $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\Gamma(\frac12)^4$, ε, $\Gamma(\frac12)^4\varepsilon$, ε^2, ε^3 *with their products with* $\sqrt{21}$*, and* $\log2,\log3,\log7$*, where* $\varepsilon=\log\frac{5+\sqrt{21}}{2}$*, the logarithm of the unit of the field of* $f^\star$.

Proof of (iii). On the slice $\Psi_N=x^3+9x^2+9x$ and $\alpha^{\mathbf{1}}=x^3$. A critical point of f on the fixed locus of a group of symmetries of f is a critical point of f; on the orthant there is one (Thm. 35.10(i), p. 309). With $x^2=3-3x$, $x^2+9x+9=6(x+2)$ and $f^\star=x/\big(36(x+2)^2\big)$. □

Proposition 20.36 (N at the centre)**.** *The law of order* 11 [computed] *modulo two primes, that of order* 12 *exactly; the native values* [derived] *where marked.*

(i) Every wall is positive and $\mathrm{Cor} = 63$*;* $f^\star = 19^{-2}$ *at* $\alpha^\star = \mathbf{1}$ [derived] *(Proposition 20.11(i)).*

(ii) The closings (to $j = 800$ *modulo* $2^{61} - 1$*, by Proposition 20.34) obey no recurrence of order at most* 10 *and degree at most* 70*, one of order* 11 *and degree* 52 *(one solution on* 790 *equations), and one of order* 12 *and degree* 31*. The symbol of the law of order* 11 *is, modulo two primes of* 61 *and* 62 *bits with rational reconstruction,*

$$\chi(\lambda) \propto (1 - 361\lambda)(1 - 25\lambda)(1 - 9\lambda)(1 - \lambda)^5(1 + 80\lambda)(361\lambda^2 - 34\lambda - 3).$$

(iii) [derived] *The native values* $\frac{1}{25}, \frac{1}{9}, 1$ *are the signs: at the centre every sign vector* σ *is a native gauge with* $f(\sigma) = K_N(\sigma)^{-2}$*, and* $|K_N(\sigma)|$ *takes the values* $19, 5, 3, 1$ *on* $2, 12, 18, 32$ *sign vectors. The remaining native values are* $-\frac{1}{80}$ *and* $\frac{17 \pm 14\sqrt{7}}{361}$ *(*0.14970 *and* -0.05550*), off the torus.*

(iv) No native value lies on $[0, f^\star]$*, and* $f^\star$ *is simple: positivity examines the ray, with the rate* $(80/361)^K$*.*

(v) [computed] *The law of order* 12 *and degree* 31 *exactly: integer coefficients of* 1305 *to* 1343 *digits, by Chinese remaindering over* 160 *primes of* 62 *bits (stable from* 150*), verified exactly on the* 221 *closings computed exactly and modulo a third prime on* 801*. Its symbol is that of (ii) times* $\lambda - \lambda_s$*, with* $\lambda_s = 0.44992\ldots$ *of height* 2581 *digits, the root that the extra order adds, off the slit;* c_0 *vanishes to order* 5 *at* $k = 0$*, so the corner relation holds.*

(vi) [computed] *Law, corner and positivity read the ray. The readings at* $K = 150, 250, 350$ *differ by* $4.1 \cdot 10^{-91}$ *and* $1.5 \cdot 10^{-156}$*, the rate* $(80/361)^K$*, and to* 220 *digits*

$$P(1) = 4.57891247888240480638020373022855744921869580872760320929687575\ldots,$$

$P(1)/\mathrm{Cor} = 0.07268$*. Check: tropical sampling gives* 4.574 ± 0.008*.*

(vii) The relations among $1, P(1), \ldots, P(12)$ *form a lattice of rank* 4*: eight masters, and the content holds* $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ *and* $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ *(Proposition 20.39(iii)–(iv)); among* $1, P(1), \ldots, P(10)$ *there are two relations, at height* 10^{33}*. There is none of height below* 10^{12} *(*210 *digits) among* $P(1)$ *alone and* 1*,* $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$*,* $\Gamma(\frac{1}{2})^4$*,* ℓ*,* $\Gamma(\frac{1}{2})^4 \log 2$*,* $\Gamma(\frac{1}{2})^4 \log 3$*,* $\Gamma(\frac{1}{2})^4 \log 7$*,* $\ell \log 2$*,* $\ell \log 3$*,* Λ_{-4}*,* Λ_{-7}*,* Λ_{-8}*,* $\sqrt{7}\,\Gamma(\frac{1}{2})^2$*,* $\log 2$*,* $\log 3$*,* $\log 7$*.*

Proposition 20.37 (The eight-mark list V)**.** [derived] *where marked; the rest* [computed]*.*

(i) The walls at $(\mathbf{1}; 0, 2)$*,* $(\mathbf{1}; 1, 1)$ *and* $(\mathbf{1}; 2, 0)$ *are all positive, with values* $1, 2, 3, 4$*, and* $\mathrm{Cor} = 300$*,* $\frac{896}{3}$*,* 300*.*

(ii) [derived] *The cobases are the four-sets other than* $P_k \cup P_l$ *with* $\{k, l\} \neq \{1, 2\}$*. On the slice* $\alpha = (x\mathbf{1}_{P_1 \cup P_2}, \mathbf{1}_{P_3 \cup P_4})$*, the fixed locus of the swaps inside the pairs and of* $P_1 \leftrightarrow P_2$*,* $P_3 \leftrightarrow P_4$*,* $\Psi_V = x(x^3 + 16x^2 + 32x + 16)$ *and* $f = x^2/(x^3 + 16x^2 + 32x + 16)^2$*, critical at*

$$x^3 + 8x^2 - 8 = 0, \qquad f = \frac{x^2}{64\,(x+1)^2(x+3)^2} \text{ there.}$$

The root $x = 0.945667817991$ *gives* $f^\star = 0.000237093045489\ldots$*, of degree three; the conjugates, at* $x = -1.07480315405$ *and* $x = -7.87086466394$*, are* 0.870337682208 *and* 0.000864231320476*, both above* $f^\star$*: off the slit.*

(iii) A census of native gauges (Newton's method from random complex starts: 31 *gauges,* 10 *values, and the slice) finds the native values* $f^\star$*,* 0.000818456*,* 0.000864231*,* 0.004487298*,* -0.013533898*,* $0.007758455 \pm 0.0136\,i$*,* 0.021107646*,* -0.026238829*,* 0.06696851*,* 0.870337682*, none in* $(0, f^\star)$*. It is a lower bound: only a symbol completes it.*

(iv) The closings $x_j = [\alpha^j \mathbf{1}]\Psi_V^{2j}$ *are* $1, 64, 34080, 39638080, 65433978400, 131819906744064, \ldots$ *(*$j \leq 9$*, by the dense expansion modulo primes,* 5 *seconds);* $x_1 = 64$ *counts the cobases whose complement is a cobasis, all but* $P_1 \cup P_2$*. They grow like* $(f^\star)^{-j} = 4217.7^j$*.*

(v) Check: tropical sampling gives $P(1) = 7.124 \pm 0.009$*, so* $P(1)/\mathrm{Cor} \approx 0.0237$*.*

(vi) The law: [not yet computed]*. The dense expansion of* Ψ_V^{2j} *in seven coordinates grows like* j^7 *per closing, and a law of the size of* N*'s needs some hundreds of closings; what* V *lacks is the structure of Proposition 20.34(ii), a splitting of the letters along classes that would make the closings a short sum.*

Remark 20.38 (What the two lists make). (i) Rays like any other. The machinery has no step at which a realization could enter, and none does: N is examinable at both directions, by the symbols, and read at $(\mathbf{1}; 0, 2)$; V has its walls, its corners, its gauge and its first closings, and fails only for want of a law. (ii) What the realization changed was arithmetic, not existence. A list of a graph whose marks are alike has $\alpha^\star = \mathbf{1}$ and a rational $f^\star$ at every balanced direction; N at its lean and V have marks of two kinds, $\alpha^\star$ is irrational, and by Proposition 20.11(ii) the conjugates of $f^\star$ are native values. For both lists they fall off the slit, below 0 for N, above $f^\star$ for V; so the mechanism of Example 20.12 is present but does not bite. (iii) Numbers of its own and of its minors: N's lean has three masters, its content holds $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$, and two masters are new (Proposition 20.39, Remark 20.40). The fixture at the same direction has one master and the number $6(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. So the list without a graph realization carries, at the direction where graphs carry their periods, a ray with a six-dimensional law, a minor's number, and numbers not among the ones the graphs of §§15–19 produce. (iv) At the centre, where N is its own and only factor and an atom of its Foreman tree (the note, Remark 5.6), the law has order 11, eight masters, and native values in $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{7})$; the period of the atom is read off its ray, $4.5789124788\ldots$, and its content holds the three-mark and the four-mark centres, with six masters new. The price is arithmetic: the law that could be made exact has coefficients of 1300 digits, against 33 at the lean.

The values of N read from its gauge end (§21.4: 500 digits at the lean, 520 at the centre) settle its masters and its content.

Proposition 20.39 (The content of N)**.** [computed]*: lattice reduction on the gauge-end values,* 480 *digits at the lean,* 520 *at the centre. Every relation below has residual under* 10^{-570}*; the generic vectors of the same lattices have residuals above* 10^{-480}*.*

(i) At $(\mathbf{1}; 0, 2)$*, the rational relations among* $1, P(1), \ldots, P(6)$ *form a lattice of rank* 3*; the next vectors have height* 10^{113}*, the generic size. The lattice contains the corner relation and the law's constant term at* $k = -1$*, where* c_0 *and* c_1 *share* $(k+1)^3$*:* $\mathrm{CT}_{k=-1}\big(c_0(k)P(k) + c_1(k)P(k+1)\big)$ *is an integer (it is* 3314356313112576 *for the law normalized as the reflection of the primitive recurrence of the closings). The constant terms at* $k = -2$ *and* $k = -3$*, where* c_0 *and* c_1 *share* $k+2$ *and* $k+3$*, are not rational: their relations are not in the lattice. The origin of the third relation:* [not yet computed]*. So the values lie in* $\mathbb{Q} + \mathbb{Q}P(1) + \mathbb{Q}P(2) + \mathbb{Q}P(3)$*: three masters, not five.*

(ii) The content of the lean holds the three-mark centre:

$$3906\,P(1) - 506772\,P(2) + 10837800\,P(3) = 15263 + 1047\,P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1),$$

residual 10^{-662}*; with the positivity values of §20, good to* 189 *digits, the residual is* 10^{-185}*. With all the numbers of §20.7 added, no further relation below heights* 10^{26} *and* 10^{23}*.*

(iii) At the centre, the relations among $1, P(1), \dots, P(12)$ *form a lattice of rank* 4*: the law at* $k = 1$*, the corner, and two more (*c_0 *and* c_1 *share* $(k+1)^2$*,* $k+2$*,* $k+3$*). Among* $1, P(1), \dots, P(10)$ *there are two, among* $1, P(1), \dots, P(9)$ *none below height* 10^{52}*. So eight masters, not ten.*

(iv) The content of the centre holds the three-mark and the four-mark centre: among 1*,* $P(1), \dots, P(8)$*,* $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ *and* $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ *the relations form a lattice of rank* 2*, heights* $1.4 \cdot 10^{25}$ *and* $1.6 \cdot 10^{25}$*. Two further searches, one adding* $\Gamma(\frac{1}{2})^4$*,* $\Gamma(\frac{1}{2})^8$*,* $(P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1))^2$ *and* $\Gamma(\frac{1}{2})^4 P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$*, the other* Λ_{-7}*,* $\sqrt{7}\,\Gamma(\frac{1}{2})^2$*,* $\log 2$*,* $\log 3$ *and* $\log 7$*, find no third relation below height* 10^{32}*.*

Remark 20.40 (Whose numbers)*.* N has the minors $\binom{\{1,2,3,6\}}{1} = N/\{4, 5\}$ and $\binom{\{1,2,3,5\}}{2} = (N/4) \setminus 6$. The four-mark rank-one list is rational at both full leans and $7(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ at its centre (the note, §1 and Example 6.3). The four-mark rank-two list is $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ at its centre (the note, Example 6.2), and also at its full lean [computed]: at $(\mathbf{1}; 0, 2)$ it has a law of order 2, $f^\star = \frac{1}{36}$, Cor $=$ 12, and its value from the gauge end is $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ to $3 \cdot 10^{-308}$. So on N, at both directions, the content holds the same-direction periods of these two minors, and nothing of the four-mark rank-one list where that list is rational; besides them, 2 masters at the lean and 6 at the centre are new. That minors' numbers enter is not a rule: the centre of H has the content $\mathbb{Q} + \mathbb{Q}P_H(\mathbf{1}; 1, 1) + \mathbb{Q}P_H(2\mathbf{1}; 2, 2)$ (§18.4), and there is no relation among 1, those two values and $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ below height 10^{94} (380 digits, the values from the gauge end), though H has the three-mark minors $\binom{[3]}{1}$ and $\binom{[3]}{2}$, whose centres are $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$; with $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and $\Gamma(\frac{1}{2})^4$ added, none below 10^{63}. N is prime (the note, Remark 5.6); H is not, its parallel pair having width one. Which minors' numbers enter a ray's content: [not yet computed] beyond these two lists.

So the answer to the seventh question: a list without a realization makes rays, laws and corners exactly as a graph does, with irrational native gauges at a lean as the one structural difference met. On N it makes the numbers of two of its minors, in the pattern their own full leans and centres dictate, and new ones besides: two at the lean and six at the centre, unrecognized in the bases tried. On V it makes everything but the law.

20.8 Which ray of a content is best?

Question 20.41 (Which ray of a content is best?)**.** Over all rays that carry one content (directions, leans, offsets, other lists with the same content), what is the frontier of the rate of positive selection against the growth of the denominators of the rational solutions, and does any ray reach a gap (Thm. 34.3, p. 300)?

This is the long shot: rate against denominator growth, over every ray that carries $\mathbb{Q} + \mathbb{Q}\ell$*.*

The *content* of a ray is the $\mathbb{Q}$-span of its values (§18.5(ii)). A ray of order two with content $\mathbb{Q} + \mathbb{Q}\theta$ is a sequence of rational approximations to θ, and rays of one content compete.

Proposition 20.42 (The merit of a one-master ray)**.** [derived] *Let the law have order* 2*, let* c *and* r *be its solutions with* $c(1) = 1$ *and zero corner, and with* $r(1) = 0$ *and corner* Cor*, so that* $P = r + P(1)\,c$*; let* D_K *be the least common denominator of* $c(K)$ *and* $r(K)$*. Then* $D_K P(K) = D_K r(K) + P(1)\,D_K c(K)$ *is a linear form in* 1 *and* $P(1)$ *with integer coefficients, positive, of size* $D_K (f^\star)^K$ *up to powers of* K*. So if the* merit

$$-\log f^\star - \limsup_K \tfrac{1}{K} \log D_K$$

is positive, $P(1)$ is irrational.

Proof. $P(K) > 0$ (Proposition 20.8, $p = 1$) and $P(K) = (f^\star)^K$ times a power of K up to a constant (Laplace's method at $\alpha^\star$ on the chart, Thm. 35.5(i), p. 308). If $P(1) = a/b$, then $b\,D_K P(K)$ is a positive integer, which is impossible once it is below 1 [recalled]. □

Proposition 20.43 (Families of rays on a banana)**.** [derived] *from Prop. 18.8. For $\binom{\mathcal{E}}{1}$ and a balanced direction $(\mathbf{m}; s, t)$ put $c_e = m_e - t$, so $\sum_e c_e = s - t$. Then*

$$\Gamma(sk)\Gamma(tk)\,P(k) = \int_0^\infty y^{\frac{(s+t)k}{2}-1} \prod_{e\in\mathcal{E}} \mathsf{G}_{c_e k;(1,-1)}(\sqrt{y}, \sqrt{y})\,dy,$$

which depends on the direction only through $S = s + t$ and the multiset $\{|c_e|\}$. Two directions with the same S and the same $\{|c_e|\}$ are a family*: their values are in the ratio*

$$\frac{P_{(\mathbf{m}';s',t')}(k)}{P_{(\mathbf{m};s,t)}(k)} = \Gamma\begin{bmatrix} s'k,\ t'k \\ Sk \end{bmatrix}^{-1}_{1} \Gamma\begin{bmatrix} sk,\ tk \\ Sk \end{bmatrix}_{1},$$

two Beta corners at the same total, rational at integer k; so a family has one content. The duality $(\mathbf{m}; s, t) \leftrightarrow ((s + t)\mathbf{1} - \mathbf{m}; t, s)$ is $c \mapsto -c$, and the multiples $(c'\mathbf{1}; c', c')$ of the centre are the centre ray read at $c'k$.

Proof. Prop. 18.8 at $(k\mathbf{m}; ks, kt)$ gives $\Gamma(sk)\Gamma(tk)P(k) = \int_0^\infty y^{tk-1} \prod_e \mathsf{G}_{c_e k;(1,-1)}(1, y)\,dy$, and shifting the line of Def. 2.1 by $\frac{1}{2}\log y$ gives $\mathsf{G}_{\nu;(1,-1)}(1, y) = y^{\nu/2}\mathsf{G}_{\nu;(1,-1)}(\sqrt{y}, \sqrt{y})$; the powers of y add up to $y^{tk-1+(s-t)k/2}$. At equal coefficients the Gramma of one mark is even in its index (reflect the line). □

Receipts [computed]: all 123 balanced directions of $\binom{[3]}{1}$ with $1 \le s, t \le 5$, positive walls and entries in decreasing order (the full leans have rational values), each with 70 closings and a search to order 4 and degree 24. A law of order two was found on 17 directions and of order three on 7; on every one of the 17, $|c_e|$ is constant, on every one of the 7 it takes two values, and every direction with $|c_e|$ constant has a law of order two. The 17 fall into nine families, five of them the centre read at $c'k$, all of content $\mathbb{Q} + \mathbb{Q}\ell$, and the merit is constant on each family (growth of D_K between $K = 200$ and $K = 240$):

$\{\lvert c_e\rvert\}$, S	directions $(\mathbf{m}; s, t)$	$f^\star$	other	$-\log f^\star$	growth	merit	$P(1)$
0, $2c'$	$(c'\mathbf{1}; c', c')$, $c' = 1$	1/9	1	2.197	3.742	−1.545	ℓ
	$c' = 2$	1/81	1	4.394	8.605	−4.211	$(\ell - 2)/3$
	$c' = 3$	1/729	1	6.592	12.781	−6.189	$(10\ell - 23)/54$
	$c' = 4$	1/6561	1	8.789	16.906	−8.117	$(62\ell - 145)/486$
	$c' = 5$	1/59049	1	10.986	20.599	−9.613	$(568\ell - 1331)/5832$
1, 5	$(3, 3, 3; 1, 4)$, $(2, 2, 2; 4, 1)$	1/243	−27/64	5.493	10.059	−4.566	$(11 - 4\ell)/54$
	$(4, 2, 2; 2, 3)$, $(3, 3, 1; 3, 2)$	64/6561	−1	4.630	9.196	−4.566	$(11 - 4\ell)/18$
1, 7	$(4, 4, 4; 2, 5)$, $(3, 3, 3; 5, 2)$	1/2187	−1728/3125	7.690	13.384	−5.694	$(19 - 8\ell)/108$
	$(5, 3, 3; 3, 4)$, $(4, 4, 2; 4, 3)$	3125/3779136	−1	7.098	12.792	−5.694	$(19 - 8\ell)/54$
1, 9	$(6, 4, 4; 4, 5)$, $(5, 5, 3; 5, 4)$	1/12500	−1	9.433	16.532	−7.098	$(751 - 320\ell)/2916$
2, 8	$(7, 3, 3; 3, 5)$, $(5, 5, 1; 5, 3)$	823543/553584375	1	6.511	15.251	−8.741	$(80\ell - 181)/648$

The growth column is the quotient $(\log D_{240} - \log D_{200})/40$, not the limit superior of $\frac{1}{K}\log D_K$ that Proposition 20.42 uses; at the centre that quotient moves between 3.74 and 4.34 from window to window, and $\frac{1}{K}\log D_K$ gives the merit −1.9 (Proposition 20.45). The other rows have not been

recomputed. Within a family the values are in the ratio of Proposition 20.43 at every k, exactly, for $k \le 30$:

$$\frac{P_{(4,2,2;2,3)}(k)}{P_{(3,3,3;1,4)}(k)} = \frac{\Gamma(4k)\Gamma(k)}{\Gamma(3k)\Gamma(2k)} = 3, 7, \tfrac{33}{2}, 39, \dots, \qquad \frac{P_{(5,3,3;3,4)}(k)}{P_{(4,4,4;2,5)}(k)} = \frac{\Gamma(5k)\Gamma(2k)}{\Gamma(4k)\Gamma(3k)} = 2, \tfrac{18}{5}, \tfrac{13}{2}, \dots;$$

the native values move by the ratio of the rates, $\frac{64}{27}$ and $\frac{3125}{1728}$, and the merit does not move. The seven rays of order three form three families in the same way, with symbols that agree up to that scaling.

Remark 20.44 (The fixture's content). At $(\mathbf{1}; 0, 2)$ the fixture has content $\mathbb{Q} + \mathbb{Q}(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ (Theorem 20.15(iii)), $-\log f^\star = \log 256 = 5.545$, and D_K grows by 7.6 to 8.4 per step for $120 \le K \le 240$: merit about -2.5 [computed]. Its denominators are $4K$-smooth: at $K = 60$ the primes in $(K, 2K]$ enter squared and those in $(2K, 4K]$ once, and at $K = 200, 240$ the largest primes are 797 and 953. The classical recurrence of order two for the same number has merit $\log(17+12\sqrt{2})-3 = +0.526$ (alias, last) [recalled]. So the ray of the fixture is a different law for that content, and it loses about three per step.

Proposition 20.45 (The centre's rate is spent on the scale of its closings)**.** *(i), (ii)* [derived]*; (iii)* [computed].

(i) At the scale of the closings the law of §19 is the recurrence of the closings:

$$9k^2Q(k) - (10k^2 + 10k + 3)\,Q(k+1) + (k+1)^2Q(k+2) = 0.$$

(ii) The zero-corner solution c of the law ($c(1) = 1$, $c(2) = \frac{1}{3}$) is $c(K) = x_{K-1}/9^{K-1}$, with $x_j = \sum_{a+b+c=j} \left(\frac{j!}{a!\,b!\,c!}\right)^2$ the closings.

(iii) Write $P(K) = r(K) + \ell\, c(K)$ and let D_K be the least common denominator of $r(K)$ and $c(K)$. For $K \le 702$,

$$v_3(D_K) = 2(K-1) - v_3(x_{K-1}), \qquad v_3(x_j) \le 13,$$

the part of D_K prime to 3 divides $\operatorname{lcm}(1, \dots, K)^2$, and $\frac{1}{K}\log D_K$ lies between 4.10 and 4.16 at $K = 200, 300, \dots, 700$.

So the rate $\log 9 = 2.197$ is spent exactly on the 3-part of D_K, up to the bounded $v_3(x_{K-1})$, and the merit is minus the growth of the part prime to 3: about -1.9 (-1.91 to -1.96 at those K). The -1.545 of the table above is the quotient $(\log D_{240} - \log D_{200})/40$; over windows of 40 steps that quotient gives values from -1.55 to -2.15. At the scale where the zero-corner solution is integral, $9^{K-1}P(K)$ decays only like a power of K: the rate of this ray is zero.

Proof. (i) Put $P(k) = 9^{1-k}Q(k)$ in $k^2P(k) - (10k^2 + 10k + 3)P(k+1) + 9(k+1)^2P(k+2) = 0$ and multiply by 9^k. With $Q(k) = x_{k-1}$ it is the closings' recurrence $N^2x_N = (10N^2 - 10N + 3)x_{N-1} - 9(N-1)^2x_{N-2}$ at $N = k+1$. (ii) Both sides solve the law and agree at $K = 1, 2$ ($x_0 = 1$, $x_1 = 3$). □

Remark 20.46 (The merit, sorted). For this ray the negative merit is absolute, a property of its law. Within the families of Prop. 20.43 it is constant (above), so no search there could change it. For the content $\mathbb{Q} + \mathbb{Q}\ell$ the question is whether ℓ is irrational, which is open (alias, last). Relative to the law: Rem. 20.44 has one content on two laws with merits -2.5 and $+0.526$. A ray of positive merit for ℓ needs a zero-corner solution whose denominators grow more slowly than the rate; at the centre they are the rate.

So the answer to the eighth question, on the content $\mathbb{Q}+\mathbb{Q}\ell$: the rays of a banana come in families (Proposition 20.43), the merit is a property of the family, and the best family met is the centre's, read at step one, with merit about -1.9 (-1.545 in the window $200 \le K \le 240$): the ray of the examination. No ray met has positive merit, and at the centre the reason is structural: the rate is spent on the scale of the closings (Proposition 20.45), which are the torus value read backwards (Corollary 22.4(ii)). Whether some ray of $\mathbb{Q}+\mathbb{Q}\ell$ has positive merit is whether ℓ is irrational by this route: open.

21 The gauge end

The examination of §19 read a ray with three inputs, the law, the corner and positivity, and §20 found where that reading stops. This section reads a ray from the other end of its stride's law, the native gauge. There the law and the Hessian at $\alpha^\star$ fix the density, and the density, continued along the open support, is the ray: neither the corner nor positivity enters, and nothing that sits at the native values in between matters. §21.1 states and proves this for every number of marks; §21.2 shows that the germ at the gauge is local data at $\alpha^\star$; §21.3 shows that a law must carry the dimension of its chart at every nondegenerate native value; §21.4 reads every ray met so far with one program; §21.5 reads the slopes and finds the arc integrals in them.

21.1 The gauge end, for every number of marks

Throughout, a ray is strictly inside its walls, $n = |\mathcal{E}|$, the chart has dimension $n-1$, $\alpha^\star$ is the native gauge on the orthant and $f^\star = f(\alpha^\star)$, and H is the Hessian of $-\log f$ at $\log \alpha^\star$ in the coordinates of the chart. The density of the stride's law (Definition 20.1) is ϱ, and $\sum_i c_i(k)P(k+i) = 0$ is a law of the values with density operator $\mathcal{L} = \sum_i t^i\, c_i(-1-i-\theta) = \sum_{j=0}^{D} a_j(t)\,\partial_t^j$, $\theta = t\,d/dt$ (Proposition 20.29), of order $D = \max_i \deg c_i$, whose leading coefficient is $a_D = (-1)^D t^D \chi(t)$, χ the symbol. Put $s = f^\star - t$, $b_j(s) = (-1)^j a_j(f^\star - s)$ (the operator in s), $b_{j,l} = [s^l]b_j$, and $\mathbf{1}_+ = \mathbf{1}_{s>0}$, the chart's side of the gauge.

Theorem 21.1 (The gauge end)**.** [derived]

(i) H is positive definite, ϱ is real-analytic on $(0, f^\star)$, and near $f^\star$

$$\varrho(t) = (f^\star - t)^{\frac{n-3}{2}}\, g(f^\star - t), \qquad g \text{ holomorphic at } 0, \qquad g(0) = \frac{\left(2\Gamma(\frac{1}{2})^2/f^\star\right)^{\frac{n-1}{2}}}{\Gamma\left(\frac{n-1}{2}\right)\sqrt{\det H}}.$$

(ii) The law holds if and only if $\varrho\,\mathbf{1}_{(0,f^\star)}$ solves $\mathcal{L}u = 0$ in the sense of distributions on $(0,\infty)$.

(iii) Let $f^\star$ be a simple root of the symbol. Then $s = 0$ is a regular singular point of $\mathcal{L}$, with exponents $0, 1, \dots, D-2$ and one more, $\mu = D - 1 - b_{D-1,0}/b_{D,1}$, and the local solutions u that extend by zero*, meaning that $u\mathbf{1}_+$ solves $\mathcal{L}$ across $s = 0$, form a line. If $\mu \notin \{0, \dots, D-2\}$ it is spanned by the solution $s^\mu(1 + O(s))$; if $\mu \in \{0, \dots, D-2\}$, by the holomorphic solution that vanishes to order exactly μ, and never by one with a logarithm. The same holds on the other side, with $\mathbf{1}_{s<0}$ and $|s|^\mu$.*

(iv) $\mu = \frac{n-3}{2}$.

(v) The germ of ϱ spans the line of (iii). So ϱ is the continuation along $(0, f^\star)$ of the solution on that line normalized by (i), and

$$P^{(j)}(k) = \int_0^{f^\star} t^k \log^j t\ \varrho(t)\, dt \qquad (j \ge 0).$$

Neither a corner nor positivity enters, nor any condition on the native values in $(0, f^\star)$ or on the parity of n.

Proof. (i) On the chart, $-\log f$ is a sum of logarithms of Laurent polynomials with positive coefficients in e^y, minus a linear form. Its Hessian at any point is a sum of covariances of exponents under positive weights, so it is semidefinite. A null direction v would make $\langle v, \cdot\rangle$ constant on the exponents of each letter, so f would change by an exponential factor along v: unbounded if the factor is not 1, against $f \le f^\star$; constant if it is, against $P(k) < \infty$. So $-\log f$ is strictly convex, $\log \alpha^\star$ is its only critical point (Thm. 35.10(i), p. 309), $-\log f \to \infty$ at infinity of the chart, and every set $\{f \ge t\}$, $t > 0$, is compact. For $0 < t < f^\star$ the level set $\{f = t\}$ is compact and smooth, and $\varrho(t) = \int_{f=t} d\sigma/|\nabla f|$ is real-analytic there. At $\log \alpha^\star$ the Morse lemma gives analytic coordinates w with $\log(f^\star/f) = \frac{1}{2}|w|^2$ and $dy = J(w)\, dw$, $J(0) = (\det H)^{-1/2}$. With $\sigma = \log(f^\star/t)$, $\mu([t, f^\star]) = \int_{|w|^2 \le 2\sigma} J\, dw$; the odd spherical moments of J vanish, so this is $\sigma^{\frac{n-1}{2}}$ times a power series in σ with leading coefficient $J(0)\, 2^{\frac{n-1}{2}}\, \Gamma(\frac{1}{2})^{n-1}/\Gamma(\frac{n+1}{2})$. As σ is holomorphic in $f^\star - t$ with $\sigma = (f^\star - t)/f^\star + \dots$, differentiating in t gives the form and the constant.

(ii) For a distribution T on $(0, \infty)$ carried by $(0, f^\star]$ and of moderate growth at 0, $\int t^k\, \theta T = -(k+1) \int t^k T$ for $\operatorname{Re} k$ large. So $\int t^k\, \mathcal{L}(\varrho \mathbf{1}_{(0,f^\star)}) = \sum_i c_i(k) P(k+i)$ (Prop. 20.2(i)), and a distribution carried by $(0, f^\star]$ is zero exactly when these moments vanish on a half-plane.

(iii) a_D has a simple zero at $f^\star$, so after dividing by a unit $s^{D-1}\mathcal{L} = \sum_{m \ge 0} s^m R_m(\theta_s)$, $\theta_s = s\,\partial_s$, with $R_m(x) = \sum_j b_{j,m-D+1+j}\, x(x-1)\cdots(x-j+1)$, and

$$R_0(x) = x(x-1)\cdots(x-D+2)\,\big(b_{D,1}(x-D+1) + b_{D-1,0}\big).$$

Now let $u = \sum_r u_r s^r$ be holomorphic. The coefficient of s^N in $s^{D-1}\mathcal{L}u$ is $\sum_m R_m(N-m) u_{N-m}$, and for $N \le D-2$ every term vanishes: the term of R_m coming from b_j carries j falling factors and needs $N - m \ge j$, while $m \ge D - 1 - j$. So $u_0, \dots, u_{D-2}$ are free, and from $N = D-1$ on $R_0(N) \ne 0$ fixes the rest. The distribution $\mathcal{L}(u\mathbf{1}_+)$ is carried by $s = 0$ and has order at most $D-2$, $\mathcal{L}(u\mathbf{1}_+) = \sum_{r=0}^{D-2} \gamma_r\, \delta^{(D-2-r)}$. The two leading terms $b_{D,1} s\, \partial^D + b_{D-1,0}\, \partial^{D-1}$ send $s^r \mathbf{1}_+$ to $r!\, b_{D,1}(r-\mu)\, \delta^{(D-2-r)}$, since $\partial^{D-1}(s^r \mathbf{1}_+) = r!\, \delta^{(D-2-r)}$ and $s\, \delta^{(m)} = -m\, \delta^{(m-1)}$; every other term raises the power of s by more than it lowers the order of differentiation, and lands on point masses of lower order. So

$$\gamma_r = r!\, b_{D,1}\, (r - \mu)\, u_r + \sum_{i<r} \kappa_{r,i} u_i,$$

a triangular system with diagonal $r!\, b_{D,1}(r-\mu)$. If $\mu \notin \{0, \dots, D-2\}$ it forces $u = 0$. The solution $s^\mu \Phi$ does extend by zero: $\lambda \mapsto s_+^\lambda$ is a holomorphic family of distributions, the Frobenius family with a parameter gives $\mathcal{L}(s_+^\lambda \Phi_\lambda) = R_0(\lambda)\, v(s)\, s_+^{\lambda - D + 1}$ with v a unit, and at $\lambda = \mu$ the factor R_0 vanishes while $s_+^{\lambda-D+1}$ is finite. If $\mu \in \{0, \dots, D-2\}$, the system forces $u_0 = \dots = u_{\mu-1} = 0$, leaves u_μ free (zero diagonal, vanishing lower terms) and fixes $u_{\mu+1}, \dots, u_{D-2}$: a line. The remaining solution has the form $s^\mu \log s\, (1 + O(s))$ plus a holomorphic one. For the two leading terms, $L_0 = b_{D,1} s\, \partial^D + b_{D-1,0}\, \partial^{D-1}$, one has $L_0(s_+^\lambda) = R_0(\lambda)\, s_+^{\lambda-D+1}$ for every λ, both sides holomorphic families of distributions. At $\lambda = \mu$, R_0 has a double zero and $s_+^{\lambda-D+1}$ a simple pole whose residue is a nonzero multiple of $\delta^{(D-2-\mu)}$; differentiating in λ, $L_0(s_+^\mu \log s_+) = \frac{1}{2} R_0''(\mu) \operatorname{Res} \cdot \delta^{(D-2-\mu)} \ne 0$. The rest of $\mathcal{L}$, and the rest of the logarithmic solution, reach only the levels $r > \mu$. At the levels $r < \mu$ the holomorphic part is forced to vanish, and then at the level $r = \mu$ nothing cancels the logarithm's point mass. The other side is the same computation with $s \mapsto -s$.

The count of the holomorphic solutions used $R_0(N) \ne 0$ for $N \ge D-1$, that is, μ not an integer $\ge D-1$. For a law of the values this always holds. The exponent -1 has multiplicity $n-1$ at $t = 0$ (Prop. 20.29), so $D \ge n-1$, and the germ of ϱ, a local solution by (ii), has the leading exponent

$\frac{n-3}{2} \le D-2$ by (i). For even n that exponent is a half-integer, so it is an indicial root, and μ is the only candidate. For odd n the germ is holomorphic and extends by zero, and if μ were not in $\{0,\dots,D-2\}$ the triangular system would force its Taylor coefficients up to order $D-2$ to vanish, among them $g(0)\neq 0$.

(iv) By (ii), $\varrho\mathbf{1}_{(0,f^\star)}$ solves $\mathcal{L}$ across $f^\star$, so its germ lies on the line of (iii); by (i) its leading exponent is $\frac{n-3}{2}$.

(v) By (iv) the germ of ϱ is on the line, and (i) fixes its scale. By (i) ϱ is analytic on $(0,f^\star)$, so it is the continuation of its germ, and the values and slopes are Prop. 20.2(i) differentiated under the integral, $0<f\le f^\star$. □

Remark 21.2 (The line from the law alone). The line of (iii) can also be reached without looking at exponents, an argument that bounds its dimension when $f^\star$ is a multiple root (Corollary 21.6). Let u be a germ at $f^\star$ that extends by zero, and ψ a cutoff, 1 on $[f^\star-\delta,f^\star]$ and 0 below $f^\star-2\delta$. By the identity in the proof of (ii), $M_u(k)=\int t^k u\,\mathbf{1}_{t<f^\star}\psi\,dt$ satisfies $\sum_i c_i(k)M_u(k+i)=\int t^k\,\mathcal{L}(u\mathbf{1}_{t<f^\star}\psi)\,dt$. The distribution $\mathcal{L}(u\mathbf{1}_{t<f^\star}\psi)$ is carried by $[f^\star-2\delta,f^\star-\delta]$, where ψ is not constant, so this defect is $O\big((f^\star-\delta)^k\big)$ times a power of k. The germ u is a finite sum of $(f^\star-t)^\lambda\log^j(f^\star-t)$ times holomorphic functions. So Laplace's method at the endpoint gives $M_u(k)$ an expansion $(f^\star)^k\sum c\,k^{-\lambda-1-m}\log^{j'}k$, whose leading term is not zero unless u is. That expansion solves the law term by term, because the defect is smaller than every term.

Now count the formal solutions $(f^\star)^kB(k)$. Put $B(k+i)=\sum_r\frac{i^r}{r!}B^{(r)}(k)$ into the law, and write $\chi(\lambda)=\sum_i[k^D]c_i\,\lambda^i$ for the symbol and $\chi_1(\lambda)=\sum_i[k^{D-1}]c_i\,\lambda^i$. The leading order is $\chi(f^\star)=0$. The next is

$$f^\star\chi'(f^\star)\,k\,B'(k)+\chi_1(f^\star)\,B(k)=0.$$

As $\chi'(f^\star)\neq 0$, this forces $B=k^\rho(a_0+a_1k^{-1}+\dots)$ with $\rho=-\chi_1(f^\star)/(f^\star\chi'(f^\star))$ and no logarithm. Each a_j is then fixed by the order at which it enters, with the factor $-jf^\star\chi'(f^\star)\neq 0$. So the expansions of two germs extending by zero are proportional; their difference has the zero expansion, so it is zero.

Remark 21.3 (Where the parity was). Only in how the line looks. For odd n the density is holomorphic at $f^\star$ from the left, $(f^\star-t)^{\frac{n-3}{2}}g$, and cut there: the line of (iii) lies inside the $D-1$ holomorphic solutions, where the exponent cannot single it out, and the logarithm of Proposition 20.29 belongs to the other solution, the one the density avoids. For even n the line is the half-integer class, since finite parts of half-integer powers carry no point masses. One mechanism covers both: the zeros of the indicial polynomial against the poles of $s_+^{\lambda-D+1}$. To compute the line, write $\mathcal{L}(s^\nu)=\sum_d T_d(\nu)s^{\nu+d}$: then T_{1-D} is the indicial polynomial and $T_d(m)=0$ whenever $m\ge 0$ and $m+d\le -1$. For a holomorphic $h=\sum h_m s^m$, $\mathcal{L}(h\,\mathbf{1}_+)$ is a sum of point masses at $s=0$: at $\nu=m$ the zero of T_d meets the simple pole of $s_+^{\nu+d}$ at $\nu+d=-j$, which leaves a point mass of order $j-1$. They vanish exactly when

$$\sum_{m=0}^{D-1-j} h_m\,T'_{-m-j}(m)=0,\qquad j=1,\dots,D-1,$$

linear conditions on the first $D-1$ Taylor coefficients, which with the recurrence of the Taylor coefficients give the germ (Example 21.8). A first proof of (iii) went through finite parts of half-integer powers alone, and so was stated for even n.

So the corners are outputs: the expansion of ϱ at $t=0$, with the exponent -1 of multiplicity $n-1$ (Prop. 20.29), is the connection of the line at $f^\star$ with the solutions at 0, and the corner relations are the law at $k=0,-1,\dots$ evaluated on the values the gauge end produces.

Remark 21.4 (What positivity sees and what it does not). The extra solution of Prop. 20.10 is $u(k) = \int t^k \nu(dt)$, with ν the jump across the slit, carried by $[0, \lambda_c]$ and of the form $(\lambda_c - t)_+^{\frac{n-3}{2}}$ times a holomorphic function at λ_c. Three inputs cannot see it. The law cannot: u solves it, so ν is itself a solution in the sense of distributions (Theorem 21.1(ii) read for ν). The corner cannot: u has none. Positivity cannot: $|\nu| \le C\varrho$ (Prop. 20.10). But $\varrho + \varepsilon\nu$ is not analytic at λ_c. So the inputs of §19, read as conditions on candidates, cut out a cone with a flat direction. The gauge end does not choose among candidates; it constructs ϱ, and analyticity on the open support is as much a property of $\mu = f_* \mathbb{I}_{\mathbb{R}} \varepsilon_{-e_0}$ as $\mu \ge 0$ is. For every ray strictly inside its walls, the "no" of §20.1 belongs to the inputs, not to the ray (with Corollary 21.6 when $f^\star$ is a multiple root).

21.2 The germ at the gauge end is local data at $\alpha^\star$

Theorem 21.1(i) uses one number of the native gauge, $\det H$. The whole germ is local data there.

Proposition 21.5 (The germ at the gauge end is local data at $\alpha^\star$)**.** [derived]*; the check* [computed]*. Expand $h = \log f$ at $\log \alpha^\star$ in the coordinates y of the chart:*

$$h = \log f^\star - \tfrac{1}{2} H_{ab}\, y^a y^b + \tfrac{1}{6} h_{abc}\, y^a y^b y^c + \tfrac{1}{24} h_{abcd}\, y^a y^b y^c y^d + \dots$$

(i) At $\alpha^\star$ each letter $L = \sum_t w_t \alpha^t$ puts the law $\sum_t w_t (\alpha^\star)^t \delta_t / L(\alpha^\star)$ on its exponents. The jet of h is the strengths times the cumulants of these laws: H from the covariances (as in Example 21.7(iii)), $-h_{abc}$ from the third cumulants, $-h_{abcd}$ from the fourth, and so on.

(ii) As $k \to \infty$,

$$P(k) = (f^\star)^k \Big(\frac{2\Gamma(\frac{1}{2})^2}{k} \Big)^{\frac{n-1}{2}} \frac{1}{\sqrt{\det H}} \Big(1 + \frac{c_1}{k} + \frac{c_2}{k^2} + \dots \Big),$$

where, with $G = H^{-1}$,

$$c_1 = \tfrac{1}{8} h_{abcd} G^{ab} G^{cd} + \tfrac{1}{8} h_{abc} h_{def} G^{ab} G^{cd} G^{ef} + \tfrac{1}{12} h_{abc} h_{def} G^{ad} G^{be} G^{cf},$$

and each c_j is a finite expression in the jet.

(iii) The Taylor coefficients of g in Theorem 21.1(i) are read off (ii) one at a time. The term $(f^\star - t)^{\frac{n-3}{2}+m}$ of ϱ contributes $(f^\star)^{k+\frac{n-1}{2}+m} B(k+1, \frac{n-1}{2}+m)$, whose expansion starts at $k^{-\frac{n-1}{2}-m}$ with the coefficient $\Gamma(\frac{n-1}{2}+m) \neq 0$.

Proof. (i) $-h = \sum_L \operatorname{str}_L \log L(e^y) - \langle \mathbf{m}, y \rangle$, and $\log L(\alpha^\star e^y) - \log L(\alpha^\star)$ is the cumulant generating function of the law in (i). (ii) Laplace's method: $e^{k(h - \log f^\star)} = e^{-\frac{k}{2} H_{ab} y^a y^b} (1 + \dots)$; expand the cubic and quartic parts and take Gaussian moments with covariance G/k. The three terms of c_1 are the quartic term and the two ways of pairing two cubic terms. (iii) $B(k+1, b) = \Gamma(b)\, k^{-b} \big(1 + O(k^{-1})\big)$. □

Corollary 21.6 (Theorem 21.1(iii) and (v) without a hypothesis on the symbol)**.** *The germs at $f^\star$ that extend by zero form a space of dimension at most the multiplicity of $f^\star$ as a root of the symbol. The germ of ϱ is the one in it whose expansion is that of Proposition 21.5(ii). When the root is simple the space is a line and $g(0)$ alone fixes the germ; otherwise finitely many c_j do.*

Proof. The argument of Remark 21.2 maps this space injectively into the formal solutions of the law at the rate $f^\star$, of which there are as many independent ones as the multiplicity of the root [recalled]. An injective map from a finite-dimensional space is inverted on its image by finitely many coordinates. □

Check [computed]. Compare c_1 from the cumulants with a_1/a_0 for the formal solution $(f^\star)^k k^\rho(a_0 + a_1 k^{-1} + \dots)$ of the law. For the five-mark centre both are $\frac{1}{2}$, to 30 digits. For Example 20.12 both are $0.33260161198559816746422402865 7\dots$ So the large-k expansion of the law at $f^\star$, the germ of ϱ, and the jet of the stride at $\alpha^\star$ are one object. The density equation is needed only to carry the germ along $(0, f^\star)$. No law met has a multiple root at $f^\star$ (the table of §21.3), so Corollary 21.6 has not yet been needed on an example.

21.3 The law knows the dimension of its chart

Theorem 21.1(iv) is a test that a law guessed from closings must pass: at the native value $f^\star$ the indicial root beyond $0, \dots, D-2$ is $\frac{n-3}{2}$, a number that the closings, integers indexed by one strength, do not contain. [computed]:

ray	n	D	simple roots of the symbol, and μ at each
three-mark centre	3	2	$\frac{1}{9}$, 1: $\mu = 0$
the 24 three-mark rays of §20.8	3	≤ 18	all 55 simple roots: $\mu = 0$
four-mark centre	4	3	$\frac{1}{16}$, $\frac{1}{4}$: $\frac{1}{2}$
H, centre	4	7	$\frac{1}{25}$, $\frac{1}{9}$: $\frac{1}{2}$; 1 double
H at $(1,1,3,3;3,1)$ (Ex. 20.12)	4	25	$f^\star$ and its conjugate on the slit: $\frac{1}{2}$; 1 double
five-mark centre	5	4	$\frac{1}{25}$, $\frac{1}{9}$, 1: 1
fixture, $(\mathbf{1}; 0, 2)$	6	5	$\frac{1}{256}$, $\frac{1}{4}$: $\frac{3}{2}$
N, $(\mathbf{1}; 0, 2)$	6	24	$f^\star$, its negative conjugate, $\frac{1}{24}$, the cubic's three roots: $\frac{3}{2}$
N, centre, law of order 12	6	31	$\frac{1}{361}, \frac{1}{25}, \frac{1}{9}, -\frac{1}{80}, \frac{17\pm14\sqrt{7}}{361}$: $\frac{3}{2}$; 1 five-fold; $\lambda_s = 0.4499215354534\dots$: $16.8575639440931\dots$

Every simple root that is a native value carries $\frac{n-3}{2}$: $f^\star$, the conjugate on the slit, negative and complex native values alike. The multiple roots are the families of native gauges (Remark 15.16). The one root that is not a native value, λ_s, which the left multiple of order 12 adds on N's centre (Prop. 20.36(v)), carries an exponent that is not a half-integer, and the law itself flags it. At $f^\star$ the value is [derived] (Theorem 21.1(iv)); at the other simple roots it is the exponent of the vanishing cycle of Thm. 35.11, p. 310, read at that gauge, and [computed] on every law here. So a law read off closings carries, at every nondegenerate native value, the dimension $n-1$ of the chart; and at $t = 0$ it carries n again, as the multiplicity $n-1$ of the exponent -1 (Prop. 20.29).

21.4 One program, every n

Receipts [computed]. One program reads every ray: law $\to$ density operator $\to$ the line at $f^\star$ (in exact rationals where $f^\star$ is rational) $\to$ its series $\to$ Beta sums at $K, \dots, K+h-1$ $\to$ the law backwards to $k = 1$. Its inputs are the law, $\alpha^\star$ and $\det H$; the corner and positivity are not used.

ray	n	μ	D	$P(1)$, gauge end	against	difference
three-mark centre	3	0	2	$2.34390723868946\dots$	§19	$6 \cdot 10^{-495}$
four-mark centre	4	$\frac{1}{2}$	3	$8.41439832211716\dots$	$7(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$	$2 \cdot 10^{-492}$
H, centre	4	$\frac{1}{2}$	7	$4.09623699833788\dots$	§18.4	all 43 digits
H, $(1,1,3,3;3,1)$	4	$\frac{1}{2}$	25	$0.48830319489527\dots$	Ex. 20.12(iii)	all 59 digits
five-mark centre	5	1	4	$39.9455880119962\dots$	§15	all 50 digits
fixture, $(\mathbf{1}; 0, 2)$	6	$\frac{3}{2}$	5	$7.21234141895757\dots$	$6(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$	$2 \cdot 10^{-489}$
N, $(\mathbf{1}; 0, 2)$	6	$\frac{3}{2}$	24	$4.75680256375823\dots$	Thm. 20.35	$2 \cdot 10^{-189}$
N, centre	6	$\frac{3}{2}$	31	$4.57891247888240\dots$	Prop. 20.36	$5 \cdot 10^{-222}$

The independent values are the closed form of §19; seven and six times $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ [recalled]; the share-side integrals of §18.4 and Ex. 20.12(iii); the value printed in §15; and the positivity readings of §20.

Readings of the same ray from two depths agree to $4\cdot10^{-387}$ (five marks, $K=260$ and 360), $5\cdot10^{-410}$ (H), $7\cdot10^{-504}$ and $3\cdot10^{-542}$ (N). The H row at $(1,1,3,3;3,1)$ passes the native value on the slit, $0.0021985\ldots$, with no special step: the density is analytic there, and the backward run only has to carry the factor $(f^\star/0.0021985)^K$ in its digits. Example 21.7 below lowered the order of the density equation with a second recurrence of order 19; the law's own equation, of order 25, does the same.

Example 21.7 (Example 20.12 read from its gauge end)**.** [computed]; the Hessian [derived]. Take H at $(1,1,3,3;3,1)$, $n=4$, as in Ex. 20.12.

(i) The closings have a recurrence of order 19 and degree 10 besides that of order 4 and degree 25: both exact, by modular nullspaces over 12 and 8 primes of 62 bits, Chinese remaindering and rational reconstruction (coefficients up to 65 and 36 digits), and exact at all 401 and 416 shifts that the 420 computed closings allow. Reflected, the first gives a density equation of order 10 and degree 19 in t that annihilates $\sum_N x_N t^{N-1}$ exactly. Its symbol is

$$(1366875\,t^2-15286\,t+27)(t-1)^2A(t),\qquad \deg A=15,$$

and the symbol of the law of order 4 is the first two factors alone, so A carries no native value.

(ii) Exponents. At $t=0$: -1, $-\frac{2}{3}$, $-\frac{1}{3}$, each three times, and 0 once. At $f^\star$: $0,1,\ldots,8$ and $\frac{1}{2}$. $f^\star$ is a simple root of the symbol, so Theorem 21.1(iii) applies; for this even n the line is the half-integer class, and $\mu=\frac{1}{2}$ as (iv) requires.

(iii) [derived] The Hessian on the chart $\alpha_3=1$, coordinates $(a,b,2)$, is $3\,\mathrm{Cov}_K+\mathrm{Cov}_\Psi$ at $\alpha^\star=(x_0,x_0,1,1)$, $x_0=\frac{\sqrt{19}-4}{3}$:

$$\det H=\frac{376096\sqrt{19}-1614848}{151875}=0.161425231735790355\ldots,\qquad g(0)=\frac{2\Gamma(\frac12)^2\,(2/f^\star)^{3/2}}{\sqrt{\det H}}.$$

(iv) The series $g(s)=\sum_m e_m s^m$ from the recurrence of $\mathcal{L}$ at $f^\star$:

m	$e_m(f^\star)^m$	$e_m(f^\star-f^{\star\prime})^m$
50	$2.4538\cdot10^{5}$	0.19761
400	$3.9255\cdot10^{5}$	$6.9444\cdot10^{-44}$
1600	$5.0797\cdot10^{5}$	$4.9749\cdot10^{-190}$

with $f^{\star\prime}=0.0021985\ldots$ the conjugate on the slit. The first column grows like $\log^2 m$: the radius of g is $f^\star$, the distance to $t=0$, not $f^\star-f^{\star\prime}$. The density continues analytically through the native value on the slit, as Theorem 21.1(i) says it must.

(v) Values. Beta sums $P(K)=\sum_m e_m(f^\star)^{K+m+3/2}B(K+1,m+\frac32)$ at $K=80,\ldots,83$, then the law of order 4 backwards:

$$P(1)=0.48830319489526888388541464396010900874354031922042897524562141926647\ldots,$$
$$P(2)=0.00132119179498773222717641714721510065075969610885135498475458271 59\ldots$$

The readings from $K=80$ and $K=120$ differ by $4\cdot10^{-89}$. The share side of Ex. 20.12(iii), recomputed here independently (the Mellin–Barnes line at $\mathrm{Re}\,v=\frac12$), agrees to all 62 digits it was computed to.

(vi) Both corners as outputs. The law at $k = 0$, not used above, gives

$$\mathrm{Cor} = -\frac{\sum_{i=1}^{4} c_i(0)P(i)}{[k^3]\,c_0} = 5.925925925\ldots = \frac{160}{27}$$

to 40 digits. The second corner of Ex. 20.12(iv),

$$12610378582200\,P(1) - 133936231480200\,P(2) + 160181043750000\,P(3) - 5981712159277,$$

is $-9.2 \cdot 10^{-101}$ from $K = 120$.

What went in: the density equation, $\alpha^\star$, $\det H$. What did not: the corner, the second corner, positivity, the share side. The backward run of the law amplifies the slit direction by $(f^\star/f^{\star\prime})^K = 10^{0.611K}$, which is why Theorem 18.5(iv) diverges on this ray. The gauge end hands the law values with no slit component, so the run only has to carry enough digits.

The same reading on the three-mark centre [computed]: $n = 3$, ϱ holomorphic at $\frac{1}{9}$ with $\varrho(\frac{1}{9}) = 2\Gamma(\frac{1}{2})^2/(f^\star\sqrt{\det H}) = 27\sqrt{3}\,\Gamma(\frac{1}{2})^2$ ($\det H = \frac{4}{27}$, §19), Beta sums at $K = 300$ with 2200 terms, the law of §19 backwards. It gives ℓ to 390 digits, and the corner as an output, $3P(1) - 9P(2) = 6.000\ldots$ to 40 digits. Here $D = 2$, $\mu = 0$, and the line is the holomorphic solution at $\frac{1}{9}$.

Example 21.8 (The five-mark centre read from its gauge end: n odd, μ an integer)**.** The line [derived]; the reading [computed]; the Hessian [derived].

(i) The closings $x_j = \sum_{a_1+\cdots+a_5=j} \left(\frac{j!}{a_1!\cdots a_5!}\right)^2 = 1, 5, 45, 545, 7885, \ldots$ have a least recurrence of order 3 and degree 4, exact at all 157 shifts of 160 closings:

$$(j+3)^4 x_{j+3} - (35j^4 + 350j^3 + 1323j^2 + 2240j + 1433)\,x_{j+2} \\ + (j+2)^2(259j^2 + 1036j + 1062)\,x_{j+1} - 225(j+1)^2(j+2)^2 x_j = 0.$$

Reflected, the density equation has order 4, symbol $(t-1)(9t-1)(25t-1)$ up to sign, and the exponent -1 four times at $t = 0$.

(ii) At $f^\star = \frac{1}{25}$ the indicial polynomial is $384\,\nu(\nu-1)^2(\nu-2)$: $D = 4$, and $\mu = \frac{n-3}{2} = 1$ falls on 1. On the holomorphic solutions u_0, u_1, u_2 are free, a space of dimension 3. The line of Theorem 21.1(iii) has $u_0 = 0$ and, by the one condition at the level $r = 2$, $u_2 = \frac{175}{4}\,u_1$, exactly: the germ vanishes to first order, as Theorem 21.1(i) says it must, and its Taylor coefficients are rational, $u_3 = \frac{48125}{32}u_1$, $u_4 = \frac{35703125}{768}u_1, \ldots$

(iii) The law has the formal solution $25^{-k}k^\rho(1 + \ldots)$ at $\frac{1}{25}$ with $\rho = -2 = -\frac{n-1}{2}$.

(iv) [derived] On the chart $\alpha_5 = 1$, $H = 2(\frac{1}{5}I - \frac{1}{25}J)$ with J the matrix of ones, so $\det H = 2^4/5^5$ (Proposition 15.13), and $u_1 = g(0) = \left(50\,\Gamma(\frac{1}{2})^2\right)^2/\sqrt{\det H} = 15625\sqrt{5}\,\Gamma(\frac{1}{2})^4$.

(v) Beta sums at $K = 150, 151, 152$, then the law backwards:

$$P(1) = 39.945588011996245698944028498423995891849782924769859572366\ldots,$$

the same from $K = 250$ and from $K = 360$. The share side,

$$\Gamma(k)^2 P(k) = \int_0^\infty y^{k-1}\mathsf{G}_{0;(1,-1)}(1,y)^5\,dy$$

(Prop. 18.8, as in the proof of Proposition 20.43), at $k = 1$ by quadrature, agrees to all 59 digits computed, and §15.4 has the value to all 50 of its digits. The corner comes out as an output: $\mathrm{Cor} = -\sum_i c_i(0)P(i)/[k^4]c_0 = 120.000\ldots$

Here the exponent alone could not find the germ; extension by zero did.

21.5 The slopes at the arc

Proposition 21.9 (Slopes from the gauge end)**.** [derived] *Under Theorem 21.1,* $P^{(j)}(K) = \int_0^{f^\star} t^K \log^j t\, \varrho(t)\, dt$ *for every* $j \ge 0$*, with* ϱ *continued along* $(0, f^\star)$*. When* $\sum_m e_m s^m$ *converges on* $[0, f^\star)$ *this is, for* K *large,*

$$P^{(j)}(K) = \sum_{m\ge 0} e_m \,\frac{\partial^j}{\partial a^j}\Big[(f^\star)^{a+m+\frac{n-1}{2}}\, B\big(a+1, m+\tfrac{n-1}{2}\big)\Big]_{a=K}.$$

The law differentiated j *times carries* $P^{(j)}$ *from* K *back to* 1*. No corner datum enters at any order.*

Proof. $\int_0^{f^\star} t^a (f^\star - t)^b\, dt = (f^\star)^{a+b+1} B(a+1, b+1)$, and $\partial_a^j t^a = t^a \log^j t$. The differentiated law is the law with an inhomogeneity built from lower slopes (Prop. 20.17(ii)). □

For the next statement put $P(k) = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ at $k\mathbf{1}$, $\ell^{(j)} = P^{(j)}(1)$, and read the ray at the scale of its closings, $Q(k) = (f^\star)^{1-k} P(k) = 9^{k-1} P(k)$. Write

$$\Lambda_{-3}(s) = \sum_{a=1}^{2} \Big(\frac{-3}{a}\Big)(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{a+3\mathbb{Z}})((0,\infty)), \qquad S_j = -2^{1-j}\int_0^{3/4} \log^{j-1}(4u)\,\frac{du}{\sqrt{u(1-u)}}.$$

Λ_{-3} is the Zamma sum with the Kronecker sign of -3 (as in §20.7). With $B(p, q; x) = \int_0^x u^{p-1}(1-u)^{q-1} du$, the Beta corner cut at x,

$$S_j = -2^{1-j}\,\frac{\partial^{j-1}}{\partial p^{j-1}}\Big[4^{p-\frac12} B\big(p, \tfrac12; \tfrac34\big)\Big]_{p=\frac12},$$

a slope of the cut corner. With $u = \sin^2\frac{\vartheta}{2}$ it is $-\int_0^{2\pi/3} \log^{j-1}(2\sin\frac{\vartheta}{2})\, d\vartheta$, an integral over the arc from 1 to $e^{2\pi i/3}$; we call the S_j the *arc integrals.* So $S_1 = -\frac23\Gamma(\frac12)^2$, and $S_2 = \frac{\sqrt3}{2}\Lambda_{-3}(2)$ [derived] (expand $-\log(2\sin\frac{\vartheta}{2}) = \sum_{m\ge1}\cos(m\vartheta)/m$).

Proposition 21.10 (The first two slopes of the three-mark centre)**.** [computed]

$$Q(1) = \ell = 3\Lambda_{-3}(2) = 2\sqrt3\, S_2,$$
$$Q'(1) = \ell' + 2\ell\log 3 = \tfrac{27}{2}\Lambda_{-3}(3) + 4\sqrt3\, S_3,$$
$$Q''(1) = \ell'' + 4\ell'\log 3 + 4\ell\log^2 3 = -\tfrac13\Gamma(\tfrac12)^4\,\ell - 27\Lambda_{-3}(4) + \tfrac{32}{\sqrt3}\Big(S_4 - \Gamma(\tfrac12)^2\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))\Big),$$

with $\Lambda_{-3}(3) = \frac{4\,\Gamma(1/2)^6}{81\sqrt3}$ [recalled]*, and*

$$S_3 = -2.144767212569494391482642112 22\ldots, \qquad S_4 = 5.95066824413978493075781012901\ldots$$

Receipts. ℓ, ℓ', ℓ'' to 390 digits from the gauge end (Proposition 21.9 at $K = 300$; the slope relation of Prop. 20.17(i), $3P'(1) - 9P'(2) + 4\ell + 12$, is $1.6\cdot10^{-395}$). They agree with §20.3's values to every digit printed there (57 and 42 significant digits); there positivity read ℓ' with no corner and ℓ'' with the corner a_2, here neither used one. S_3 by the trilogarithm formula and by quadrature, S_4 by quadrature, both to 400 digits. Integer relations by reduction of lattices at 370 digits: ℓ' against 29 numbers (the 24 of §20.3 together with $\ell\Gamma(\frac12)^2$, $\ell\Gamma(\frac12)^2/\sqrt3$, $\Gamma(\frac12)^6$, S_3, $S_3/\sqrt3$) returns the relation above with coefficients $-3, -6, 2, 36$ and residual $3\cdot10^{-391}$. $Q''(1)$ against 23 numbers of weight at most four returns coefficients $-3, -1, -96, -81, 96$, residual $2\cdot10^{-390}$; written for ℓ'' itself, with $\ell'\log3$ and $\ell\log^2 3$ moved across, the relation has height 144. Neither S_3 nor S_4 lies in the span of the other numbers: the smallest relations have heights 10^{12} and 10^{14}, the size expected when there is none.

Remark 21.11 (What the slopes carry, sorted). (i) §20.3's negative was a property of the basis: none of its numbers was an arc integral of weight three ($\ell = 2\sqrt{3}\,S_2$ and $\Gamma(\frac12)^2/\sqrt{3} = -\frac{\sqrt{3}}{2}S_1$ were there). Rem. 20.19(i)'s "new numbers, one per order" is borne out, and the numbers have names: order one brings S_3, order two brings S_4, each beside a value $\Lambda_{-3}(j+2)$ of the same weight. (ii) "At the price of one corner" belongs to the positivity reading (Proposition 21.9). (iii) The scale $(f^\star)^{1-k}$ removes every $\log 3$: Q is the ray read where its zero-corner solution is integral (Proposition 20.45). (iv) [derived] from Proposition 21.10: the arc parts through order two are those of one function,

$$Q(1+\varepsilon) = \frac{\sqrt{3}}{2\varepsilon}\Big[B\big(\tfrac12,\tfrac12;\tfrac34\big) - 16^{\varepsilon}B\big(\tfrac12+2\varepsilon,\tfrac12;\tfrac34\big)\Big] + \tfrac{27}{2}\Lambda_{-3}(3)\,\varepsilon \\ - \tfrac12\Big[\Gamma(\tfrac12)^4\Lambda_{-3}(2) + 27\Lambda_{-3}(4) + \tfrac{32}{\sqrt{3}}\Gamma(\tfrac12)^2\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))\Big]\varepsilon^2 + O(\varepsilon^3),$$

where $B(\frac12,\frac12;\frac34) = \frac23\Gamma(\frac12)^2$. The bracket is $\sum_{j\ge1}(4\varepsilon)^j S_{j+1}/j!$, because $16^{\varepsilon}u^{2\varepsilon} = (4u)^{2\varepsilon}$; so the arc part is the cut corner of the definition of S_j, read at $p = \frac12+2\varepsilon$. Why the cut sits at $\frac34$ is answered on the torus side of the same law: the arc runs to the zero $(1, e^{2\pi i/3}, e^{4\pi i/3})$ of the letter $K = \alpha_1+\alpha_2+\alpha_3$ on the torus, where one mark's torus average changes form (Corollary 22.4(iv)). Whether Q near $k=1$ is a cut Beta corner at $\frac34$ plus Gamma corners: [not yet computed].

22 The torus: two chains through one gauge

At the centre of a list the native gauge is $\mathbf{1}$, a point of the orthant and of the torus, and the law of the values is also a law of the torus value. §22.1 sets the two chains side by side: their densities are the two cut solutions on the line of Theorem 21.1(iii), one on each side of the gauge. §22.2 ties them at three marks by a connection with trigonometric coefficients, which reads the corner as the torus's residue at the zeros of the letter, the zero-corner solution as the torus read backwards, and the arc integrals of §21.5 as the torus's log-moments of the letter.

22.1 Two chains through one gauge

At the centre of a list, $\alpha^\star = \mathbf{1}$ lies on the orthant and on the torus $\{|\alpha_e| = 1\}$. Since $\Psi(\alpha) = \alpha^{\mathbf{1}}K(\alpha^{-1})$ for every list, the stride on the torus is $f = |K(\alpha)|^{-2}$, real and at least $f^\star$ (Prop. 20.11(i)).

Definition 22.1 (The torus value)**.** At the centre, the *torus value* is $T(k) = \int f^k \prod_{e\neq e_0}\frac{d\vartheta_e}{2\pi}$ over the torus $\alpha_e = e^{i\vartheta_e}$, $\alpha_{e_0} = 1$, and the *torus law* ν_T is the image of the normalized Haar measure under f, carried by $[f^\star,\infty)$.

Proposition 22.2 (The torus is the second chain)**.** [derived] *At the centre of a list with a law:*

(i) T converges on a half-plane $\operatorname{Re}k < k_T$ with $k_T > 0$, $T(-j) = x_j$ are the closings, and T solves the law there.

(ii) ν_T has a density ϱ_T; $\varrho_T\mathbf{1}_{(f^\star,\infty)}$ solves the density equation across $f^\star$; and near $f^\star$

$$\varrho(t) = g(0)\,(f^\star - t)^{\frac{n-3}{2}}\,\Phi(f^\star - t), \qquad \varrho_T(t) = (2\pi)^{1-n}\,g(0)\,(t-f^\star)^{\frac{n-3}{2}}\,\Phi(f^\star - t),$$

with one real-analytic Φ, $\Phi(0) = 1$: the chart's density and the torus's are the two cut solutions on the line of Theorem 21.1(iii), one on each side of the gauge.

(iii) The corners read the torus: by Prop. 18.1 the pole of P at $k = -j$ has the leading coefficient $\operatorname{Cor}\cdot T(-j)$.

Proof. (i) f^k is bounded on the torus for $\operatorname{Re} k \le 0$, and $|K|^{-2k}$ is integrable near the zeros of K for small $\operatorname{Re} k > 0$. The constant term of $(1/f)^j$ is its torus average. The relations along a ray (Thm. 35.3, p. 308) are exact forms; on the torus, which has no boundary, they integrate to zero wherever f^k is smooth, which is $\operatorname{Re} k \ll 0$, and the law continues to the half-plane. (ii) The argument of Thm. 21.1(ii) on $[f^\star, \infty)$, with Mellin transforms on a left half-plane, and Theorem 21.1(iii) on the side $s < 0$. On the torus near $\mathbf{1}$, $\log f(e^{i\vartheta}) = \log f^\star + \frac{1}{2}\vartheta^{\mathsf{T}} H \vartheta + O(\vartheta^3)$ with the same H, so the Morse constant is that of Thm. 21.1(i) with $d\vartheta/(2\pi)^{n-1}$ in place of dy. (iii) Prop. 18.1 with (i). □

Receipts at three marks [computed] (checks, by quadrature). The torus law of three marks has a closed density (alias, last); at the gauge it gives $\varrho_T(\frac{1}{9}+) = \frac{27\sqrt{3}}{4\Gamma(1/2)^2} = (2\pi)^{-2} \cdot 27\sqrt{3}\,\Gamma(\frac{1}{2})^2 = (2\pi)^{-2}\varrho(\frac{1}{9}-)$. And the moments over $(0, \frac{1}{9})$ of $(2\pi)^2$ times its continuation through $\frac{1}{9}$ are $P(1)$, $P(2)$, $P(3)$ to 19 digits: at three marks the chart's density is the torus's density continued across the gauge.

22.2 Three marks: the connection, and where the arc comes from

Put $S(k) = x^{(1)}(k)/\Gamma(k)^2$, the sign-class solution of Prop. 20.24 (the lowered setting with one minus sign, Remark 15.8).

Theorem 22.3 (The connection of chart, torus and sign class)**.** *(i)–(ii)* [computed]*; (iii)* [derived]*. At the three-mark centre:*

(i) $S(k) = \frac{1}{3}\Gamma(\frac{1}{3})\Gamma(\frac{2}{3})\,9^{1-k}\,T(1-k)$*: the sign class is the torus read backwards.*

(ii) The chart is a combination of the torus at k and at $1-k$:

$$P(k) = 2\,\Gamma(\tfrac{1}{2})^4\Big(T(k) + \frac{\cot(\pi k)}{\sqrt{3}}\,9^{1-k}\,T(1-k)\Big), \qquad \textit{equivalently} \qquad T = \frac{P}{2\Gamma(\frac{1}{2})^4} - \frac{3\cot(\pi k)}{2\Gamma(\frac{1}{2})^2}\,S.$$

(iii) T has a simple pole at $k = 1$ with residue $-1/\big(\sqrt{3}\,\Gamma(\frac{1}{2})^2\big)$. It comes from the two zeros $(1, \omega, \omega^2)$, $(1, \omega^2, \omega)$ of K on the torus, $\omega = e^{2\pi i/3}$, at each of which $|K|^2$ is a quadratic form of determinant $\frac{3}{4}$ in ϑ.

Receipts. The coefficients of $T = \alpha P + \beta S$, solved from k and $k+1$ at $k = 0.1, 0.2, \ldots, 0.9$ (P from the gauge end, T by its one-mark form below, S by its integral; 14 digits): $\alpha = 0.0506605918212 = 1/(2\Gamma(\frac{1}{2})^4)$ at all nine, and $\beta\tan(\pi k) = -0.47746482927 = -3/(2\Gamma(\frac{1}{2})^2)$; (i) at four points to 14 digits. (iii): near a zero, $\int |K|^{-2k}$ over a disc is $\frac{2\pi}{\sqrt{3/4}}\int r^{1-2k}dr$, and the two zeros and the factor $(2\pi)^{-2}$ give the residue.

Corollary 22.4 (What the connection carries)**.** [derived] *from Theorem 22.3; the moments m_j* [computed].

(i) The corner is the torus's residue at the zeros of the letter. *Near $k = 0$, (ii) and (iii) give $\frac{2\Gamma(1/2)^4}{\sqrt{3}} \cdot \frac{1}{\pi k} \cdot 9 \cdot \frac{1}{\sqrt{3}\,\Gamma(1/2)^2 k} = \frac{6}{k^2}$: Cor = 6. At $k = -j$ the same reading with Prop. 18.1 gives $\operatorname{Res}_{k=1+j} T = -\frac{3\sqrt{3}}{\Gamma(1/2)^2}\,9^{-1-j}\,T(-j)$: the torus value's poles carry its own values at the negative integers.*

(ii) The zero-corner solution is the torus read backwards. *$u_3(k) = 9^{1-k}T(1-k)$, which at the integers is $9^{1-k}x_{k-1}$, the solution $c(K)$ of Prop. 20.45. So the rate of the chart is the normalization of a chain that does not decay, and every rational solution of the law is that chain read backwards or carries the corner; this is why the merit of this law is negative.*

(iii) The slopes at $k=1$ are torus data. *Write* $T(-\varepsilon)=\sum_j \frac{(2\varepsilon)^j}{j!}m_j$, $m_j=\int \log^j|K|$ *over the torus, and* $T(1+\varepsilon)=-\frac{1}{\sqrt{3}\Gamma(1/2)^2\varepsilon}+T_0+T_1\varepsilon+\dots$ *Then (ii) at* $k=1+\varepsilon$ *gives*

$$P_{\binom{[3]}{1}}(\mathbf{1};1,1)=2\Gamma(\tfrac12)^4\,T_0+\tfrac{4\Gamma(1/2)^2}{\sqrt{3}}\,(m_1-\log 3),$$
$$P'(1)+2P_{\binom{[3]}{1}}(\mathbf{1};1,1)\log 3=2\Gamma(\tfrac12)^4\,T_1+\tfrac{4\Gamma(1/2)^2}{\sqrt{3}}\,(m_2+\log^2 3)-\tfrac{2\Gamma(1/2)^6}{3\sqrt{3}},$$

with $m_1=\frac{\sqrt{3}}{4\Gamma(1/2)^2}P_{\binom{[3]}{1}}(\mathbf{1};1,1)$ *and* $m_2=\frac{3}{\Gamma(1/2)^2}S_3+\frac{\Gamma(1/2)^4}{4}$ [computed] *(30 digits;* S_3 *the arc integral of §21.5). The first line gives* $T_0=\frac{2\log 3}{\sqrt{3}\,\Gamma(1/2)^2}$, *and the second, with Prop. 21.10, gives* $T_1=\frac{\Gamma(1/2)^2}{6\sqrt{3}}-\frac{2\log^2 3}{\sqrt{3}\,\Gamma(1/2)^2}$; *computed directly from the one-mark form of* T *near* $k=1$ *they agree to 9 and 6 digits (a check). So the term* $4\sqrt{3}\,S_3$ *of Prop. 21.10 is* $\frac{4\Gamma(1/2)^2}{\sqrt{3}}m_2-\frac{\Gamma(1/2)^6}{\sqrt{3}}$, *the torus's second log-moment of the letter, and the rest of that relation is the torus's Laurent data at its pole, made of* $\Gamma(\frac12)^2$, $\sqrt{3}$ *and* $\log 3$.

(iv) Why the cut sits at $\frac34$. *Integrate one mark of the letter over its circle: with* $A=\alpha_1+\alpha_2$,

$$\int_0^{2\pi}|A+e^{i\vartheta_3}|^{-2k}\frac{d\vartheta_3}{2\pi}=\begin{cases}\sum_{m\ge0}\left(\frac{\Gamma(k+m)}{\Gamma(k)\,m!}\right)^2|A|^{2m}, & |A|<1,\\ |A|^{-2k}\sum_{m\ge0}\left(\frac{\Gamma(k+m)}{\Gamma(k)\,m!}\right)^2|A|^{-2m}, & |A|>1,\end{cases}$$

two Beta-type series of one mark. On $\alpha_1=1$, $\alpha_2=e^{i\vartheta}$, $|A|^2=4\cos^2\frac{\vartheta}{2}$, *and the two regimes meet at* $|A|=1$*:* $\vartheta=\frac{2\pi}{3}$, $\sin^2\frac{\vartheta}{2}=\frac34$, *the angle of the zero* $(1,\omega,\omega^2)$ *of* K. *So* $T(k)$ *is an integral over* $\vartheta\in(0,\pi)$ *split at the arc; its log-moments* m_j *are arc integrals (*$m_1=\frac{1}{\pi}\int_0^{2\pi/3}\log(2\cos\frac{\vartheta}{2})\,d\vartheta$*); and its pole at* $k=1$ *sits at the arc's endpoint. The cut Beta corner of Remark 21.11(iv) is this arc.*

So the arc integrals that §21.5 found in the slopes of the chart are the torus side of the same law, and the connection carries them across.

L Ledger

The running state of §§15–22 in one place. L.1 sorts every negative and every hypothesis met; L.2 is what is open, merged from the eight lists of the separate entries; L.3 logs every correction and where it now stands; L.4 gives the old numbers against the new; L.5 says where the receipts are. A later entry updates this ledger rather than adding a list of its own.

L.1 What is absolute and what is relative

One test throughout: change the arena and see whether the negative survives. A negative is *absolute* (a hypothesis *genuine*) when it is a property of the ray or of the number; it is *relative* when it is a property of the inputs read, the basis searched, the route of a proof, or the digits. A hypothesis that only reflects the route of a proof should go when the route changes; one that stays should have a reason that can be said in a sentence.

statement, and where	sorted	what decides it
The negatives of §20		
"The answer to the first question is no" (§20.1, Ex. 20.12)	relative to the inputs	Theorem 21.1 reads every ray strictly inside its walls, for every n (a multiple root at $f^\star$: Corollary 21.6). Example 21.7 reads Ex. 20.12 to 68 digits with both corners as outputs, and §21.4 reads every ray met.
Positivity cannot see a root on the slit (Prop. 20.10)	true; relative	Positivity is $\mu \geq 0$. The slit direction is not analytic at λ_c; the density of μ is analytic on $(0, f^\star)$ (Remark 21.4).
Each order of slope costs a corner (Prop. 20.17(iii), Rem. 20.19(i))	relative to the inputs	From the gauge end no order costs one (Proposition 21.9).
No relation for ℓ' among the numbers of §20.3	relative to the basis	With the arc integrals S_3, S_4 the slopes ℓ', ℓ'' have relations of heights 36 and 144 (Proposition 21.10), and the arc integrals are the torus's log-moments of the letter (Corollary 22.4(iii)).
The five-mark frame is no product of Beta corners (Rem. 20.25, first form)	false	Proposition 20.26. Balance is a property of a written monomial, not of a number (Remark 20.27).
A corner must supply the condition that positivity misses on the slit (Hyp. 20.22, last clause, first form)	not needed	The second corner of Ex. 20.12 is an output of the gauge end (Example 21.7(vi)).
A transform from H to $\binom{[5]}{1}$ must turn $\frac{1}{2}$ into 1 with a logarithm, and a triple 0 into a single one (Rem. 20.30)	partly relative	At $\frac{1}{25}$ and $\frac{1}{9}$ that is what adding a mark does. At $t = 1$ the curve of native gauges of H must also become a point, and that part stands (Remark 20.31).
N: five masters at the lean, ten at the centre, numbers recognized by nothing (Thm. 20.35, Prop. 20.36, first form)	wrong: the digits	The searches read digits the values did not have. From the gauge end: three masters and eight; the contents hold $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$, the centre's also $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ (Proposition 20.39, Remark 20.40). Two masters at the lean and six at the centre are new (L.2, O10).
No law of V (Prop. 20.37)	computational	A law from a certificate or from a splitting of Ψ_V; then Theorem 21.1 reads V.

statement, and where	sorted	what decides it
No ray of positive merit (§20.8)	structural for the ray; open for the content	At the centre the rate $\log 9$ is spent on the 3-part of the denominators, and the merit is about -1.9 (Proposition 20.45); the zero-corner solution is the torus read backwards (Corollary 22.4(ii)). For $\mathbb{Q}+\mathbb{Q}\ell$ the question is whether ℓ is irrational. A merit belongs to a law, not to a content (Rem. 20.44).
The procedure of Thm. 18.5(iv) needs $\lvert\lambda_c\rvert > f^\star$	true; absolute for that procedure	Theorem 21.1 does not use the procedure.
The hypotheses of the readings		
n even (the first proof of the gauge end)	route	Extension by zero separates the germ for every n (Theorem 21.1(iii), Remark 21.3); the five-mark centre is read (Example 21.8).
The exponent at $f^\star$ is a check to be run (Ex. 21.7(ii), first form)	route	It is forced, $\mu = \frac{n-3}{2}$ (Theorem 21.1(iv)), and so a free test of every guessed law at every nondegenerate native value (§21.3).
$f^\star$ a simple root of the symbol	route	The whole germ is local data at $\alpha^\star$ (Proposition 21.5, Corollary 21.6). No law met has a multiple root at $f^\star$.
The series at $f^\star$ converges on $[0, f^\star)$	route	ϱ is analytic on $(0, f^\star)$; a complex singularity near $f^\star$ limits one series, not the continuation.
No root on the slit (§20.1); $\lvert\lambda_c\rvert > f^\star$	route	Those readings see only $\mu \geq 0$, or the large-k expansion truncated, and a solution carried below $f^\star$ is smaller than every term of that expansion. The germ is that expansion summed, and its continuation sees everything.
Rays through the origin	route	The gauge end sees only $\alpha^\star$: for $\int f^k \varphi \, dy$ with $\varphi > 0$ analytic, $g(0)$ gains the factor $\varphi(\alpha^\star)$ and the jet gains $\log \varphi$.
Strictly inside the walls	genuine	On a wall the maximum of f runs off to a face and $P(k)$ diverges; the reading moves to the face. That is the tree, not a defect.
Balance; integral directions	genuine	The chart integral needs degree 0, and the law needs integer shifts of the exponents. With irrational exponents the level sets are not algebraic and there is no law.
A law of the values, certified	genuine	Closings are periods over the torus, values over the orthant; a closing recurrence is a law of the values when the orthant period solves it, which is what a certificate checks (L.2, O1). At a centre the torus value solves the law (Proposition 22.2(i)), which is why closings find it.
Native values on the slit exist	genuine; costless	They are conjugates of an irrational $f^\star$ (Prop. 20.11(ii)).
The merit of the centre is negative	genuine for that ray	The values decay exactly as fast as the closings grow (Proposition 20.45), and the closings are a chain that does not decay (Corollary 22.4(ii)).
The mismatch at $t = 1$ between H and $\binom{[5]}{1}$	genuine for mark-adding	The doubled mark gives H a curve of native gauges; the banana has an isolated sign class (Remark 20.31).

statement, and where	sorted	what decides it
The cut at $\frac{3}{4}$ in the slopes (Rem. 21.11(iv), first form: unexplained)	explained	One mark's torus average changes form at $\lvert A\rvert = 1$, the end of the arc at the zero of the letter (Corollary 22.4(iv)).
N's new masters; Hyp. 20.22 beyond three marks; which minors' numbers enter a content	not yet explained	Questions of basis or structure. The slopes and the torus point to integrals over the zero loci of the letters before more constants of the kinds already tried (L.2, O6, O10).

What stands as absolute: native values on the slit exist, and cost nothing; a law of the values is needed and must be certified, though every guessed law is now tested for free, at every native value, against the dimension of its chart; at $t = 1$, the curve of native gauges of H, which adding a mark does not remove; the walls, where the reading moves to a face; and the arithmetic of a content, which no ray of the families met settles (whether ℓ is irrational is open). Everything else posed as a negative or a hypothesis in §§18–21 was a property of the inputs, the basis, the route of a proof, or the digits.

L.2 Open

Each item ends with the old list items it replaces: 15.8, 16.4, 17.7, 18.6 and 20.9 were the lists "Open" of §§15–18 and of the answers of §20, 19 the list at the end of §19, 21.7 and 22.8 those of §§21–22 as first written.

O1 *Certificates* (Thm. 35.3, p. 308) for the laws read from closings: the banana for every n by an operator proof, which would derive every banana ray law from the stacked Gamma, the law of each list produced by the one before (at each $n \le 6$ the same A_s appeared; from five to six marks the symbol of the five-mark law vanishes at its native value 1, L_5^* raises degrees by less than four, and the block of six marks is built differently), and one certificate with s a variable, as Theorem 15.25 has for the banana; a series root, where the input is a product and the laws have orders 6 and 5; the laws of §§18.3–18.4, among them the lean $(2\mathbf{1}; 4, 1)$ and H's law of order 4; the fixture at $(\mathbf{1}; 0, 2)$, the ray of Example 20.12, N at both directions, and V; the converse of Corollary 18.2 in general. Each with bounds on the pieces down to $k \to 0^+$, as §19, Step 5, has for three marks (the four-mark banana ray needs the same bounds, with its own tropical stride), and with the corner as the boundary term of the certificate at $k = 0$, where $g_e^{(0)}$ does not vanish at infinity, instead of a limit. Certified so far: the lines of Proposition 17.15 and the three-mark law of §19. Theorem 21.1 needs a law of the values; every law guessed from closings passes the test of §21.3. [not yet computed] [15.8(i), 16.4(iii), 17.7(i), 18.6(i), 18.6(vii), 19(i), 19(iii), 20.9(i), 21.7(i), 22.8(i)]

O2 *Certified brackets.* Positivity polytopes for the two masters of H (§18.4) and for the lean $(2\mathbf{1}; 4, 1)$, whose rational solutions have unproved signs; from the gauge end, tail bounds for the series at $f^\star$ (majorant series), which would make every reading of §21.4 a bracket. [not yet computed] [19(i), 19(ii), 21.7(ii)]

O3 *The slit.* Which algebraic native values sit on $(0, f^\star)$ for a ray strictly inside its walls, beyond the conjugates of an irrational $f^\star$ (Proposition 20.11); along walls, whether they are exactly the faces' native values, carrying the faces' masters. That there are none strictly inside walls, the first half of the old 18.6(ii), is false (Example 20.12). For reading a ray the question is moot (Theorem 21.1); it stays a question about native values. [not yet computed] [18.6(ii), 20.9(ii)]

O4 *Rates.* The lower bound of Theorem 20.9 on every ray; it is attained on the three-mark centre. [not yet computed] [20.9(iii)]

O5 *Slopes.* The three-mark centre near $k = 1$: a closed form of $Q(1+\varepsilon)$ (Remark 21.11(iv)), whether it is a cut Beta corner at $\frac{3}{4}$ plus Gamma corners, and the slopes of every order against S_{j+2} and $\Lambda_{-3}(j+2)$; the corner a_5 in the family's terms, and the relation of degree two carried by a_4 (Remark 20.19(iv)). Through order two the slopes are identified (Proposition 21.10) and placed on the torus (Corollary 22.4(iii)). [not yet computed] [20.9(iv), 21.7(iv)]

O6 *Second corners and channels.* The second corners from the list: the boundary term at $\sigma = -1$ as a constant term of the continued period (Thm. 1.6), assembled from its Laurent coefficients of orders $-(n-1)$, $-(n-2)$, $-(n-3)$ at $\sigma = 0$ and $\sigma = -1$: the leading ones are the corner and the corner times $|\mathcal{B}|$ (Proposition 15.4), the next vanishes at 0 (Proposition 15.3) and the ray law fixes it at -1, and on the share side the third ones are Laurent data of the Gammas at the lattice points of the closing, heads of the counting input and Trailmoments of the counting input cut at the data (Prop. 5.13, p. 32; Prop. 5.14, p. 33), so that the rationality of the two boundary terms says that the Trailmoments cancel and the heads remain; which lists put $(\sigma+1)^2$ into both c_0 and c_1 (in §15.4 the two triangles $\binom{[3]}{2}$ with parallel classes, not $\binom{[4]}{2}$ with a mark doubled); the second corner of Example 20.12 as a sum of face numbers; Proposition 20.21 at the other poles and at the walls of a split; Hypothesis 20.22 beyond three marks. Every second corner met is an output of the gauge end (Example 21.7(vi)); what is open is its structure. [not yet computed] [15.8(v), 20.9(v)]

O7 *The coaction.* The block of the note's Theorem A that gives the Beta corners of the five-mark frame (Proposition 20.26, Remark 20.27); the number that multiplies the zero-corner solution at the fixture's full lean (Remark 20.25); the period matrix of the centre of H, that is, the solutions selected at its other native gauges, and whether they are stacked Gammas, as the lowered settings of $\binom{[5]}{1}$ are (Proposition 15.27). [not yet computed] [18.6(vi), 20.9(vi)]

O8 *A transform between H and $\binom{[5]}{1}$.* Whether the two laws are related by adding a mark together with a transform that shrinks the curves of native gauges of H at $t = 1$ to a point; their frames (Prop. 35.8, p. 309). The local exponents are known (Proposition 20.29, Remark 20.30), and adding a mark accounts for $\frac{1}{25}$ and $\frac{1}{9}$ (Remark 20.31). [not yet computed] [18.6(v), 20.9(vii), 21.7(vi)]

O9 *Connections between the chains.* The connection data of the density between $t = 0$ and $t = f^\star$, which carry the numbers (Remark 20.30); off the bananas, what replaces the identity $\mathrm{Cor}\cdot\det H = \prod_{c\neq\star}(1-\lambda^\star/\lambda_c)$ of §18.5(i), which ties the two ends of strength. At three marks Theorem 22.3 gives them, computed; open are a derivation inside the family (the chart, the torus and the sign class are three real forms of one pencil, and the route is the rotation $y \mapsto -y$ of the one-mark Gramma $\mathsf{G}_{0;(1,-1)}(1,y)$ on the share side, Proposition 15.2); the connection at four and five marks, with periodic coefficients, and what replaces the arc at four marks, where K vanishes on the torus along a curve; the second chain at a lean, where the stride on the torus through $\alpha^\star$ is not real and the second real form through the gauge is not a torus. [not yet computed] [20.9(vii), 21.7(iii), 22.8(iii), 22.8(iv), 22.8(vii)]

O10 *N and V.* The law of V, by a splitting of Ψ_V along classes or by a certificate, after which Theorem 21.1 reads it; the law of order 11 of N's centre, exactly; the two new masters at N's lean and the six at its centre, against N's torus (the log-moments of K_N and the Laurent data of its torus value) and the arc integrals of its letters' zeros; the third relation of Proposition 20.39(i);

which minors' numbers enter a ray's content, beyond the two lists of Remark 20.40. [not yet computed] [20.9(viii), 21.7(v), 22.8(v), 22.8(vi)]

O11 *The content of a list.* Whether every ray strictly inside its walls without the list's symmetries spans the whole content (the hypothesis of §18.5(ii)): an offset ray, a ray breaking the symmetries $a \leftrightarrow b$, $2 \leftrightarrow 3$ of H, and an offset ray of N. [not yet computed] [18.6(iii), 20.9(viii)]

O12 *H at its hinge.* The masters of H through its hinge at a mark, as in §§17.1–17.3, with the weights; what the tree expects for H (the end of §18.4). [not yet computed] [18.6(iv)]

O13 *Arithmetic.* A ray of positive merit; a ray of content $\mathbb{Q} + \mathbb{Q}\ell$ whose zero-corner solution is not the closings at the scale $(f^\star)^{k-1}$ (Remark 20.46, Corollary 22.4(ii)); families on lists other than bananas, from a share side as in Proposition 20.43; the denominators of the brackets of the whole moment problem, which are rational at every degree (§20.1); the other rows of the table of §20.8 recomputed with $\frac{1}{K}\log D_K$. [not yet computed] [20.9(ix), 21.7(vii)]

O14 *The three-mark centre inside the family.* Reduce the live Trailmoment of the receipts of Proposition 15.19 at $n = 2$ to the Trailmoment of the odd periodic density modulo 3 at index 2, inside the family; Thm. 5.17 does not reach it, the parity being mismatched. The identity $\ell = 3\Lambda_{-3}(2)$ is computed to 390 digits (Proposition 21.10), and on the torus $m_1 = \frac{\sqrt{3}}{4\Gamma(1/2)^2}\ell$ (Corollary 22.4(iii)). [not yet computed] [15.8(iii)]

O15 *The direction $x-1$ of the block.* $(\mathsf{T}_{0;(1)}[x]\,(x-1)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (q\partial_q)G_0(q)|_{q=1} = \int dy/(X-1)^2 = 0.16106970227667485885\ldots$, which the ray reaches only through the value term of its second master; expanding h as in Proposition 15.22, it is the lowered setting with h replaced by $y\,\partial_y h$, $\int_0^\infty g^4\,y\,h'(y)\,dy$ [derived]. Which member of the family it is, and whether a lean of the five-mark list reads it as a period. [not yet computed] [15.8(ii)]

O16 *Relations of degree two; pairings.* The last minor of §14.17, $M_5(1)x^{(1)}(2) - M_5(2)x^{(1)}(1) = 4.34000293153566\ldots$, is the minimal solution of the exterior square of the ray law, of native value $\frac{1}{225}$ [derived], and it is the masters over the stacked Gamma (Remark 15.29(iii)), so a relation of degree two fixing it would tie the masters to $x^{(1)}(1)$. Whether pairings of the sign classes with closings of the share side twisted by the signs fix such relations, beside the exact pairing of Proposition 15.13, a relation of degree one between solutions fixed by the residue at zero shares; the products of values that the corner a_4 writes linearly in the slopes (Remark 20.19(iv)) are relations of this kind. The pairing for the other lists: the partner of the period in the adjoint law, when the closings are not. [not yet computed] [15.8(vi), 15.8(vii)]

O17 *Lean rays.* At a lean with uniform masses only $\eta = \mathbf{1}$ stays native for $\binom{\mathcal{E}}{1}$; at a sign the native equation reads $(m-t)K(\eta) = (s-t)\eta_e$ for every e [derived]. The gauges that move off the signs, and their values. [not yet computed] [15.8(viii)]

O18 *Families of native gauges.* In both cases of Proposition 15.15 the letter collapses at the gauge to the letter of a minor that leaves directions free: a class whose contraction letter vanishes is free, and a class whose own letter vanishes leaves a deletion with coloops, whose relative scales are free. A rule for every list, and the exponents -3 and $-\frac{5}{2}$ of Remark 15.16. The multiple roots of the symbols met are the families (§21.3), and at a family of dimension d the density has the exponent $\frac{n-3-d}{2}$ (Remark 20.31). [not yet computed] [15.8(ix)]

O19 $\mathrm{M}(K_4)$. The six-mark atom is self-dual only up to a relabelling of its marks, so Cor. 22.8(iii) does not split it (as the handbook notes after Cor. 22.8). Its centre ray, read opened at a mark

into $K_4 \setminus e$ and K_4/e, the latter the triangle with two marks doubled of §17.4. Its full lean $(\mathbf{1}; 0, 2)$ is the fixture of Theorem 20.15. [not yet computed] [15.8(x), 16.4(iv), 17.7(vi)]

O20 *The boundaries of §17.* The finite part $\frac{21}{2} + \frac{3}{4}P_{\binom{[4]}{1}}(\mathbf{1}; 1, 1)$ at the wall $d = 2$ from the collapse of Cor. 8.8 as a regularized lean of the four-mark banana, and the third relation of Proposition 17.7 from the next order; the relation of Proposition 17.11 that carries $\Gamma(\frac{1}{2})^4$, from the next Laurent order at zero shares; Theorem 17.12(i) and (ii) as boundaries, (i) tying the lines 1 and 2 with the coefficient $8 = |\mathcal{B}|$, and which insertion reads it (a mean share as in Proposition 15.24?); the other weights A_s, B_s, R_s, and C_s for $s \geq 2$, of $\binom{[5]}{2}$, as periods of the surface $e_2^2 = e_1 e_3$ at its $\mathbb{Q}(\sqrt{-5})$ point. [not yet computed] [17.7(ii)–(v)]

Closed since they were posed.

- The three masters of $\binom{[5]}{2}$, and the law in m of the ray of $\binom{[4]}{2}$ with its closing: §§17.1–17.3 (Propositions 17.2 and 17.1, Theorem 17.8). [15.8(iv), 16.4(i)]
- Two marks doubled, closed, with its second corner as a boundary: §17.4 (Theorem 17.12, Remark 17.13(ii)). [16.4(ii)]
- Certificates for the lines used in §17 (Proposition 17.15). [17.7(i), first half]
- The local exponents of H and $\binom{[5]}{1}$ (Proposition 20.29, Remark 20.30). [18.6(v), first half]
- A list on six marks that is not a graph, with native gauges off the signs and off the torus: N (§20.7). Its certificate is in O1. [18.6(vii)]
- The eight questions of §20: each is answered at the head of §§20.1–20.8; what stays open of them is above.
- The gauge end for odd $n \geq 5$: Theorem 21.1, Example 21.8. [21.6(iii) of the version of §21 that §22 cites]
- Why the arc ends where K vanishes on the torus: Corollary 22.4(iv). [21.7(iii), first half]
- A multiple root at $f^\star$: Corollary 21.6; no example met. [22.8(ii)]
- The masters of N: three at the lean and eight at the centre, not five and ten (Proposition 20.39). [20.9(viii), in part]

L.3 Corrections

Every correction made by a later entry, with the entry that made it. All are applied in place in this file; the old wording survives only here.

C1 [17.6(i)] §15.4, the masters column, counts over $\mathbb{Q}$. Over $\mathbb{Q} + \mathbb{Q}P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ the list $\binom{[5]}{2}$ has two masters, and over $\mathbb{Q} + \mathbb{Q}P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) + \mathbb{Q}\Gamma(\frac{1}{2})^4$ the triangle with two marks doubled has one (Theorems 17.8 and 17.12).

C2 [17.6(ii)] Remark 16.9: the three-mark centre enters every row of §15.4 but the banana, row four included, through the contraction of H at its single mark; not only the rows whose trees carry a three-mark piece.

C3 [17.6(iii)] Remark 16.8: two of the three masters of $\binom{[5]}{2}$ are slope Trailmoments of the deletion's ray, and the third is $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$; not all three.

C4 [19.7(i)] The receipts of Proposition 17.15: Thm. 35.3 as stated gives the relations for large k; down to $k \to 0^+$ it needs bounds on the pieces, which §19, Step 5, supplies for three marks. Not "applies as stated".

C5 [19.7(ii)] Remark 17.17: the ratios $-r/c$ are lower bounds at every q, since $c(q) > 0$ at every q (§19, Step 8), not only at the q computed.

C6 [19.7(iii)] Theorem 18.5: a root of the symbol need not be a native value (the extra root $\frac{131003}{4771109}$ of §18.4). Its (ii) speaks of roots, and the count of one condition in (iii) is for a simple root that is the value of a nondegenerate native gauge.

C7 [20.10(i)] The open item 18.6(ii), first half ("no native value in $(0, f^\star)$ strictly inside the walls"): false; Example 20.12 has one, the conjugate of an irrational $f^\star$. Now O3.

C8 [20.10(ii)] Remark 18.7 stands as written; a sentence pointing to Example 20.12 and to §21 is added.

C9 [20.10(iii)] Theorem 18.5(iv) needs $|\lambda_c| > f^\star$ for every root $\lambda_c \neq f^\star$ that carries a local solution not holomorphic at q_c, not only λ_c off the slit; added to its hypotheses. The whole moment problem needs only $\lambda_c \notin [0, f^\star]$ (Theorem 20.9).

C10 [20.10(iv)] The open items 18.6(vii) and 18.6(v) are answered in part: by N (§20.7) and by the local exponents (Proposition 20.29). The rest is in O1 and O8.

C11 [21.8(i)] Remark 20.25: "which is not balanced and is no product of Beta corners [derived]" is false (Proposition 20.26, Remark 20.27, which now follow it in §20.5).

C12 [21.8(ii)] §20.1, last paragraph: "So the answer to the first question is no" holds for the inputs of §19 only (Theorem 21.1).

C13 [21.8(iii)] Hypothesis 20.22: its last clause ("wherever a seen root sits on the slit a corner of that ray supplies the missing condition") is not needed, and is removed (Example 21.7).

C14 [21.8(iv)] Proposition 20.17(iii) and Remark 20.19(i): "each order of slope costs one corner" holds for positivity only (Proposition 21.9).

C15 [21.8(v)] Proposition 20.29, proof: "the boundary terms at 0 and $f^\star$ cancel by themselves" is replaced by Theorem 21.1(ii).

C16 [21.8(vi)] §20.8, the table: the growth column is the quotient $(\log D_{240} - \log D_{200})/40$, not $\limsup \frac{1}{K} \log D_K$; at the centre the merit is about -1.9, not -1.545 (Proposition 20.45). The other rows are in O13.

C17 [21.8(vii)] §20.3: the list announced as 25 numbers has 24.

C18 [22.9(i)] Theorem 20.35(iv): the positivity reading at $K = 200$ is good to 189 digits, not 400; its rate is $10^{-0.973K}$.

C19 [22.9(ii)] Theorem 20.35(v) and Remark 20.38(iii): "five masters, independent over $\mathbb{Q}$ with 1 as far as height 10^{64}" is false; there are three, and the content holds $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ (Proposition 20.39(i)–(ii)).

C20 [22.9(iii)] Proposition 20.36(vii) and Remark 20.38(iv): "ten masters" is false; there are eight, and the content holds $P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0, \infty))$ (Proposition 20.39(iii)–(iv)). "No relation among $1, P(1), \ldots, P(10)$ below 10^{18}" was true and too weak: two exist at height 10^{33}.

C21 [22.9(iv)] §20.7, the answer to the seventh question: revised.

C22 [22.9(v)] The gauge end "for even n" (its first form): the parity belongs to that proof; Theorem 21.1 covers every n (Remark 21.3).

C23 [22.9(vi)] Example 21.7(ii) and the first sorting of §21: the exponent at $f^\star$ is forced (Theorem 21.1(iv)); it is not a check.

C24 [22.9(vii)] Remark 21.11(iv): "why the cut sits at $\frac{3}{4}$: [not yet computed]" is answered by Corollary 22.4(iv).

C25 [this file] Theorem 21.1(iii), proof: the count of holomorphic solutions assumed that μ is not an integer $\geq D-1$. That this holds for every law of the values is now proved, in the last paragraph of that proof.

C26 [this file] Remarks 16.8 and 16.9: three tags [not yet computed] that §17 answered are replaced by the answers (Propositions 17.2 and 17.1; Theorem 17.12; Remark 17.13(ii)). §19.6: the rate of the polytope now points to Theorem 20.9.

C27 [this file] Remarks 20.25 and 20.30: a tag [not yet computed] each, restated by the remarks that now follow them (Remarks 20.27 and 20.31), is folded into those.

L.4 Old numbers and new

Numbers 15.x–19.x are unchanged. The entries that carried 20.x–22.x are merged and renumbered; their numbered statements are listed here by their printed numbers in the entry where they first appeared. *Questions* is the entry of 23 October ending the file of §§15–20; *Answers* is the entry of 23 October answering it; §21 is the entry of 24 October in its last form, with the numbers of the earlier form that §22 cites in brackets where they differ; §22 is the entry of 24 October. A statement merged into another points to the one that stands.

old number → number in this file					
Questions (§20 of 23 October)					
Prop. 20.1	→ 20.3	Prop. 20.2	→ 20.4	Prop. 20.3	→ 20.5
Prop. 20.4	→ 20.6	Q. 20.5	→ 20.7	Q. 20.6	→ 20.13
Q. 20.7	→ 20.16	Q. 20.8	→ 20.20	Q. 20.9	→ 20.23
Q. 20.10	→ 20.28	Q. 20.11	→ 20.32	Q. 20.12	→ 20.41
Answers (§20 of 23 October)					
Def. 20.1	→ 20.1	Prop. 20.2	→ 20.2	Prop. 20.3	→ 20.8
Thm. 20.4	→ 20.9	Prop. 20.5	→ 20.10	Prop. 20.6	→ 20.11
Ex. 20.7	→ 20.12	Prop. 20.8	→ 20.14	Thm. 20.9	→ 20.15
Prop. 20.10	→ 20.17	Prop. 20.11	→ 20.18	Rem. 20.12	→ 20.19
Prop. 20.13	→ 20.21	Hyp. 20.14	→ 20.22	Prop. 20.15	→ 20.24
Rem. 20.16	→ 20.25	Prop. 20.17	→ 20.29	Rem. 20.18	→ 20.30
Def. 20.19	→ 20.33	Prop. 20.20	→ 20.34	Thm. 20.21	→ 20.35
Prop. 20.22	→ 20.36	Prop. 20.23	→ 20.37	Rem. 20.24	→ 20.38
Prop. 20.25	→ 20.42	Prop. 20.26	→ 20.43	Rem. 20.27	→ 20.44
§21 (24 October; in brackets the numbers of the earlier form, which §22 cites)					
Thm. 21.1	→ 21.1	Rem. 21.2	→ 21.3	Rem. 21.3 [21.2]	→ 21.4
Ex. 21.4 [21.3]	→ 21.7	Ex. 21.5	→ 21.8	Prop. 21.6 [21.4]	→ 21.9
Prop. 21.7 [21.5]	→ 21.10	Rem. 21.8 [21.6]	→ 21.11	Prop. 21.9 [21.7]	→ 20.26

old number → number in this file					
Rem. 21.10 [21.8]	→ 20.27	Prop. 21.11 [21.9]	→ 20.45	Rem. 21.12 [21.10]	→ 20.46
Rem. 21.13 [21.11]	→ 20.31	Prop. 21.14	→ 21.5	Cor. 21.15	→ 21.6
Rem. 21.16	→ L.1				
§22 (24 October)					
Thm. 22.1	→ 21.1	Rem. 22.2	→ 21.3	Ex. 22.3	→ 21.8
Prop. 22.4	→ 20.39	Rem. 22.5	→ 20.40	Def. 22.6	→ 22.1
Prop. 22.7	→ 22.2	Thm. 22.8	→ 22.3	Cor. 22.9	→ 22.4

Merged statements: Theorem 22.1 of 24 October is Theorem 21.1(iii)–(iv), Remark 22.2 is Remark 21.3, and Example 22.3 is Example 21.8; the proof by moments of Theorem 21.1(iii) of 24 October is Remark 21.2; the sorting tables (§21.0, Remark 21.16, §22.0 of 24 October), the lists "Open" and "Corrections" of every entry, and the sections "Sorted again" and "Where §21 stands" are L.1–L.3. Proposition 20.2 of the questions entry, "The values are the moments of the law of the stride", is Proposition 20.4, now titled "The two ends of the stride's law".

L.5 Receipts

One archive, `receipts_s15_s22.zip`, with a folder per section and a `MANIFEST.txt` that maps each numbered statement to the script that prints it. §§15–17 were printed by `centre_ray.py`, which was delivered with those entries and is not in the archive; §18 by `ray_reading_s18.py`; §19 by `exam_s19.py`; the questions of §20 by `questions_s20.py`; the answers of §20 by the receipts of that entry, whose toolkit and laws read from closings are carried in `s22/s20/`; §21.1–21.2, Example 21.7, the three-mark reading, §21.5 and the propositions of §20 that came from the entry of 24 October (Propositions 20.26 and 20.45) by `receipts_s21.py`; §21.3, §21.4, Example 21.8(ii), Proposition 20.39, Remark 20.40 and §22 by the scripts of `s22/`. Every script uses python-flint and the standard library only, and in every entry the checks (tropical sampling, quadrature) are never part of a reading.

Aliases, last

Constants and functions. $\Gamma(\frac{1}{2})^2 = \pi$; $(\mathsf{T}_{m;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \zeta(m)$, so $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \zeta(3)$; $\ell = P_{\binom{[3]}{1}}(\mathbf{1};1,1) = 3L(\chi_{-3},2)$ [recalled], which agrees with the check of §19 to 40 digits and with the gauge end to 390; $P_{\binom{[4]}{1}}(\mathbf{1};1,1) = 7\zeta(3)$; $\Lambda_D = L(\chi_D,2)$, the Kronecker sign being the quadratic character of discriminant D, and $\Lambda_{-3}(s) = L(\chi_{-3},s)$, so $\Lambda_{-3}(3) = \frac{4\pi^3}{81\sqrt{3}}$, which $\Gamma(\frac{1}{2})^6/\sqrt{3}$ stood for in §18; $S_j = \mathrm{Ls}_j(\frac{2\pi}{3})$, the log-sine integrals, and $S_2 = \mathrm{Cl}_2(\frac{2\pi}{3})$, the Clausen value; $B(p,q;x)$ is the incomplete Beta function; the ε of §20.7 is the regulator of $\mathbb{Q}(\sqrt{21})$. $\mathsf{G}_{\nu;(1,-1)}(1,y) = 2y^{\nu/2}K_\nu(2\sqrt{y})$, so $\mathsf{G}_{\nu;(1,-1)}(a,a) = 2K_\nu(2a)$, the leans are moments of products of Bessel functions K_ν of integer orders, and Proposition 18.8 reads the lean as the order of the Bessel functions. In Bessel form the three-mark centre is $P(k) = 4^{2-k}\Gamma(k)^{-2}\int_0^\infty x^{2k-1}K_0(x)^3dx$, the equal-mass two-loop vacuum (sunrise) integral with propagator powers k in dimension $2k$; log-sine integrals at $\frac{2\pi}{3}$ are the constants of the ε-expansions of that integral [recalled].
Lists. $\binom{[n]}{1}$ is the banana; H is the cycle matroid of a triangle with one edge doubled; $\binom{[4]}{1}$ and $\binom{[4]}{2}$ are $U_{1,4}$ and $U_{2,4}$; $\mathrm{M}(K_4)$ is the cycle matroid of K_4, the wheel with three spokes. N is the matroid of six points of a plane with exactly three of them collinear, $U_{3,6}$ with one basis removed; it has a $U_{2,4}$ minor, so it is neither binary nor graphic (Tutte). V, the eight-mark list, is the Vámos matroid,

realizable over no field. At the full lean $(\mathbf{1}; 0, 2)$ of a graph the corner is the Hepp bound and $P(1)$ the Feynman period, $6\zeta(3)$ for the fixture; the observed correlation between the two, with the conjecture that equal Hepp bounds mean equal periods, is the classical side of Question 20.13 (Panzer); tropical sampling is tropical Monte Carlo over Hepp sectors.

§15. $x^{(1)}(1) = 8\int_0^\infty tI_0(t)K_0(t)^4\,dt$, whose evaluation by the Gammas at fifteenths is the classical one through the singular modulus of $\mathbb{Q}(\sqrt{-15})$ [recalled]; Theorem 15.25 gives the recurrences of the moments $\int_0^\infty t^{2s-1}K_0(t)^n\,dt$, $n \le 7$, by adding marks. The direction $x - 1$ of the block (O15) is $4\int_0^\infty t^2 I_1(t)K_0(t)^4\,dt$, to 25 digits [computed].

§17. C_1 is the Chowla–Selberg period of $\mathbb{Q}(\sqrt{-5})$; $X_\backslash - X_/ = (e_2^2 - e_1e_3)/e_4$, so $x_1 = \int dy/(X_\backslash - X_/)$ is a period of the quartic surface $e_2^2 = e_1e_3$.

§18. The x_j are the classical periods of the Laurent polynomial $1/f$, and the law is the Picard–Fuchs operator of its pencil read on coefficients; Theorem 18.5(iv) is the mechanism of Apéry limits, the selection read off the slit (Pincherle's minimal solution, Perron's theorem).

§19. The closings are the sums of squares of trinomial coefficients, the even moments of the planar walk of three unit steps; (L) is an Apéry-like recurrence, and Step 9 is Apéry's limit made two-sided by positivity (Pincherle's minimal solution, read with signs); the certificate is a creative-telescoping certificate for the integral; φ is the tropicalization of $-\log f$, and $\int e^{-k\varphi}dy = 6/k^2$ its volume.

§20. For $\binom{\mathcal{E}}{1}$ the body $\mathcal{P} - \mathcal{P}$ of Proposition 20.3 is the root polytope of type A_{n-1}. The stride's law μ is the pushforward of Lebesgue measure under f, the density of states of the stride, and its density a period of the fibres; the values are its moments, and $\log P$ is the log-partition function of an exponential family. At the centre the level sets are anticanonical hypersurfaces of $\mathbb{P}^{n-1}$ (elliptic curves, K3 surfaces and Calabi–Yau threefolds for $n = 3, 4, 5$), and the equation in t is their Picard–Fuchs equation; for three marks the pencil $(x + y + z)(xy + yz + zx) = \lambda xyz$ is modular of level 6. Proposition 20.29 is the Mellin transform of the law, and its exponents are the Frobenius exponents. The top piece of a certificate is a Nullstellensatz certificate for the ideal of critical points and the whole a creative-telescoping certificate. Proposition 20.8 is the Hausdorff moment problem on $[0, f^\star]$, generated by the four families of the Markov–Lukács theorem; e^{-g} in Theorem 20.9 is e^{-G} for the Green's function G of $\mathbb{C} \setminus [0, f^\star]$ with pole at infinity (the growth of Chebyshev polynomials); Proposition 20.10 is the indeterminacy of a moment problem along a Stieltjes transform. Proposition 20.6 is the Gumbel law of Chapter G. The coaction of Theorem A is Brown's, and the group acting on rays is the mechanism of Rhin and Viola. Proposition 20.43 writes $\Gamma(sk)\Gamma(tk)P(k)$ as a Mellin transform of a product of Bessel functions of orders $k|c_e|$. In §20.5, the frame X: $\Gamma(\frac{1}{15})\Gamma(\frac{2}{15})\Gamma(\frac{4}{15})\Gamma(\frac{8}{15})$ is a Chowla–Selberg period of $\mathbb{Q}(\sqrt{-15})$; the F of Remark 20.27 is the Koblitz–Ogus function, that a constant F makes a Gamma monomial algebraic up to a power of π is Deligne's theorem, and the converse is the Rohrlich–Lang conjecture. In §20.6, Remark 20.31: at a nondegenerate gauge Katz's middle convolution with the quadratic character exchanges the local monodromy -1 with a unipotent block of size two, as adding a mark does; on a unipotent block of size $m \ge 2$ it gives the sign with a block of size $m - 1$, so it does not produce the exponent 1 at $t = 1$. In §20.8: Proposition 20.42 is Apéry's criterion, and Remark 20.44 compares with Apéry's recurrence for $\zeta(3)$, of characteristic roots $17 \pm 12\sqrt{2}$; the recurrence of Proposition 20.45(i) is Zagier's sporadic case C, $x_j = \sum_k \binom{j}{k}^2\binom{2k}{k}$, and ℓ its Apéry limit up to a rational factor; the irrationality of $L(\chi_{-3}, 2)$ is open [recalled].

§21. ϱ is the period of the vanishing cycle of the Morse critical point $\alpha^\star$, and Theorem 21.1(i) is Laplace's method with the Morse lemma. Theorem 21.1(iii) is the Frobenius method at a regular singular point together with the extension of a solution by zero; the line is the image of the variation $T - 1$ of the local monodromy at $f^\star$ (Picard–Lefschetz), and the forced exponent of (iv) is the local monodromy of a Morse point in dimension $n - 1$: T is a reflection when n is even and a transvection when n is odd, which is all Remark 21.3 sees of the parity. The formal solution at a simple root is Birkhoff's, Laplace's method at an endpoint is Watson's lemma, and the reading is the numerical

solution of a connection problem by Frobenius series. Proposition 21.5: (i) says that the Taylor series of $\log f$ at $\alpha^\star$ is the cumulant series of the letters' exponent laws; (ii) is Laplace's method with Wick's theorem; the germ at $f^\star$ is the Borel sum of the large-k expansion; the count of formal solutions at a multiple root in Corollary 21.6 is the Birkhoff–Adams theory of linear recurrences. The share side of Ex. 20.12 is a Mellin–Barnes integral, and that of Example 21.8 is $16\int_0^\infty xK_0(x)^5\,dx$, a Bessel moment.

§22. The torus value is the zeta Mahler measure of the letter (Akatsuka). At three marks $T(k) = W_3(-2k)$, the moments of the three-step uniform random walk; its density is that of the walk's distance in closed form, $\frac{2\sqrt{3}}{\pi}\frac{x}{3+x^2}\,{}_2F_1\big(\frac{1}{3},\frac{2}{3};1;\frac{x^2(9-x^2)^2}{(3+x^2)^3}\big)$ with $x = t^{-1/2}$ (Borwein, Straub, Wan, Zudilin); the closings are Zagier's sporadic sequence C. The m_j are higher Mahler measures of $1+x+y$: m_1 is Smyth's $\frac{3\sqrt{3}}{4\pi}L(\chi_{-3},2)$, and $m_2 = \frac{3}{\pi}\mathrm{Ls}_3(\frac{2\pi}{3}) + \frac{\pi^2}{4}$ (Borwein, Straub). The one-mark average in Corollary 22.4(iv) is Jensen's formula with the hypergeometric function ${}_2F_1(k,k;1;\cdot)$. Theorem 22.3 is, in Bessel form, an identity between the moments of $K_0(x)^3$, of $J_0(x)^3$ and of $K_0(x)^2I_0(x)$ (a Wick rotation).